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Three-dimensional memory devices and methods for forming the same

Abstract

In certain aspects, a three-dimensional (3D) memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first and second semiconductor structures. The first semiconductor structure includes an array of NAND memory strings, a first peripheral circuit of the array of NAND memory strings including a first transistor, a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, and a first semiconductor layer in contact with the first transistor. The polysilicon layer is in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a second peripheral circuit of the array of NAND memory strings including a second transistor, and a second semiconductor layer in contact with the second transistor. The second semiconductor layer is between the bonding interface and the second peripheral circuit. The polysilicon layer is between the first semiconductor layer and the second semiconductor layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10283493	12/2018	Nishida	N/A	N/A
2008/0049519	12/2007	Horch	365/185.28	G11C 16/12
2008/0153200	12/2007	Sitaram	N/A	N/A
2009/0168482	12/2008	Park et al.	N/A	N/A
2010/0302848	12/2009	Mikhalev	N/A	N/A
2011/0159682	12/2010	Yang	N/A	N/A
2014/0175637	12/2013	Stuber et al.	N/A	N/A
2018/0277517	12/2017	Kim et al.	N/A	N/A
2018/0358371	12/2017	Hwang et al.	N/A	N/A
2019/0221557	12/2018	Kim	N/A	H01L 24/80
2020/0105721	12/2019	Park	N/A	N/A
2020/0194452	12/2019	Xiao	N/A	H01L 23/5226
2020/0203328	12/2019	Park et al.	N/A	N/A
2020/0328176	12/2019	Liu	N/A	N/A
2020/0328180	12/2019	Cheng et al.	N/A	N/A
2020/0328186	12/2019	Liu	N/A	N/A
2020/0350320	12/2019	Cheng et al.	N/A	N/A
2020/0350322	12/2019	Liu et al.	N/A	N/A
2020/0363983	12/2019	Li et al.	N/A	N/A
2021/0104534	12/2020	Huo et al.	N/A	N/A
2021/0111089	12/2020	Liu	N/A	N/A
2021/0111122	12/2020	Liu	N/A	N/A
2021/0134753	12/2020	Lee	N/A	N/A

2021/0134778	12/2020	Huang et al.	N/A	N/A
2022/0375887	12/2021	Watanabe	N/A	H10B 80/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101147258	12/2007	CN	N/A
101179077	12/2007	CN	N/A
101286480	12/2007	CN	N/A
102237314	12/2010	CN	N/A
104282629	12/2014	CN	N/A
106169307	12/2015	CN	N/A
110192269	12/2018	CN	N/A
110249427	12/2018	CN	N/A
110291631	12/2018	CN	N/A
110546762	12/2018	CN	N/A
110620117	12/2018	CN	N/A
110720145	12/2019	CN	N/A
110731012	12/2019	CN	N/A
110876281	12/2019	CN	N/A
110945652	12/2019	CN	N/A
110970439	12/2019	CN	N/A
111211126	12/2019	CN	N/A
112271191	12/2020	CN	N/A
112614831	12/2020	CN	N/A
112802855	12/2020	CN	N/A
2018117102	12/2017	JP	N/A
2021-086411	12/2020	JP	H01L 27/115
20080024764	12/2007	KR	N/A
2020211271	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report issued in corresponding International Application No. PCT/CN2021/103418, mailed Mar. 28, 2022, 5 pages. cited by applicant

International Search Report issued in corresponding International Application No. PCT/CN2021/103411, mailed Apr. 6, 2022, 4 pages. cited by applicant

International Search Report issued in corresponding International Application No. PCT/CN2021/103408, mailed Mar. 28, 2022, 4 pages. cited by applicant

International Search Report issued in corresponding International Application No. PCT/CN2021/103420, mailed Mar. 28, 2022, 4 pages. cited by applicant

International Search Report issued in corresponding International Application No. PCT/CN2021/103497, mailed Mar. 28, 2022, 4 pages. cited by applicant

Supplemental European Search Report issued in corresponding EP Application No. 21 94 7510, dated Jan. 15, 2024, 8 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/103420, filed on Jun. 30, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” which is hereby incorporated by reference in its entirety. This application is also related to U.S. application Ser. No. 17/481,803, filed on Sep. 22, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/481,875, filed on Sep. 22, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” U.S. application Ser. No. 17/481,902, filed on Sep. 22, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES HAVING POLYSILICON LAYER AND METHODS FOR FORMING THE SAME,” and U.S. application Ser. No. 17/481,943, filed on Sep. 22, 2021, entitled “THREE-DIMENSIONAL MEMORY DEVICES AND METHODS FOR FORMING THE SAME,” all of which are hereby incorporated by reference in their entireties.

BACKGROUND

- (1) The present disclosure relates to memory devices and fabrication methods thereof.
- (2) Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and costly. As a result, memory density for planar memory cells approaches an upper limit.
- (3) A three-dimensional (3D) memory architecture can address the density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral circuits for facilitating operations of the memory array.

SUMMARY

- (4) In one aspect, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes an array of NAND memory strings, a first peripheral circuit of the array of NAND memory strings including a first transistor, a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, and a first semiconductor layer in contact with the first transistor. The polysilicon layer is in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a second peripheral circuit of the array of NAND memory strings including a second transistor, and a second semiconductor layer in contact with the second transistor. The second semiconductor layer is between the bonding interface and the second peripheral circuit. The polysilicon layer is between the first semiconductor layer and the second semiconductor layer.
- (5) In another aspect, a system includes a memory device configured to store data. The memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes an array of NAND memory strings, a first peripheral circuit of the array of NAND memory strings including a first transistor, a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, and a first semiconductor layer in contact with the first transistor. The polysilicon layer is in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a second peripheral circuit of the array of NAND memory strings including a second transistor, and a second semiconductor layer in contact with the second transistor. The second semiconductor layer is between the bonding interface and the second peripheral circuit. The polysilicon layer is between the first semiconductor layer and the second semiconductor layer. The system also includes a memory controller coupled to

the memory device and configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit.

(6) In still another aspect, a method for forming a 3D memory device is disclosed. A first transistor is formed on a first substrate. A polysilicon layer is formed above the first transistor on the first substrate. An array of NAND memory strings is formed on the polysilicon layer. A second transistor is formed on a front side a second substrate. The first substrate and the second substrate are bonded in a face-to-back manner.

(7) In yet another aspect, a method for forming a 3D memory device is disclosed. A first transistor is formed on a first substrate. A polysilicon layer is formed above the first transistor on the first substrate. An array of NAND memory strings is formed on the polysilicon layer. A semiconductor layer is formed on above the array of NAND memory strings. The semiconductor layer includes single crystalline silicon. A second transistor is formed on the semiconductor layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

(2) FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device, according to some aspects of the present disclosure.

(3) FIG. 1B illustrates a schematic view of a cross-section of another 3D memory device, according to some aspects of the present disclosure.

(4) FIG. 2 illustrates a schematic circuit diagram of a memory device including peripheral circuits, according to some aspects of the present disclosure.

(5) FIG. 3 illustrates a block diagram of a memory device including a memory cell array and peripheral circuits, according to some aspects of the present disclosure.

(6) FIG. 4A illustrates a block diagram of peripheral circuits provided with various voltages, according to some aspects of the present disclosure.

(7) FIG. 4B illustrates a schematic diagram of peripheral circuits provided with various voltages arranged in separate semiconductor structures, according to some aspects of the present disclosure.

(8) FIGS. 5A and 5B illustrate a perspective view and a side view, respectively, of a planar transistor, according to some aspects of the present disclosure.

(9) FIGS. 6A and 6B illustrate a perspective view and a side view, respectively, of a 3D transistor, according to some aspects of the present disclosure.

(10) FIG. 7 illustrates a circuit diagram of a word line driver and a page buffer, according to some aspects of the present disclosure.

(11) FIG. 8 illustrates a side view of a NAND memory string in 3D memory devices, according to some aspects of the present disclosure.

(12) FIGS. 9A and 9B illustrate schematic views of cross-sections of 3D memory devices having two stacked semiconductor structures, according to various aspects of the present disclosure.

(13) FIG. 10 illustrates a schematic view of a cross-section of the 3D memory devices in FIGS. 9A and 9B, according to various aspects of the present disclosure.

(14) FIGS. 11A and 11B illustrate side views of various examples of the 3D memory device in FIG. 10, according to various aspects of the present disclosure.

(15) FIGS. 12A-12G illustrate a fabrication process for forming the 3D memory device in FIG. 10, according to some aspects of the present disclosure.

(16) FIGS. 13A and 13B illustrate another fabrication process for forming the 3D memory devices

in FIG. 10, according to some aspects of the present disclosure.

(17) FIG. 14 illustrates a flowchart of a method for forming the 3D memory device in FIG. 10, according to some aspects of the present disclosure.

(18) FIGS. 15A and 15B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 9A and 9B, according to various aspects of the present disclosure.

(19) FIGS. 16A and 16B illustrate side views of various examples of the 3D memory devices in FIGS. 15A and 15B, according to various aspects of the present disclosure.

(20) FIGS. 17A-17H illustrate a fabrication process for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure.

(21) FIGS. 18A-18F illustrate another fabrication process for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure.

(22) FIG. 19 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure.

(23) FIG. 20 illustrates a flowchart of a method for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure.

(24) FIGS. 21A and 21B illustrate schematic views of cross-sections of 3D memory devices having two stacked semiconductor structures, according to various aspects of the present disclosure.

(25) FIGS. 22A and 22B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 21A and 21B, according to some aspects of the present disclosure.

(26) FIGS. 23A and 23B illustrate side views of various examples of the 3D memory devices in FIGS. 22A and 22B, according to various aspects of the present disclosure.

(27) FIGS. 24A-24F illustrate a fabrication process for forming the 3D memory device in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(28) FIGS. 25A-25G illustrate another fabrication process for forming the 3D memory device in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(29) FIG. 26 illustrates a flowchart of a method for forming the 3D memory device in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(30) FIG. 27 illustrates a flowchart of another method for forming the 3D memory device in FIGS. 22A and 22B, according to some aspects of the present disclosure.

(31) FIGS. 28A and 28B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 21A and 21B, according to some aspects of the present disclosure.

(32) FIGS. 29A and 29B illustrate side views of various examples of the 3D memory devices in FIGS. 28A and 28B, according to various aspects of the present disclosure.

(33) FIGS. 30A-30G illustrate a fabrication process for forming the 3D memory device in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(34) FIGS. 31A-31H illustrate another fabrication process for forming the 3D memory device in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(35) FIG. 32 illustrates a flowchart of a method for forming the 3D memory device in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(36) FIG. 33 illustrates a flowchart of another method for forming the 3D memory device in FIGS. 28A and 28B, according to some aspects of the present disclosure.

(37) FIGS. 34A-34D illustrate a fabrication process of transfer bonding, according to some aspects of the present disclosure.

(38) FIGS. 35A-35D illustrate another fabrication process of transfer bonding, according to some aspects of the present disclosure.

(39) FIG. 36 illustrates a block diagram of an exemplary system having a memory device, according to some aspects of the present disclosure.

(40) FIG. 37A illustrates a diagram of an exemplary memory card having a memory device, according to some aspects of the present disclosure.

(41) FIG. 37B illustrates a diagram of an exemplary solid-state drive (SSD) having a memory

device, according to some aspects of the present disclosure.

(42) The present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

(43) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the present disclosure can also be employed in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, and modified with one another and in ways not specifically depicted in the drawings, such that these combinations, adjustments, and modifications are within the scope of the present disclosure.

(44) In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(45) It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

(46) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(47) As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layers thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductors and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

(48) With the development of 3D memory devices, such as 3D NAND Flash memory devices, the more stacked layers (e.g., more word lines and the resulting more memory cells) require more peripheral circuits (and the components, e.g., transistors, forming the peripheral circuits) for operating the 3D memory devices. For example, the number and/or size of page buffers needs to increase to match the increased number of memory cells. In another example, the number of string drivers in the word line driver is proportional to the number of word lines in the 3D NAND Flash memory. Thus, the continuous increase of the word lines also increases the area occupied by the

word line driver, as well as the complexity of metal routings, sometimes even the number of metal layers. Moreover, in some 3D memory devices in which the peripheral circuits are fabricated under the memory cell array, sometimes known as “periphery under cell” (PUC) architecture or “complementary metal-oxide-semiconductor (CMOS) under the array” (CuA) architecture, the continuous increase of peripheral circuits' areas makes it the bottleneck for reducing the total chip size since the memory cell array can be scaled up vertically by increasing the number of levels instead of increasing the planar size.

(49) Thus, it is desirable to reduce the planar areas occupied by the peripheral circuits of the 3D memory devices with the increased numbers of peripheral circuits and the transistors thereof. However, scaling down the transistor size of the peripheral circuits following the advanced CMOS technology node trend used for the logic devices would cause a significant cost increase and higher leakage current, which are undesirable for memory devices. Moreover, because the 3D NAND Flash memory devices require a relatively high voltage (e.g., above 5 V) in certain memory operations, such as program and erase, unlike logic devices, which can reduce its working voltage as the CMOS technology node advances, the voltage provided to the memory peripheral circuits cannot be reduced. As a result, scaling down the memory peripheral circuit sizes by following the trend for advancing the CMOS technology nodes, like the normal logic devices, becomes infeasible.

(50) To address one or more of the aforementioned issues, the present disclosure introduces various solutions in which the peripheral circuits of a memory device are disposed in different planes (levels, tiers) in the vertical direction, i.e., stacked over one another, to reduce the planar chip size of the peripheral circuits, as well as the total chip size of the memory device. In some implementations, the memory cell array (e.g., NAND memory strings), the memory peripheral circuits provided with a relatively high voltage (e.g., above 3.3 V), and the memory peripheral circuits provided with a relatively low voltage (e.g., below 2 V) are disposed in different planes in the vertical direction, i.e., stacked over one another, to further reduce the chip size. The 3D memory device architectures and fabrication processes disclosed in the present disclosure can be easily scaled up vertically to stack more peripheral circuits in different planes to further reduce the chip size. Moreover, the 3D memory device architectures and fabrication processes disclosed herein can be compatible with the PUC/CuA architecture and process. In some implementations, the memory cell array (e.g., NAND memory strings) can be formed on a deposited polysilicon (a.k.a. polycrystalline silicon) layer (e.g., in contact with a polysilicon source plate), as opposed to a single crystalline silicon substrate, which is suitable for “floating gate” type of NAND memory strings or certain designs of channel structures in “charge trap” type of NAND memory strings, for example, that are suitable for gate-induced drain leakage (GIDL) erase operations.

(51) The peripheral circuits can be separated into different planes in the vertical direction based on different performance requirements, for example, the voltages applied to the transistors thereof, which affect the dimensions of the transistors (e.g., gate dielectric thickness), dimensions of the substrates in which the transistors are formed (e.g., substrate thickness), and thermal budgets (e.g., the interconnect material). Thus, peripheral circuits with different dimension requirements (e.g., gate dielectric thickness and substrate thickness) and thermal budgets can be fabricated in different processes to reduce the design and process constraints from each other, thereby improving the device performance and fabrication complexity.

(52) According to some aspects of the present disclosure, the memory cell array and various peripheral circuits with different performance and dimension requirements can be fabricated in parallel on different substrates and then stacked over one another using various joining technologies, such as hybrid bonding, transfer bonding, etc. As a result, the fabrication cycle of the memory device can be further reduced. Moreover, since the thermal budgets of the different devices become independent to each other, interconnect materials with desirable electric performance but low thermal budget, such as copper, can be used in interconnecting the memory

cells and transistors of the peripheral circuits, thereby further improving the device performance. Bonding technologies can introduce additional benefits as well. In some implementations, hybrid bonding in a face-to-face manner achieves millions of parallel short interconnects between the bonded semiconductor structures to increase the throughput and input/output (I/O) speed of the memory devices. In some implementations, transfer bonding re-uses a single wafer to transfer thin semiconductor layers thereof onto different memory devices for forming transistors thereon, which can reduce the cost of the memory devices.

(53) The 3D memory device architectures and fabrication processes disclosed in the present disclosure also have the flexibility to allow various device pad-out schemes to meet different needs and different designs of the memory cell array. In some implementations, the pad-out interconnect layer is formed from the side of the semiconductor structure that has the peripheral circuits to shorten the interconnect distance between the pad-out interconnect layer and the transistors of the peripheral circuits to reduce the parasitic capacitance from the interconnects and improve the electric performance. In some implementations, the pad-out interconnect layer is formed on a thinned substrate to enable inter-layer vias (LLVs, e.g., submicron-level) for pad-out interconnects with high I/O throughput and low fabrication complicity.

(54) FIG. 1A illustrates a schematic view of a cross-section of a 3D memory device **100**, according to some aspects of the present disclosure. 3D memory device **100** represents an example of a bonded chip. In some implementations, the components of 3D memory device **100** (e.g., memory cell array and peripheral circuits) are formed separately on different substrates in parallel and then jointed to form a bonded chip (a process referred to herein as a “parallel process”). In some implementations, a semiconductor layer (e.g., single crystalline silicon) is attached onto another semiconductor structure using transferring bonding, then some of the components of 3D memory device **100** (e.g., some peripheral circuits) are formed on the attached semiconductor layer (a process referred to herein as a “series process”).

(55) It is noted that x- and y-axes are added in FIG. 1A to further illustrate the spatial relationships of the components of a semiconductor device. A substrate of a semiconductor device, e.g., 3D memory device **100**, includes two lateral surfaces (e.g., a top surface and a bottom surface) extending laterally in the x-direction (the lateral direction or width direction). As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of a semiconductor device is determined relative to the substrate of the semiconductor device in the y-direction (the vertical direction or thickness direction) when the substrate is positioned in the lowest plane of the semiconductor device in the y-direction. The same notion for describing the spatial relationships is applied throughout the present disclosure.

(56) 3D memory device **100** can include a first semiconductor structure **102** including an array of memory cells (also referred to herein as a “memory cell array”). In some implementations, the memory cell array includes an array of NAND Flash memory cells. For ease of description, a NAND Flash memory cell array may be used as an example for describing the memory cell array in the present disclosure. But it is understood that the memory cell array is not limited to NAND Flash memory cell array and may include any other suitable types of memory cell arrays, such as NOR Flash memory cell array, phase change memory (PCM) cell array, resistive memory cell array, magnetic memory cell array, spin transfer torque (STT) memory cell array, to name a few.

(57) First semiconductor structure **102** can be a NAND Flash memory device in which memory cells are provided in the form of an array of 3D NAND memory strings and/or an array of two-dimensional (2D) NAND memory cells. NAND memory cells can be organized into pages or fingers, which are then organized into blocks in which each NAND memory cell is coupled to a separate line called a bit line (BL). All cells with the same vertical position in the NAND memory cell can be coupled through the control gates by a word line (WL). In some implementations, a memory plane contains a certain number of blocks that are coupled through the same bit line. First semiconductor structure **102** can include one or more memory planes, and the peripheral circuits

that are needed to perform all the read/program (write)/erase operations can be included in a second semiconductor structure **104** and first semiconductor structure **102**.

(58) In some implementations, the array of NAND memory cells is an array of 2D NAND memory cells, each of which includes a floating-gate transistor. The array of 2D NAND memory cells includes a plurality of 2D NAND memory strings, each of which includes a plurality of memory cells connected in series (resembling a NAND gate) and two select transistors, according to some implementations. Each 2D NAND memory string is arranged in the same plane (i.e., referring to herein a flat, two-dimensional (2D) surface, different from the term “memory plane” in the present discourse) on the substrate, according to some implementations. In some implementations, the array of NAND memory cells is an array of 3D NAND memory strings, each of which extends vertically above the substrate (in 3D) through a stack structure, e.g., a memory stack. Depending on the 3D NAND technology (e.g., the number of layers/tiers in the memory stack), a 3D NAND memory string typically includes a certain number of NAND memory cells, each of which includes a floating-gate transistor or a charge-trap transistor.

(59) Consistent with the scope of the present disclosure, first semiconductor structure **102** can also include a polysilicon layer **106** on which the memory cell array is formed. In some implementations, the memory cell array includes an array of NAND memory strings, and polysilicon layer **106** is in contact with the sources of the NAND memory strings. That is, polysilicon layer **106** can serve as a common source plate of multiple NAND memory strings. As described below in detail, polysilicon layer **106** can be formed in first semiconductor structure **102** using one or more thin film deposition processes including, but not limited to, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or any combination thereof, which are compatible with PUC/CuA processes. It is understood that in some examples, polysilicon layer **106** may be a semiconductor layer in general that is not limited to polysilicon.

(60) As shown in FIG. 1A, first semiconductor structure **102** can further include some of the peripheral circuits of the memory cell array, and the memory cell array and the peripheral circuits in first semiconductor structure **102** can be separated by polysilicon layer **106** in the vertical direction. That is, polysilicon layer **106** can be disposed vertically between the memory cell array and the peripheral circuits in first semiconductor structure **102**. In some implementations, the peripheral circuits are disposed under polysilicon layer **106** and the memory cell array formed thereon. Depending on the thickness of polysilicon layer **106**, interconnects (e.g., inter-layer vias (ILVs) in the submicron-level or through substrate vias (TSVs) in the micron- or tens micron-level) can be formed through polysilicon layer **106** to make direct, short-distance (e.g., submicron- to tens micron-levels) electrical connections between the memory cell array and the peripheral circuits in first semiconductor structure **102**.

(61) As shown in FIG. 1A, 3D memory device **100** can also include a second semiconductor structure **104** including some of the peripheral circuits of the memory cell array in first semiconductor structure **102**. That is, the peripheral circuits of the memory cell array can be separated into at least two semiconductor structures (e.g., **102** and **104** in FIG. 1A). The peripheral circuits (a.k.a. control and sensing circuits) can include any suitable digital, analog, and/or mixed-signal circuits used for facilitating the operations of the memory cell array. For example, the peripheral circuits can include one or more of a page buffer, a decoder (e.g., a row decoder and a column decoder), a sense amplifier, a driver (e.g., a word line driver), an I/O circuit, a charge pump, a voltage source or generator, a current or voltage reference, any portions (e.g., a sub-circuit) of the functional circuits mentioned above, or any active or passive components of the circuit (e.g., transistors, diodes, resistors, or capacitors). The peripheral circuits in first and second semiconductor structures **102** and **104** can use CMOS technology, e.g., which can be implemented with logic processes in any suitable technology nodes.

(62) As shown in FIG. 1A, first and second semiconductor structures **102** and **104** are stacked over

one another in different planes, according to some implementations. As a result, the memory cell array in first semiconductor structure **102**, the peripheral circuits in first semiconductor structure **102**, and the peripheral circuits in second semiconductor structure **104** can be stacked over one another in different planes to reduce the planar size of 3D memory device **100**, compared with memory devices in which all the peripheral circuits are disposed in the same plane.

(63) As shown in FIG. **1A**, 3D memory device **100** further includes a bonding interface **103** vertically between first semiconductor structure **102** and second semiconductor structure **104**.

Bonding interface **103** can be an interface between two semiconductor structures formed by any suitable bonding technologies as described below in detail, such as hybrid bonding, anodic bonding, fusion bonding, transfer bonding, adhesive bonding, eutectic bonding, to name a few.

(64) It is understood that the relative positions of stacked first and second semiconductor structures **102** and **104** are not limited and may vary in different examples. FIG. **1B** illustrates a schematic view of a cross-section of another exemplary 3D memory device **101**, according to some implementations. Different from 3D memory device **100** in FIG. **1A** in which the memory cell array in first semiconductor structure is vertically between the peripheral circuits in first semiconductor structure **102** and the peripheral circuits in second semiconductor structure **104**, in 3D memory device **101** in FIG. **1B**, the peripheral circuits in first semiconductor structure **102** are vertically between the memory cell array in first semiconductor structure **102** and the peripheral circuits in second semiconductor structures. That is, second semiconductor structure **102** can be bonded to first semiconductor structure **102** on either side thereof, such as the side on which the memory cell array is formed in 3D memory device **100** in FIG. **1A**, or the side on which the peripheral circuits are formed in 3D memory device **101** in FIG. **1B**. As a result, in contrast to 3D memory device **100** in which bonding interface **103** is formed vertically between second semiconductor structure **104** and the memory cell array of first semiconductor structure **102**, a bonding interface **105** is formed vertically between second semiconductor structure **104** and the peripheral circuits of first semiconductor structure **102**. Similar to bonding interface **103**, bonding interface **105** can be an interface between two semiconductor structures formed by any suitable bonding technologies as described below in detail, such as hybrid bonding, anodic bonding, fusion bonding, transfer bonding, adhesive bonding, eutectic bonding, to name a few.

(65) As described below in detail, in some implementations, first and second semiconductor structures **102** and **104** can be fabricated separately (and in parallel in some implementations) by the parallel process, such that the thermal budget of fabricating one of first and second semiconductor structures **102** and **104** does not limit the processes of fabricating the other one. Moreover, a large number of interconnects (e.g., bonding contacts and/or ILVs/TSVs) can be formed across bonding interface **103** or **105** to make direct, short-distance (e.g., micron- or submicron-level) electrical connections between semiconductor structures **102** and **104**, as opposed to the long-distance (e.g., millimeter or centimeter-level) chip-to-chip data bus on the circuit board, such as printed circuit board (PCB), thereby eliminating chip interface delay and achieving high-speed I/O throughput with reduced power consumption. Data transfer among the memory cell array and the different peripheral circuits in different semiconductor structures **102** and **104** can be performed through the interconnects (e.g., bonding contacts and/or ILVs/TSVs) across bonding interfaces **103** or **105** and through polysilicon layer **106**. By vertically integrating first and second semiconductor structures **102** and **104**, as well as vertically separating the memory cell array and the peripheral circuits into different planes in first semiconductor structure **102**, the chip size can be reduced, and the memory cell density can be increased.

(66) FIG. **2** illustrates a schematic circuit diagram of a memory device **200** including peripheral circuits, according to some aspects of the present disclosure. Memory device **200** can include a memory cell array **201** and peripheral circuits **202** coupled to memory cell array **201**. 3D memory devices **100** and **101** may be examples of memory device **200** in which memory cell array **201** and at least two portions of peripheral circuits **202** may be included in different stacked semiconductor

structures **102** and **104**. Memory cell array **201** can be a NAND Flash memory cell array in which memory cells **206** are provided in the form of an array of NAND memory strings **208** each extending vertically above a substrate (not shown). In some implementations, each NAND memory string **208** includes a plurality of memory cells **206** coupled in series and stacked vertically. Each memory cell **206** can hold a continuous, analog value, such as an electrical voltage or charge, that depends on the number of electrons trapped within a region of memory cell **206**. Each memory cell **206** can be either a floating gate type of memory cell including a floating-gate transistor or a charge trap type of memory cell including a charge-trap transistor.

(67) In some implementations, each memory cell **206** is a single-level cell (SLC) that has two possible memory states and thus, can store one bit of data. For example, the first memory state “0” can correspond to a first range of voltages, and the second memory state “1” can correspond to a second range of voltages. In some implementations, each memory cell **206** is a multi-level cell (MLC) that is capable of storing more than a single bit of data in more than four memory states. For example, the MLC can store two bits per cell, three bits per cell (also known as triple-level cell (TLC)), or four bits per cell (also known as a quad-level cell (QLC)). Each MLC can be programmed to assume a range of possible nominal storage values. In one example, if each MLC stores two bits of data, then the MLC can be programmed to assume one of three possible programming levels from an erased state by writing one of three possible nominal storage values to the cell. A fourth nominal storage value can be used for the erased state.

(68) As shown in FIG. 2, each NAND memory string **208** can include a source select gate (SSG) transistor **210** at its source end and a drain select gate (DSG) transistor **212** at its drain end. SSG transistor **210** and DSG transistor **212** can be configured to activate selected NAND memory strings **208** (columns of the array) during read and program operations. In some implementations, SSG transistors **210** of NAND memory strings **208** in the same block **204** are coupled through a same source line (SL) **214**, e.g., a common SL, for example, to the ground. DSG transistor **212** of each NAND memory string **208** is coupled to a respective bit line **216** from which data can be read or programmed via an output bus (not shown), according to some implementations. In some implementations, each NAND memory string **208** is configured to be selected or deselected by applying a select voltage (e.g., above the threshold voltage of DSG transistor **212**) or a deselect voltage (e.g., 0 V) to respective DSG transistor **212** through one or more DSG lines **213** and/or by applying a select voltage (e.g., above the threshold voltage of SSG transistor **210**) or a deselect voltage (e.g., 0 V) to respective SSG transistor **210** through one or more SSG lines **215**.

(69) As shown in FIG. 2, NAND memory strings **208** can be organized into multiple blocks **204**, each of which can have a common source line **214**. In some implementations, each block **204** is the basic data unit for erase operations, i.e., all memory cells **206** on the same block **204** are erased at the same time. Memory cells **206** of adjacent NAND memory strings **208** can be coupled through word lines **218** that select which row of memory cells **206** is affected by read and program operations. In some implementations, each word line **218** is coupled to a page **220** of memory cells **206**, which is the basic data unit for program and read operations. The size of one page **220** in bits can correspond to the number of NAND memory strings **208** coupled by word line **218** in one block **204**. Each word line **218** can include a plurality of control gates (gate electrodes) at each memory cell **206** in respective page **220** and a gate line coupling the control gates.

(70) FIG. 8 illustrates a side view of NAND memory string **208** in 3D memory devices, according to some aspects of the present disclosure. As shown in FIG. 8, NAND memory string **208** can extend vertically through a memory stack **804** on a polysilicon layer **805**. Polysilicon layer **805** may be an example of polysilicon layer **106** in FIGS. 1A and 1B. Memory stack **804** can include interleaved gate conductive layers **806** and dielectric layers **808**. The number of the pairs of gate conductive layers **806** and dielectric layers **808** in memory stack **804** can determine the number of memory cells **206** in memory cell array **201**. Gate conductive layer **806** can include conductive materials including, but not limited to, tungsten (W), cobalt (Co), copper (Cu), aluminum (Al),

polysilicon, doped silicon, silicides, or any combination thereof. In some implementations, each gate conductive layer **806** includes a metal layer, such as a tungsten layer. In some implementations, each gate conductive layer **806** includes a doped polysilicon layer. Each gate conductive layer **806** can include control gates surrounding the memory cells, the gates of DSG transistors **212**, or the gates of SSG transistors **210**, and can extend laterally as DSG line **213** at the top of memory stack **804**, SSG line **215** at the bottom of memory stack **804**, or word line **218** between DSG line **213** and SSG line **215**.

(71) As shown in FIG. **8**, NAND memory string **208** includes a channel structure **812** extending vertically through memory stack **804**. In some implementations, channel structure **812** includes a channel hole filled with semiconductor material(s) (e.g., as a semiconductor channel **820**) and dielectric material(s) (e.g., as a memory film **818**). In some implementations, semiconductor channel **820** includes silicon, such as polysilicon. In some implementations, memory film **818** is a composite dielectric layer including a tunneling layer **826**, a storage layer **824** (also known as a “charge trap/storage layer”), and a blocking layer **822**. Channel structure **812** can have a cylinder shape (e.g., a pillar shape). Semiconductor channel **820**, tunneling layer **826**, storage layer **824**, blocking layer **822** are arranged radially from the center toward the outer surface of the pillar in this order, according to some implementations. Tunneling layer **826** can include silicon oxide, silicon oxynitride, or any combination thereof. Storage layer **824** can include silicon nitride, silicon oxynitride, silicon, or any combination thereof. Blocking layer **822** can include silicon oxide, silicon oxynitride, high dielectric constant (high-k) dielectrics, or any combination thereof. In one example, memory film **818** may include a composite layer of silicon oxide/silicon oxynitride/silicon oxide (ONO). Channel structure **812** can further include a channel plug **816** on the drain end of NAND memory string **208**. Channel plug **816** can include polysilicon and be in contact with semiconductor channel **820**.

(72) In some implementations, polysilicon layer **805** is in contact with semiconductor channel **820** of channel structure **812** on the source end of NAND memory string **208**. Parts of memory film **818** of channel structure **812** on the source end can be removed to expose semiconductor channel **820** to contact polysilicon layer **805**. In some implementations, part of semiconductor channel **820** on the source end of NAND memory string **208** is doped to form a doped region **832** that is in contact with polysilicon layer **805**. It is understood that in some examples, polysilicon layer **805** may be doped with the same dopant as doped region **832**, and the dopants may diffuse to part of semiconductor channel **820** to form doped region **832**. In some implementations, polysilicon layer **805** includes N-type doped polysilicon to enable GILD erase operations.

(73) As shown in FIG. **8**, a slit structure **828** does not include any conductors therein (e.g., a source contact) and thus, does not function as part of source line **214**, according to some implementations. Instead, source contacts (not shown) may be formed on an opposite side of polysilicon layer **805** with respect to channel structure **812**, such that the source contacts and parts of polysilicon layer **805** may function as parts of source line **214** coupled to the source of NAND memory string **208**, for example, for applying an erase voltage to the source of NAND memory string **208** during erase operations.

(74) Referring back to FIG. **2**, peripheral circuits **202** can be coupled to memory cell array **201** through bit lines **216**, word lines **218**, source lines **214**, SSG lines **215**, and DSG lines **213**. As described above, peripheral circuits **202** can include any suitable circuits for facilitating the operations of memory cell array **201** by applying and sensing voltage signals and/or current signals through bit lines **216** to and from each target memory cell **206** through word lines **218**, source lines **214**, SSG lines **215**, and DSG lines **213**. Peripheral circuits **202** can include various types of peripheral circuits formed using CMOS technologies. For example, FIG. **3** illustrates some exemplary peripheral circuits **202** including a page buffer **304**, a column decoder/bit line driver **306**, a row decoder/word line driver **308**, a voltage generator **310**, control logic **312**, registers **314**, an interface (I/F) **316**, and a data bus **318**. It is understood that in some examples, additional

peripheral circuits **202** may be included as well.

(75) Page buffer **304** can be configured to buffer data read from or programmed to memory cell array **201** according to the control signals of control logic **312**. In one example, page buffer **304** may store one page of program data (write data) to be programmed into one page **220** of memory cell array **201**. In another example, page buffer **304** also performs program verify operations to ensure that the data has been properly programmed into memory cells **206** coupled to selected word lines **218**.

(76) Row decoder/word line driver **308** can be configured to be controlled by control logic **312** and select block **204** of memory cell array **201** and a word line **218** of selected block **204**. Row decoder/word line driver **308** can be further configured to drive memory cell array **201**. For example, row decoder/word line driver **308** may drive memory cells **206** coupled to the selected word line **218** using a word line voltage generated from voltage generator **310**.

(77) Column decoder/bit line driver **306** can be configured to be controlled by control logic **312** and select one or more 3D NAND memory strings **208** by applying bit line voltages generated from voltage generator **310**. For example, column decoder/bit line driver **306** may apply column signals for selecting a set of N bits of data from page buffer **304** to be outputted in a read operation.

(78) Control logic **312** can be coupled to each peripheral circuit **202** and configured to control operations of peripheral circuits **202**. Registers **314** can be coupled to control logic **312** and include status registers, command registers, and address registers for storing status information, command operation codes (OP codes), and command addresses for controlling the operations of each peripheral circuit **202**.

(79) Interface **316** can be coupled to control logic **312** and configured to interface memory cell array **201** with a memory controller (not shown). In some implementations, interface **316** acts as a control buffer to buffer and relay control commands received from the memory controller and/or a host (not shown) to control logic **312** and status information received from control logic **312** to the memory controller and/or the host. Interface **316** can also be coupled to page buffer **304** and column decoder/bit line driver **306** via data bus **318** and act as an I/O interface and a data buffer to buffer and relay the program data received from the memory controller and/or the host to page buffer **304** and the read data from page buffer **304** to the memory controller and/or the host. In some implementations, interface **316** and data bus **318** are parts of an I/O circuit of peripheral circuits **202**.

(80) Voltage generator **310** can be configured to be controlled by control logic **312** and generate the word line voltages (e.g., read voltage, program voltage, pass voltage, local voltage, and verification voltage) and the bit line voltages to be supplied to memory cell array **201**. In some implementations, voltage generator **310** is part of a voltage source that provides voltages at various levels of different peripheral circuits **202**, as described below in detail. Consistent with the scope of the present disclosure, in some implementations, the voltages provided by voltage generator **310**, for example, to row decoder/word line driver **308**, column decoder/bit line driver **306**, and page buffer **304** are above certain levels that are sufficient to perform the memory operations. For example, the voltages provided to the page buffer circuits in page buffer **304** and/or the logic circuits in control logic **312** may be between 2 V and 3.3 V, such as 3.3 V, and the voltages provided to the driving circuits in row decoder/word line driver **308** and/or column decoder/bit line driver **306** may be between 5 V and 30 V.

(81) Different from logic devices (e.g., microprocessors), memory devices, such as 3D NAND Flash memory, require a wide range of voltages to be supplied to different memory peripheral circuits. For example, FIG. 4A illustrates a block diagram of peripheral circuits provided with various voltages, according to some aspects of the present disclosure. In some implementations, a memory device (e.g., memory device **200**) includes a low low voltage (LLV) source **401**, a low voltage (LV) source **403**, and a high voltage (HV) source **405**, each of which is configured to provide a voltage at a respective level (Vdd1, Vdd2, or Vdd3). For example, Vdd3>Vdd2>Vdd1.

Each voltage source **401**, **403**, or **405** can receive a voltage input at a suitable level from an external power source (e.g., a battery). Each voltage source **401**, **403**, or **405** can also include voltage converters and/or voltage regulators to convert the external voltage input to the respective level (Vdd1, Vdd2, or Vdd3) and maintain and output the voltage at the respective level (Vdd1, Vdd2, or Vdd3) through a corresponding power rail. In some implementations, voltage generator **310** of memory device **200** is part of voltage sources **401**, **403**, and **405**.

(82) In some implementations, LLV source **401** is configured to provide a voltage below 2 V, such as between 0.9 V and 2 V (e.g., 0.9 V, 0.95 V, 1 V, 1.05 V, 1.1 V, 1.15 V, 1.2 V, 1.25 V, 1.3 V, 1.35 V, 1.4 V, 1.45V, 1.5V, 1.55 V, 1.6 V, 1.65 V, 1.7 V, 1.75 V, 1.8 V, 1.85 V, 1.9 V, 1.95 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). In one example, the voltage is 1.2 V. In some implementations, LV source **403** is configured to provide a voltage between 2 V and 3.3 V (e.g., 2 V, 2.1 V, 2.2 V, 2.3 V, 2.4 V, 2.5 V, 2.6 V, 2.7 V, 2.8 V, 2.9 V, 3 V, 3.1 V, 3.2 V, 3.3 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). In one example, the voltage is 3.3 V. In some implementations, HV source **405** is configured to provide a voltage greater than 3.3 V, such as between 5 V and 30 V (e.g., 5 V, 6 V, 7 V, 8 V, 9 V, 10 V, 11 V, 12 V, 13V, 14V, 15V, 16V, 17V, 18V, 19 V, 20 V, 21 V, 22 V, 23 V, 24 V, 25 V, 26 V, 27 V, 28 V, 29 V, 30 V, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the voltage ranges described above with respect to HV source **405**, LV source **403**, and LLV source **401** are for illustrative purposes and non-limiting, and any other suitable voltage ranges may be provided by HV source **405**, LV source **403**, and LLV source **401**.

(83) Based on their suitable voltage levels (Vdd1, Vdd 2, or Vdd3), the memory peripheral circuits (e.g., peripheral circuits **202**) can be categorized into LLV circuits **402**, LV circuits **404**, and HV circuits **406**, which can be coupled to LLV source **401**, LV source **403**, and HV source **405**, respectively. In some implementations, HV circuits **406** include one or more driving circuits that are coupled to the memory cell array (e.g., memory cell array **201**) through word lines, bit lines, SSG lines, DSG lines, source lines, etc., and configured to drive the memory cell array by applying a voltage at a suitable level to the word lines, bit lines, SSG lines, DSG lines, source lines, etc., when performing memory operations (e.g., read, program, or erase). In one example, HV circuit **406** may include word line driving circuits (e.g., in row decoder/word line driver **308**) that are coupled to word lines and apply a program voltage (Vprog) or a pass voltage (Vpass) in the range of, for example, 5 V and 30 V, to the word lines during program operations. In another example, HV circuit **406** may include bit line driving circuits (e.g., in column decoder/bit line driver **306**) that are coupled to bit lines and apply an erase voltage (Veras) in the range of, for example, 5 V and 30 V, to bit lines during erase operations. In some implementations, LV circuits **404** include page buffer circuits (e.g., in latches of page buffer **304**) and are configured to buffer the data read from or programmed to the memory cell array. For example, the page buffer may be provided with a voltage of, for example, 3.3 V, by LV source **403**. LV circuits **404** can also include logic circuits (e.g., in control logic **312**). In some implementations, LLV circuits **402** include an I/O circuit (e.g., in interface **316** and/or data bus **318**) configured to interface the memory cell array with a memory controller. For example, the I/O circuit may be provided with a voltage of, for example, 1.2 V, by LLV source **401**.

(84) As described above, to reduce the total area occupied by the memory peripheral circuits, peripheral circuits **202** can be separately formed in different planes based on different performance requirements, such as the applied voltages. For example, FIG. **4B** illustrates a schematic diagram of peripheral circuits provided with various voltages arranged in separate semiconductor structures, according to some aspects of the present disclosure. In some implementations, LLV circuits **402** and HV circuits **406** are separated, for example, in semiconductor structures **408** and **410**, respectively, due to their significant difference in voltages and the resulting difference in device dimensions, such as different substrate thicknesses and different gate dielectric thicknesses. In one

example, the thickness of the semiconductor layer (e.g., a substrate or a thinned substrate) in which HV circuits **406** are formed in semiconductor structure **410** may be larger than the thickness of the semiconductor layer (e.g., a substrate or a thinned substrate) in which LLV circuits **402** are formed in semiconductor structure **408**. In another example, the thickness of the gate dielectric of transistors forming HV circuits **406** may be larger than the thickness of the gate dielectric of transistors forming LLV circuits **402**. For example, the thickness difference may be at least 5-fold. It is understood that stacked LLV circuits **402** and HV circuits **406** in different planes may be formed in two semiconductor structure **408** or **410** separated by a bonding interface (e.g., in FIGS. **1A** and **1B**).

(85) LV circuits **404** can be formed in either semiconductor structure **408** or **410**, or in another semiconductor, i.e., in the same plane as LLV circuits **402** or HV circuits **406**, or a different plane from LLV circuits **402** and HV circuits **406**. As shown in FIG. **4B**, in some implementations, some of LV circuits **404** are formed in semiconductor structure **408**, i.e., in the same plane as LLV circuits **402**, while some of LV circuits **404** are formed in semiconductor structure **410**, i.e., in the same plane as HV circuits **406**. That is, LV circuits **404** can be separated into different planes as well. The thickness of the gate dielectric of transistors forming LV circuits **404** in semiconductor structure **408** can be the same as the thickness of the gate dielectric of transistors forming LV circuits **404** in semiconductor structure **410**, for example, when the same voltage is applied to LV circuits **404** in different semiconductor structures **408** and **410**. In some implementations, the same voltage is applied to both LV circuits **404** in semiconductor structure **408** and the LV circuits **404** in semiconductor structure **410**, such that the voltage applied to HV circuits **406** in semiconductor structure **410** is higher than the voltage applied to LV circuits **404** in semiconductor structure **408** or **410**, which is in turn higher than the voltage applied to LLV circuits **402** in semiconductor structure **408**. Moreover, since the voltage applied to LV circuits **404** is between the voltages applied to HV circuits **406** and LLV circuits **402**, the thickness of the gate dielectric of transistors forming LV circuits **404** is between the thickness of the gate dielectric of transistors forming HV circuits **406** and the thickness of the gate dielectric of transistors forming LLV circuits **402**, according to some implementations. For example, the gate dielectric thickness of transistors forming LV circuits **404** may be larger than the gate dielectric thickness of transistors forming LLV circuits **402**, but smaller than the gate dielectric thickness of transistors forming HV circuits **406**.

(86) Based on the different performance requirements (e.g., associated with different applied voltages), peripheral circuits **202** can be separated into at least two stacked semiconductor structures **408** and **410** in different planes. In some implementations, the I/O circuits in interface **316** and/or data bus **318** (as LLV circuits **402**) and logic circuits in control logic **312** (as part of LV circuits) are disposed in semiconductor structure **408**, while the page buffer circuits in page buffer **304** and driving circuits in row decoder/word line driver **308** and column decoder/bit line driver **306** are disposed in semiconductor structure **410**. For example, FIG. **7** illustrates a circuit diagram of word line driver **308** and page buffer **304**, according to some aspects of the present disclosure.

(87) In some implementations, page buffer **304** includes a plurality of page buffer circuits **702** each coupled to one NAND memory string **208** via a respective bit line **216**. That is, memory device **200** can include bit lines **216** respectively coupled to NAND memory strings **208**, and page buffer **304** can include page buffer circuits **702** respectively coupled to bit lines **216** and NAND memory strings **208**. Each page buffer circuit **702** can include one or more latches, switches, supplies, nodes (e.g., data nodes and I/O nodes), current mirrors, verify logic, sense circuits, etc. In some implementations, each page buffer circuit **702** is configured to store sensing data corresponding to read data, which is received from a respective bit line **216**, and output the stored sensing data to at the time of the read operation; each page buffer circuit **702** is also configured to store program data and output the stored program data to a respective bit line **216** at the time of the program operation.

(88) In some implementations, word line driver **308** includes a plurality of string drivers **704** (a.k.a. driving circuits) respectively coupled to word lines **218**. Word line driver **308** can also include a

plurality of local word lines **706** (LWLs) respectively coupled to string drivers **704**. Each string driver **704** can include a gate coupled to a decoder (not shown), a source/drain coupled to a respective local word line **706**, and another source/drain coupled to a respective word line **218**. In some memory operations, the decoder can select certain string drivers **704**, for example, by applying a voltage signal greater than the threshold voltage of string drivers **704**, and a voltage (e.g., program voltage, pass voltage, or erase voltage) to each local word line **706**, such that the voltage is applied by each selected string driver **704** to a respective word line **218**. In contrast, the decoder can also deselect certain string drivers **704**, for example, by applying a voltage signal smaller than the threshold voltage of string drivers **704**, such that each deselected string driver **704** floats a respective word line **218** during the memory operation.

(89) In some implementations, page buffer circuits **702** include parts of LV circuits **404** disposed in semiconductor structures **408** and/or **410**. In one example, since the number of page buffer circuits **702** increases as the number of bit numbers increases, which may occupy a large area for memory devices with large numbers of memory cells, page buffer circuits **702** may be to semiconductor structures **408** and **410**. In some implementations, string drivers **704** include parts of HV circuits **406** disposed in semiconductor structure **410**.

(90) Consistent with the scope of the present disclosure, each peripheral circuit **202** can include a plurality of transistors as the basic building units thereof. The transistors can be metal-oxide-semiconductor field-effect-transistors (MOSFETs) in 2D (2D transistors, a.k.a. planar transistors) or 3D (3D transistors). For example, FIGS. 5A and 5B illustrate a perspective view and a side view, respectively, of a planar transistor **500**, according to some aspects of the present disclosure, and FIGS. 6A and 6B illustrate a perspective view and a side view, respectively, of a 3D transistor **600**, according to some aspects of the present disclosure. FIG. 5B illustrates the side view of the cross-section of planar transistor **500** in FIG. 5A in the BB plane, and FIG. 6B illustrates the side view of the cross-section of 3D transistor **600** in FIG. 6A in the BB plane.

(91) As shown in FIGS. 5A and 5B, planar transistor **500** can be a MOSFET on a substrate **502**, which can include silicon (e.g., single crystalline silicon, c-Si), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), or any other suitable semiconductor materials. Trench isolations **503**, such as shallow trench isolations (STI), can be formed in substrate **502** and between adjacent planar transistors **500** to reduce current leakage. Trench isolations **503** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high dielectric constant (high-k) dielectrics (e.g., aluminum oxide, hafnium oxide, zirconium oxide, etc.). In some implementations, high-k dielectric materials include any dielectrics having a dielectric constant, or k-value, higher than that of silicon nitride ($k > 7$). In some implementations, trench isolation **503** includes silicon oxide.

(92) As shown in FIGS. 5A and 5B, planar transistor **500** can also include a gate structure **508** on substrate **502**. In some implementations, gate structure **508** is on the top surface of substrate **502**. As shown in FIG. 5B, gate structure **508** can include a gate dielectric **507** on substrate **502**, i.e., above and in contact with the top surface of substrate **502**. Gate structure **508** can also include a gate electrode **509** on gate dielectric **507**, i.e., above and in contact with gate dielectric **507**. Gate dielectric **507** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. In some implementations, gate dielectric **507** includes silicon oxide, i.e., a gate oxide. Gate electrode **509** can include any suitable conductive materials, such as polysilicon, metals (e.g., W, Cu, Al, etc.), metal compounds (e.g., TiN, TaN, etc.), or silicides. In some implementations, gate electrode **509** includes doped polysilicon, i.e., a gate poly.

(93) As shown in FIGS. 5A, planar transistor **500** can further include a pair of a source and a drain **506** in substrate **502**. Source and drain **506** can be doped with any suitable P-type dopants, such as boron (B) or Gallium (Ga), or any suitable N-type dopants, such as phosphorus (P) or arsenic (As). Source and drain **506** can be separated by gate structure **508** in the plan view. In other words, gate structure **508** is formed between source and drain **506** in the plan view, according to some

implementations. The channel of planar transistor **500** in substrate **502** can be formed laterally between source and drain **506** under gate structure **508** when a gate voltage applied to gate electrode **509** of gate structure **508** is above the threshold voltage of planar transistor **500**. As shown in FIGS. 5A and 5B, gate structure **508** can be above and in contact with the top surface of the part of substrate **502** in which the channel can be formed (the active region). That is, gate structure **508** is in contact with only one side of the active region, i.e., in the plane of the top surface of substrate **502**, according to some implementations. It is understood, although not shown in FIGS. 5A and 5B, planar transistor **500** may include additional components, such as wells and spacers.

(94) As shown in FIGS. 6A and 6B, 3D transistor **600** can be a MOSFET on a substrate **602**, which can include silicon (e.g., single crystalline silicon, c-Si), SiGe, GaAs, Ge, silicon on insulator SOI, or any other suitable materials. In some implementations, substrate **602** includes single crystalline silicon. Trench isolations **603**, such as STI, can be formed in substrate **602** and between adjacent 3D transistors **600** to reduce current leakage. Trench isolations **603** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics (e.g., aluminum oxide, hafnium oxide, zirconium oxide, etc.). In some implementations, trench isolation **603** includes silicon oxide.

(95) As shown in FIGS. 6A and 6B, different from planar transistor **500**, 3D transistor **600** can further include a 3D semiconductor body **604** above substrate **602**. That is, in some implementations, 3D semiconductor body **604** at least partially extends above the top surface of substrate **602** to expose not only the top surface, but also the two side surfaces, of 3D semiconductor body **604**. As shown in FIGS. 6A and 6B, for example, 3D semiconductor body **604** may be in a 3D structure, which is also known as a “fin,” to expose three sides thereof. 3D semiconductor body **604** is formed from substrate **602** and thus, has the same semiconductor material as substrate **602**, according to some implementations. In some implementations, 3D semiconductor body **604** includes single crystalline silicon. Since the channels can be formed in 3D semiconductor body **604**, as opposed to substrate **602**, 3D semiconductor body **604** may be viewed as the active region for 3D transistor **600**.

(96) As shown in FIGS. 6A and 6B, 3D transistor **600** can also include a gate structure **608** on substrate **602**. Different from planar transistors **500** in which gate structure **508** is in contact with only one side of the active region, i.e., in the plane of the top surface of substrate **502**, gate structure **608** of 3D transistor **600** can be in contact with a plurality of sides of the active region, i.e., in multiple planes of the top surface and side surfaces of the 3D semiconductor body **604**. In other words, the active region of 3D transistor **600**, i.e., 3D semiconductor body **604**, can be at least partially surrounded by gate structure **608**.

(97) Gate structure **608** can include a gate dielectric **607** over 3D semiconductor body **604**, e.g., in contact with the top surface and two side surfaces of 3D semiconductor body **604**. Gate structure **608** can also include a gate electrode **609** over and in contact with gate dielectric **607**. Gate dielectric **607** can include any suitable dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or high-k dielectrics. In some implementations, gate dielectric **607** includes silicon oxide, i.e., a gate oxide. Gate electrode **609** can include any suitable conductive materials, such as polysilicon, metals (e.g., W, Cu, Al, etc.), metal compounds (e.g., TiN, TaN, etc.), or silicides. In some implementations, gate electrode **609** includes doped polysilicon, i.e., a gate poly.

(98) As shown in FIG. 6A, 3D transistor **600** can further include a pair of a source and a drain **606** in 3D semiconductor body **604**. Source and drain **606** can be doped with any suitable P-type dopants, such as B or Ga, or any suitable N-type dopants, such as P or Ar. Source and drain **606** can be separated by gate structure **608** in the plan view. In other words, gate structure **608** is formed between source and drain **606** in the plan view, according to some implementations. As a result, multiple channels of 3D transistor **600** in 3D semiconductor body **604** can be formed laterally between source and drain **606** surrounded by gate structure **608** when a gate voltage applied to gate

electrode **609** of gate structure **608** is above the threshold voltage of 3D transistor **600**. Different from planar transistor **500** in which only a single channel can be formed on the top surface of substrate **502**, multiple channels can be formed on the top surface and side surfaces of 3D semiconductor body **604** in 3D transistor **600**. In some implementations, 3D transistor **600** includes a multi-gate transistor. It is understood, although not shown in FIGS. **6A**, and **6B**, 3D transistor **600** may include additional components, such as wells, spacers, and stressors (a.k.a. strain elements) at source and drain **606**.

(99) It is further understood that FIGS. **6A** and **6B** illustrate one example of 3D transistors that can be used in memory peripheral circuits, and any other suitable 3D multi-gate transistors may be used in memory peripheral circuits as well, including, for example, a gate all around (GAA) silicon on nothing (SON) transistor, a multiple independent gate FET (MIGET), a trigate FET, a H-gate FET, and a Ω -FET, a quadruple gate FET, a cylindrical FET, or a multi-bridge/stacked nanowire FET.

(100) Regardless of planar transistor **500** or 3D transistor **600**, each transistor a memory peripheral circuit can include a gate dielectric (e.g., gate dielectrics **507** and **607**) having a thickness T (gate dielectric thickness, e.g., shown in FIGS. **5B** and **6B**). The gate dielectric thickness T of a transistor can be designed to accommodate the voltage applied to the transistor. For example, referring back to FIGS. **4A** and **4B**, the gate dielectric thickness of transistors in HV circuits **406** (e.g., driving circuits such as string drivers **704**) may be larger than the gate dielectric thickness of transistors in LV circuits **404** (e.g., page buffer circuits **702** or logic circuits in control logic **312**), which may be in turn larger than the gate dielectric thickness of transistors in LLV circuits **402** (e.g., I/O circuits in interface **316** and data bus **318**). In some implementations, the difference between the gate dielectric thickness of transistors in HV circuits **406** and the dielectric thickness of transistors in LLV circuits **402** is at least 5-fold, such as between 5-fold and 50-fold. For example, the gate dielectric thickness of transistors in HV circuits **406** may be at least 5 times larger than the gate dielectric thickness of transistors in LLV circuits **402**.

(101) In some implementations, the dielectric thickness of transistors in LLV circuits **402** is between 2 nm and 4 nm (e.g., 2 nm, 2.1 nm, 2.2 nm, 2.3 nm, 2.4 nm, 2.5 nm, 2.6 nm, 2.7 nm, 2.8 nm, 2.9 nm, 3 nm, 3.1 nm, 3.2 nm, 3.3 nm, 3.4 nm, 3.5 nm, 3.6 nm, 3.7 nm, 3.8 nm, 3.9 nm, 4 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the LLV voltage range applied to LLV circuits **402**, as described above in detail, such as below 2 V (e.g., 1.2 V). In some implementations, the dielectric thickness of transistors in LV circuits **404** is between 4 nm and 10 nm (e.g., 4 nm, 4.5 nm, 5 nm, 5.5 nm, 6 nm, 6.5 nm, 7 nm, 7.5 nm, 8 nm, 8.5 nm, 9 nm, 9.5 nm, 10 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the LV voltage range applied to LV circuits **404**, as described above in detail, such as between 2 V and 3.3 V (e.g., 3.3 V). In some implementations, the dielectric thickness of transistors in HV circuits **406** is between 20 nm and 100 nm (e.g., 20 nm, 21 nm, 22 nm, 23 nm, 24 nm, 25 nm, 26 nm, 27 nm, 28 nm, 29 nm, 30 nm, 31 nm, 32 nm, 33 nm, 34 nm, 35 nm, 36 nm, 37 nm, 38 nm, 39 nm, 40 nm, 45 nm, 50 nm, 55 nm, 60 nm, 65 nm, 70 nm, 75 nm, 80 nm, 85 nm, 90 nm, 95 nm, 100 nm, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). It is understood that the thickness may be commensurate with the HV voltage range applied to HV circuits **406**, as described above in detail, such as greater than 3.3 V (e.g., between 5 V and 30 V).

(102) FIGS. **9A** and **9B** illustrate schematic views of cross-sections of 3D memory devices **900** and **901** having two stacked semiconductor structures, according to various aspects of the present disclosure. 3D memory devices **900** and **901** may be examples of 3D memory device **100** in FIG. **1A** in which the memory array cell of first semiconductor structure **102** is disposed vertically between the peripheral circuits of first semiconductor structure **102** and the peripheral circuits of second semiconductor structure **104**. That is, the two separate portions of the peripheral circuits can

be disposed on the two sides of 3D memory device **900** or **901**, respectively, in the vertical direction. As shown in FIGS. **9A** and **9B**, second semiconductor structure **104** including some of the peripheral circuits is bonded to first semiconductor structure **102** on the side having the memory cell array to form bonding interface **103** between second semiconductor structure **104** and the memory cell array of first semiconductor structure **102**, according to some implementations. (103) Moreover, as shown in FIGS. **9A** and **9B**, 3D memory device **900** or **901** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **9A**, second semiconductor structure **104** including some of the peripheral circuits on one side of 3D memory device **900** may include pad-out interconnect layer **902**. In another example shown in FIG. **9B**, first semiconductor structure **102** including the memory cell array and some of the peripheral circuits on another side of 3D memory device **901** may include pad-out interconnect layer **902**, such that 3D memory device **901** may be pad-out from another peripheral circuit side. In either example, 3D memory device **900** or **901** may be pad-out from a peripheral circuit side to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **900** or **901**.

(104) FIG. **10** illustrates a schematic view of a cross-section of the 3D memory devices in FIGS. **9A** and **9B**, according to some aspects of the present disclosure. 3D memory device **1000** may be an example of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **10**, 3D memory device **1000** can include stacked first and second semiconductor structures **102** and **104**. In some implementations, first semiconductor structure **102** includes a semiconductor layer **1002**, polysilicon layer **106**, a bonding layer **1008**, a memory cell array vertically between polysilicon layer **106** and bonding layer **1008**, and some of the peripheral circuits vertically between semiconductor layer **1002** and polysilicon layer **106**.

(105) The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with polysilicon layer **106** (e.g., as shown in FIG. **8**). Polysilicon layer **106** can be a deposited polysilicon layer (e.g., either N-type doped, P-type doped, or undoped), which is suitable for “floating gate” type of NAND memory strings or certain designs of channel structures (e.g., channel structure **812** in FIG. **8**) in “charge trap” type of NAND memory strings, for example, that are suitable for suitable for GIDL erase operations. Bonding layer **1008** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(106) In some implementations, the peripheral circuits in first semiconductor structure **102** are in contact with semiconductor layer **1002**, but not polysilicon layer **106**. That is, the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1002** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Through contacts (e.g., ILVs/TSVs) through polysilicon layer **106** can make direct, short-distance (e.g., submicron- or micron-level) electrical connections between the memory cell array and the peripheral circuits in first semiconductor structure **102**.

(107) In some implementations, second semiconductor structure **104** includes a semiconductor layer **1004**, a bonding layer **1010**, and some of the peripheral circuits of the memory cell array that are vertically between semiconductor layer **1004** and bonding layer **1010**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Similar to semiconductor layer **1002**, semiconductor layer **1004** can

include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance.

(108) Similar to bonding layer **1008** in first semiconductor structure **102**, bonding layer **1010** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. That is, bonding layers **1008** and **1010** can be disposed on opposite sides of bonding interface **103**, and the bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, different from through contacts (e.g., ILVs/TSVs), a large number (e.g., millions) of bonding contacts across bonding interface **103** can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(109) As shown in FIG. **10**, since first and second semiconductor structures **102** and **104** are bonded in a face-to-face manner (e.g., semiconductor layer **1002** being disposed on the bottom side of first semiconductor structure **102**, while semiconductor layer **1004** being disposed on the top side of second semiconductor structure **104** in FIG. **10**), the transistors of peripheral circuits in first and second semiconductor structures **102** and **104** are disposed facing toward each other, according to some implementations. Moreover, within first semiconductor structure **102**, since polysilicon layer **106** is vertically between the memory cell array and the peripheral circuits, and the memory cell array and the peripheral circuits formed on polysilicon layer **106** and semiconductor layer **1002**, respectively, the memory cell array and the peripheral circuits face toward the same direction (e.g., in the positive y-direction in FIG. **10**). It is understood that pad-out interconnect layer **902** in FIGS. **9A** and **9B** is omitted from 3D memory device **1000** in FIG. **10** for ease of illustration and may be included in 3D memory device **1000** as described above with respect to FIGS. **9A** and **9B**.

(110) As described above, first and second semiconductor structures **102** and **104** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and first semiconductor structure **102** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1002** and **1004** in first and second semiconductor structures **102** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1002** in first semiconductor structure **102** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in first and second semiconductor structures **102** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in first semiconductor structure **102** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**. The thicker gate dielectric can sustain a higher working voltage applied to the transistors in first semiconductor structure **102** than the transistors in second semiconductor structure **104** to avoid break down during high voltage operations.

(111) FIGS. **11A** and **11B** illustrate side views of various examples of 3D memory device **1000** in FIG. **10**, according to various aspects of the present disclosure. As shown in FIG. **11A**, as one example of 3D memory device **1000** in FIG. **10**, 3D memory device **1100** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104**, which are

stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. 11A), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, according to some implementations.

(112) As shown in FIG. 11A, first semiconductor structure **102** can include semiconductor layer **1002** having semiconductor materials. In some implementations, semiconductor layer **1002** is a silicon substrate having single crystalline silicon. First semiconductor structure **102** can also include a device layer **1102** above and in contact with semiconductor layer **1002**. In some implementations, device layer **1102** includes a first peripheral circuit **1104** and a second peripheral circuit **1106**. First peripheral circuit **1104** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **1106** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **1104** includes a plurality of transistors **1108** in contact with semiconductor layer **1002**, and second peripheral circuit **1106** includes a plurality of transistors **1110** in contact with semiconductor layer **1002**. Transistors **1108** and **1110** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1108** or **1110** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1108** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1110** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **1108** than transistor **1110**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1108** and **1110**) can be formed on or in semiconductor layer **1002** as well.

(113) In some implementations, first semiconductor structure **102** further includes an interconnect layer **1112** above device layer **1102** to transfer electrical signals to and from peripheral circuits **1106** and **1104**. As shown in FIG. 11A, interconnect layer **1112** can be vertically between polysilicon layer **106** and device layer **1102** (including transistors **1108** and **1110** of peripheral circuits **1104** and **1106**). Interconnect layer **1112** can include a plurality of interconnects (also referred to herein as “contacts”), including lateral lines and vias. As used herein, the term “interconnects” can broadly include any suitable types of interconnects, such as middle-end-of-line (MEOL) interconnects and back-end-of-line (BEOL) interconnects. The interconnects in interconnect layer **1112** can be coupled to transistors **1108** and **1110** of peripheral circuits **1104** and **1106** in device layer **1102**. Interconnect layer **1112** can further include one or more interlayer dielectric (ILD) layers (also known as “intermetal dielectric (IMD) layers”) in which the lateral lines and vias can form. That is, interconnect layer **1112** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1102** are coupled to one another through the interconnects in interconnect layer **1112**. For example, peripheral circuit **1104** may be coupled to peripheral circuit **1106** through interconnect layer **1112**. The interconnects in interconnect layer **1112** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1112** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low dielectric constant (low-k) dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1112** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(114) As shown in FIG. 11A, first semiconductor structure can further include polysilicon layer **106** above and in contact with interconnect layer **1112**. Polysilicon layer **106** is a doped polysilicon layer on interconnect layer **1112**, as described below in detail with respect to the fabrication process, according to some implementations. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in polysilicon layer **106** as well.

(115) As shown in FIG. 11A, first semiconductor structure **102** can also include a memory cell

array, such as an array of NAND memory strings **208** above and in contact with polysilicon layer **106**. The sources of the NAND memory strings **208** can be in contact with polysilicon layer **106**. In some implementations, polysilicon layer **106** is vertically between NAND memory strings **208** and device layer **1102** including transistors **1108** and **1110**. In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as channel structure **812**, described above in detail with respect to FIG. **8**. In some implementations, NAND memory string **208** is a “floating gate” type of NAND memory string, and polysilicon layer **106** is the source plate of the floating gate type of NAND memory strings.

(116) Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **1127**. Memory stack **1127** may be an example of memory stack **804** in FIG. **8**, and the conductive layer and dielectric layer in memory stack **1127** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **1127** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **1127**.

(117) As shown in FIG. **11A**, first semiconductor structure **102** can further include an interconnect layer **1128** above and in contact with NAND memory strings **208** to transfer electrical signals to and from NAND memory strings **208**. In some implementations, memory stack **1127** and NAND memory strings **208** are vertically between interconnect layer **1128** and polysilicon layer **106**. Interconnect layer **1128** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1128** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **1128** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **1128** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1128** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(118) As shown in FIG. **11A**, first semiconductor structure **102** can further include one or more contacts **1124** extending vertically through polysilicon layer **106**. In some implementations, contact **1124** couples the interconnects in interconnect layer **1128** to the interconnects in interconnect layer **1112** to make an electrical connection through polysilicon layer **106** between NAND memory strings **208** and transistors **1108** and **1110**. Contact **1124** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1124** includes W. In some implementations, contact **1124** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from polysilicon layer **106**. Depending on the thickness of polysilicon layer **106**, contact **1124** can be an ILV having a depth (in the vertical direction) in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth (in the vertical direction) in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(119) As shown in FIG. **11A**, first semiconductor structure **102** can further include a bonding layer **1008** at bonding interface **103** and above and in contact with interconnect layer **1128**. Bonding layer **1008** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, the bonding contacts of bonding layer **1008** include Cu. The remaining area of bonding layer **1008** can be

formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. The bonding contacts and surrounding dielectrics in bonding layer **1008** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO₂-to-SiO₂) bonding simultaneously. (120) As shown in FIG. **11A**, second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **103**. Second semiconductor structure **104** can also include a bonding layer **1010** at bonding interface **103**, e.g., on the opposite side of bonding interface **103** with respect to bonding layer **1008** in first semiconductor structure **102**. Bonding layer **1010** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials, such as Cu. The remaining area of bonding layer **1010** can be formed with dielectric materials, such as silicon oxide. The bonding contacts and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1008** of first semiconductor structure **102** and the bottom surface of bonding layer **1010** of second semiconductor structure **104**.

(121) As shown in FIG. **11A**, second semiconductor structure **104** can further include an interconnect layer **1126** above bonding layer **1010** to transfer electrical signals. Interconnect layer **1126** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. Interconnect layer **1126** can further include one or more ILD layers in which the lateral lines and vias can be formed. The interconnects in interconnect layer **1126** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1126** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1126** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **1126** can occur after the high-temperature processes in forming device layer **1114** in second semiconductor structure **104**, as well as being separated from the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **1126** having Cu can become feasible.

(122) As shown in FIG. **11A**, second semiconductor structure **104** can include a device layer **1114** above and in contact with interconnect layer **1126**. In some implementations, interconnect layer **1126** is vertically between bonding interface **103** and device layer **1114**. In some implementations, device layer **1114** includes a third peripheral circuit **1116** and a fourth peripheral circuit **1118**. Third peripheral circuit **1116** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **1118** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **1116** includes a plurality of transistors **1120**, and fourth peripheral circuit **1118** includes a plurality of transistors **1122** as well. Transistors **1120** and **1122** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1120** or **1122** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1120** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **1122** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **1120** than transistor **1122**.

(123) Moreover, the different voltages applied to different transistors **1120**, **1122**, **1108**, and **1110** in second and first semiconductor structures **104** and **102** can lead to differences of device dimensions between second and first semiconductor structures **104** and **102**. In some implementations, the thickness of the gate dielectric of transistor **1108** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1120** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **1108** than transistor **1120**. In some implementations, the thickness of the gate dielectric of transistor **1122** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **1110** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **1122** and transistor **1110**. In some implementations, the thickness of semiconductor layer **1002** in which transistor **1108** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **1120** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **1108** than transistor **1120**.

(124) As shown in FIG. **11A**, second semiconductor structure **104** can further include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a thinned silicon substrate having single crystalline silicon on which transistors **1120** and **1122** can be formed. Semiconductor layer **1004** can be disposed above and in contact with transistors **1120** and **1122** of peripheral circuits **1116** and **1118** in device layer **1114**. In some implementations, transistors **1120** and **1122** are disposed vertically between bonding interface **103** and semiconductor layer **1004**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1120** and **1122**) can be formed on or in semiconductor layer **1004** as well.

(125) As shown in FIG. **11A**, second semiconductor structure **104** can further include a pad-out interconnect layer **902** above and in contact with semiconductor layer **1004**. In some implementations, semiconductor layer **1004** is disposed vertically between pad-out interconnect layer **902** and transistors **1120** and **1122**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **1132**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **1126** can be formed on opposite sides of semiconductor layer **1004**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **1100** and external devices, e.g., for pad-out purposes.

(126) As shown in FIG. **11A**, second semiconductor structure **104** can further include one or more contacts **1130** extending vertically through semiconductor layer **1004**. In some implementations, contact **1130** couples the interconnects in interconnect layer **1126** to contact pads **1132** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1004**. Contact **1130** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1130** includes W. In some implementations, contact **1130** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **1130** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(127) As a result, peripheral circuits **1104**, **1106**, **1116**, and **1118** in first and second semiconductor structures **102** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **1112**, **1126**, and **1128**, bonding layers **1008** and **1010**, as well as contacts **1124**. Moreover, peripheral circuits **1104**, **1106**, **1116**, and **1118** and NAND memory strings **208** in 3D memory device **1100** can be further coupled to external devices through contacts **1130** and pad-out interconnect layer **902**.

(128) It is also understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having peripheral circuit **1116** as shown in FIG. **11A** (corresponding to FIG. **9A**) and may be from first semiconductor structure **102** having peripheral circuit **1104** (corresponding to FIG. **9B**). For example, as shown in FIG. **11B**, a 3D memory device **1101** may

include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1002** of first semiconductor structure **102** on which transistors **1108** of peripheral circuit **1104** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **1134** extending vertically through semiconductor layer **1002**. In some implementations, contact **1134** couples the interconnects in interconnect layer **1112** in first semiconductor structure **102** to contact pads **1132** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **1134** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1134** includes W. In some implementations, contact **1134** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **1134** can be an ILV having a thickness in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm). It is understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1100** and **1101** are not repeated for ease of description.

(129) FIGS. **12A-12G** illustrate a fabrication process for forming the 3D memory devices in FIG. **10**, according to some aspects of the present disclosure. FIGS. **13A** and **13B** illustrate another fabrication process for forming the 3D memory devices in FIG. **10**, according to some aspects of the present disclosure. FIG. **14** illustrates a flowchart of a method **1400** for forming the 3D memory devices in FIG. **10**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **12A-12G**, **13A**, **13B**, and **14** include 3D memory devices **1100** and **1101** depicted in FIGS. **11A** and **11B**. FIGS. **12A-12G**, **13A**, **13B**, and **14** will be described together. It is understood that the operations shown in method **1400** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **14**.

(130) Referring to FIG. **14**, method **1400** starts at operation **1402**, in which a first transistor is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **12A**, a plurality of transistors **1204** and **1206** are formed on a silicon substrate **1202**. Transistors **1204** and **1206** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, chemical mechanical polishing (CMP), and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1202** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1204** and **1206**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1202** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1204** is different from the thickness of gate dielectric of transistor **1206**, for example, by depositing a thicker silicon oxide film in the region of transistor **1204** than the region of transistor **1206**, or by etching back part of the silicon oxide film deposited in the region of transistor **1206**. It is understood that the details of fabricating transistors **1204** and **1206** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(131) In some implementations, an interconnect layer **1208** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **12A**, an interconnect layer **1208** can be formed above transistors **1204** and **1206**. Interconnect layer **1208** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1204** and **1206**. In some implementations, interconnect layer **1208** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1208** can

include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 12A can be collectively referred to as interconnect layer **1208**. In some implementations, the interconnects in interconnect layer **1208** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(132) Method **1400** proceeds to operation **1404**, as illustrated in FIG. 14, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. 12A, a polysilicon layer **1211** is formed above interconnect layer **1208** and transistors **1204** and **1206** on first silicon substrate **1202**. Polysilicon layer **1211** can be formed by depositing polysilicon on interconnect layer **1208** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **1211** is doped with P-type or N-type dopant using in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(133) Method **1400** proceeds to operation **1406**, as illustrated in FIG. 14, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. 12C, a stack structure, such as a memory stack **1226** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **1211**. To form memory stack **1226**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **1211**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1226** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1226** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1226** and polysilicon layer **1211**.

(134) As illustrated in FIG. 12C, NAND memory strings **1228** are formed above polysilicon layer **1211**, each of which extends vertically through memory stack **1226** to be in contact with polysilicon layer **1211**. In some implementations, fabrication processes to form NAND memory string **1228** include forming a channel hole through memory stack **1226** (or the dielectric stack) and into polysilicon layer **1211** using dry etching/and or wet etching, such as deep reactive-ion etching (DRIE), followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1228** may vary depending on the types of channel structures of NAND memory strings **1228** (e.g., channel structure **812** in FIG. 8) and thus, are not elaborated for ease of description.

(135) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. 12C, an interconnect layer **1230** is formed above memory stack **1226** and NAND memory strings **1228**. Interconnect layer **1230** can include interconnects of MEOL

and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1228**. In some implementations, interconnect layer **1230** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1230** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **12D** can be collectively referred to as interconnect layer **1230**.

(136) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. **12C**, one or more contacts **1224** each extending vertically through polysilicon layer **1211** is formed. Contacts **1224** can couple the interconnects in interconnect layers **1230** and **1208**. Contacts **1224** can be formed by first patterning contact holes through polysilicon layer **1211** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(137) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **12C**, a bonding layer **1232** is formed above interconnect layer **1230**. Bonding layer **1232** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1230** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1230** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(138) Method **1400** proceeds to operation **1408**, as illustrated in FIG. **14**, in which a second transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **12D**, a plurality of transistors **1214** and **1216** are formed on a silicon substrate **1210** having single crystalline silicon. Transistors **1214** and **1216** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1210** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1214** and **1216**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1210** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1214** is different from the thickness of gate dielectric of transistor **1216**, for example, by depositing a thicker silicon oxide film in the region of transistor **1214** than the region of transistor **1216**, or by etching back part of the silicon oxide film deposited in the region of transistor **1216**. It is understood that the details of fabricating transistors **1214** and **1216** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(139) In some implementations, an interconnect layer **1220** is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **12D**, an interconnect layer **1220** can be formed above transistors **1214** and **1216**. Interconnect layer **1220** can include interconnects of MEOL and/or BEOL in a plurality

of ILD layers to make electrical connections with transistors **1214** and **1216**. In some implementations, interconnect layer **1220** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1220** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **12C** can be collectively referred to as interconnect layer **1220**. Different from interconnect layer **1208**, in some implementations, the interconnects in interconnect layer **1220** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **1220** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **1220**.

(140) In some implementations, a second bonding layer is formed above the interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **12D**, a bonding layer **1222** is formed above interconnect layer **1220**. Bonding layer **1222** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1220** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1220** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(141) Method **1400** proceeds to operation **1410**, as illustrated in FIG. **14**, in which the first substrate and the second substrate are bonded in a face-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(142) As illustrated in FIG. **12E**, silicon substrate **1210** and components formed thereon (e.g., transistors **1214** and **1216**) are flipped upside down. Bonding layer **1222** facing down is bonded with bonding layer **1232** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **1212**. That is, silicon substrate **1210** and components formed thereon can be bonded with silicon substrate **1202** and components formed thereon in a face-to-face manner, such that the bonding contacts in bonding layer **1232** are in contact with the bonding contacts in bonding layer **1222** at bonding interface **1212**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. Although not shown in FIG. **12E**, it is understood that in some examples, silicon substrate **1202** and components formed thereon (e.g., transistors **1204** **1206**, memory stack **1226** and NAND memory strings **1228**) can be flipped upside down, and bonding layer **1232** facing down can be bonded with bonding layer **1222** facing up, i.e., in a face-to-face manner, thereby forming bonding interface **1212** as well.

(143) As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1212** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1232** and the bonding contacts in bonding layer **1222** are aligned and in contact with one another, such that memory stack **1226** and NAND memory strings **1228** formed therethrough as well as transistors **1204** and **1206** can be coupled to transistors **1214** and **1216** through the bonded

bonding contacts across bonding interface **1212**, according to some implementations.

(144) Method **1400** proceeds to operation **1412**, as illustrated in FIG. **14**, in which the first substrate or the second substrate is thinned. In some implementations, as illustrated in FIG. **12F**, silicon substrate **1210** (shown in FIG. **12E**) is thinned to become a semiconductor layer **1234** having single crystalline silicon. Silicon substrate **1210** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. In some implementations, as illustrated in FIG. **13A**, silicon substrate **1202** (shown in FIG. **12E**) is thinned to become a semiconductor layer **1302** having single crystalline silicon. Similarly, silicon substrate **1202** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. (145) Method **1400** proceeds to operation **1414**, as illustrated in FIG. **14**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned second substrate or on the thinned first substrate.

(146) In some implementations, as illustrated in FIG. **12G**, a pad-out interconnect layer **1236** is formed on semiconductor layer **1234** (the thinned silicon substrate **1210**). Pad-out interconnect layer **1236** can include interconnects, such as contact pads **1238**, formed in one or more ILD layers. Contact pads **1238** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1235** are formed, extending vertically through semiconductor layer **1234**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1235** can couple contact pads **1238** in pad-out interconnect layer **1236** to the interconnects in interconnect layer **1220**. It is understood that in some examples, contacts **1235** may be formed in silicon substrate **1210** before thinning (the formation of semiconductor layer **1234**, e.g., shown in FIG. **12D**) and be exposed from the backside of silicon substrate **1210** (where the thinning occurs) after the thinning.

(147) In some implementations, as illustrated in FIG. **13B**, a pad-out interconnect layer **1306** is formed on semiconductor layer **1302** (the thinned silicon substrate **1202**). Pad-out interconnect layer **1306** can include interconnects, such as contact pads **1308**, formed in one or more ILD layers. Contact pads **1308** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1304** are formed, extending vertically through semiconductor layer **1302**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1304** can couple contact pads **1308** in pad-out interconnect layer **1306** to the interconnects in interconnect layer **1208**. It is understood that in some examples, contacts **1304** may be formed in silicon substrate **1202** before thinning (the formation of semiconductor layer **1234**) and be exposed from the backside of silicon substrate **1202** (where the thinning occurs) after the thinning.

(148) FIGS. **15A** and **15B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **9A** and **9B**, according to various aspects of the present disclosure. 3D memory devices **1500** and **1501** may be examples of 3D memory devices **900** and **901** in FIGS. **9A** and **9B**. As shown in FIG. **15A**, 3D memory device **1500** can include stacked first and second semiconductor structures **102** and **104**. In some implementations, first semiconductor structure **102** includes semiconductor layer **1002**, polysilicon layer **106**, bonding layer **1008**, a memory cell array vertically between polysilicon layer **106** and bonding layer **1008**, and some of the peripheral circuits vertically between semiconductor layer **1002** and polysilicon layer **106**.

(149) The memory cell array can include an array of NAND memory strings (e.g., NAND memory

strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with polysilicon layer **106** (e.g., as shown in FIG. **8**). Polysilicon layer **106** can be a deposited polysilicon layer (e.g., either N-type doped, P-type doped, or undoped), which is suitable for “floating gate” type of NAND memory strings or certain designs of channel structures (e.g., channel structure **812** in FIG. **8**) in “charge trap” type of NAND memory strings, for example, that are suitable for suitable for GIDL erase operations. Bonding layer **1008** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(150) In some implementations, the peripheral circuits in first semiconductor structure **102** are in contact with semiconductor layer **1002**, but not polysilicon layer **106**. That is, the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1002** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. Through contacts (e.g., ILVs/TSVs) through polysilicon layer **106** can make direct, short-distance (e.g., submicron- or micron-level) electrical connections between the memory cell array and the peripheral circuits in first semiconductor structure **102**.

(151) In some implementations, second semiconductor structure **104** includes a semiconductor layer **1004**, a bonding layer **1010**, and some of the peripheral circuits of the memory cell array. Semiconductor layer **1004** can be vertically between the peripheral circuits and bonding layer **1010**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Similar to semiconductor layer **1002**, semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. The peripheral circuits and bonding layer **1010** can be formed on opposite sides of semiconductor layer **1004**, such that semiconductor layer **1004** is disposed vertically between the peripheral circuits and bonding layer **1010**. In some implementations, the transistors of the peripheral circuits are formed on the front side of semiconductor layer **1004**, and the bonding contacts of bonding layer **1010** are formed on the backside of semiconductor layer **1004**.

(152) Similar to bonding layer **1008** in first semiconductor structure **102**, bonding layer **1010** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **103** is vertically between and in contact with bonding layers **1008** and **1010**, respectively, according to some implementations. That is, bonding layers **1008** and **1010** can be disposed on opposite sides of bonding interface **103**, and the bonding contacts of bonding layer **1008** can be in contact with the bonding contacts of bonding layer **1010** at bonding interface **103**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **103**, in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1004**, can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(153) It is understood that in some examples, first and second semiconductor structures **102** and **104** may not include bonding layers **1008** and **1010**, respectively, disposed on opposite sides of bonding interface **103** as shown in FIG. **15A**. In FIG. **15B**, semiconductor layer **1004** in second semiconductor structure **104** of 3D memory device **1501** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of first semiconductor structure **102** by transfer bonding, and bonding interface **103** between first and

second semiconductor structures **102** and **104** can result from transfer bonding, as opposed to hybrid bonding. Through contacts (e.g., ILVs/TSVs) through semiconductor layer **1004** vertically between first and second semiconductor structures **102** and **104** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(154) As shown in FIGS. **15A** and **15B**, since first and second semiconductor structures **102** and **104** are bonded in a face-to-back manner (e.g., semiconductor layers **1002** and **1004** being disposed on the bottom sides of first and second semiconductor structures **102** and **104**, respectively, in FIGS. **15A** and **15B**), the transistors of peripheral circuits in first and second semiconductor structures **102** and **104** are disposed facing toward the same direction (e.g., the positive y-direction in FIGS. **15A** and **15B**), according to some implementations. Moreover, within first semiconductor structure **102**, since polysilicon layer **106** is vertically between the memory cell array and the peripheral circuits, and the memory cell array and the peripheral circuits formed on polysilicon layer **106** and semiconductor layer **1002**, respectively, the memory cell array and the peripheral circuits face toward the same direction (e.g., in the positive y-direction in FIGS. **15A** and **15B**). It is understood that pad-out interconnect layer **902** in FIGS. **9A** and **9B** is omitted from 3D memory devices **1500** and **1501** in FIGS. **15A** and **15B** for ease of illustration and may be included in 3D memory devices **1500** and **1501** as described above with respect to FIGS. **9A** and **9B**.

(155) As described above, first and second semiconductor structures **102** and **104** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and first semiconductor structure **102** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1002** and **1004** in first and second semiconductor structures **102** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1002** in first semiconductor structure **102** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in first and second semiconductor structures **102** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in first semiconductor structure **102** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**.

(156) FIGS. **16A** and **16B** illustrate side views of various examples of 3D memory devices **1500** and **1501** in FIGS. **15A** and **15B**, according to various aspects of the present disclosure. As shown in FIG. **16A**, as one example of 3D memory devices **1500** and **1501** in FIGS. **15A** and **15B**, 3D memory device **1600** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **16A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **103** therebetween, according to some implementations.

(157) As shown in FIG. **16A**, first semiconductor structure **102** can include semiconductor layer **1002** having semiconductor materials. In some implementations, semiconductor layer **1002** is a silicon substrate having single crystalline silicon. First semiconductor structure **102** can also include a device layer **1602** above and in contact with semiconductor layer **1002**. In some implementations, device layer **1602** includes a first peripheral circuit **1604** and a second peripheral circuit **1606**. First peripheral circuit **1604** can include HV circuits **406**, such as driving circuits

(e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and second peripheral circuit **1606** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **1604** includes a plurality of transistors **1608** in contact with semiconductor layer **1002**, and second peripheral circuit **1606** includes a plurality of transistors **1610** in contact with semiconductor layer **1002**. Transistors **1608** and **1610** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1608** or **1610** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1608** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1610** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **1608** than transistor **1610**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **1608** and **1610**) can be formed on or in semiconductor layer **1002** as well.

(158) In some implementations, first semiconductor structure **102** further includes an interconnect layer **1612** above device layer **1602** to transfer electrical signals to and from peripheral circuits **1606** and **1604**. As shown in FIG. **16A**, interconnect layer **1612** can be vertically between polysilicon layer **106** and device layer **1602** (including transistors **1608** and **1610** of peripheral circuits **1604** and **1606**). Interconnect layer **1612** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnects in interconnect layer **1612** can be coupled to transistors **1608** and **1610** of peripheral circuits **1604** and **1606** in device layer **1602**. Interconnect layer **1612** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **1612** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **1602** are coupled to one another through the interconnects in interconnect layer **1612**. For example, peripheral circuit **1604** may be coupled to peripheral circuit **1606** through interconnect layer **1612**. The interconnects in interconnect layer **1612** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1612** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1612** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(159) As shown in FIG. **16A**, first semiconductor structure can further include polysilicon layer **106** above and in contact with interconnect layer **1612**. Polysilicon layer **106** is a doped polysilicon layer on interconnect layer **1612**, as described below in detail with respect to the fabrication process, according to some implementations. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in polysilicon layer **106** as well.

(160) As shown in FIG. **16A**, first semiconductor structure **102** can also include a memory cell array, such as an array of NAND memory strings **208** above and in contact with polysilicon layer **106**. The sources of the NAND memory strings **208** can be in contact with polysilicon layer **106**. In some implementations, polysilicon layer **106** is vertically between NAND memory strings **208** and device layer **1602** including transistors **1608** and **1610**. In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as channel structure **812**, described above in detail with respect to FIG. **8**. In some implementations, NAND memory string **208** is a “floating gate” type of NAND memory string, and polysilicon layer **106** is the source plate of the floating gate type of NAND memory strings.

(161) Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure,

e.g., a memory stack **1627**. Memory stack **1627** may be an example of memory stack **804** in FIG. 8, and the conductive layer and dielectric layer in memory stack **1627** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **1627** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **1627**.

(162) As shown in FIG. 16A, first semiconductor structure **102** can further include an interconnect layer **1628** above and in contact with NAND memory strings **208** to transfer electrical signals to and from NAND memory strings **208**. In some implementations, memory stack **1627** and NAND memory strings **208** are vertically between interconnect layer **1628** and polysilicon layer **106**. Interconnect layer **1628** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **1628** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **1628** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **1628** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1628** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(163) As shown in FIG. 16A, first semiconductor structure **102** can further include one or more contacts **1624** extending vertically through polysilicon layer **106**. In some implementations, contact **1624** couples the interconnects in interconnect layer **1628** to the interconnects in interconnect layer **1612** to make an electrical connection through polysilicon layer **106** between NAND memory strings **208** and transistors **1608** and **1610**. Contact **1624** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1624** includes W. In some implementations, contact **1624** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from polysilicon layer **106**. Depending on the thickness of polysilicon layer **106**, contact **1624** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(164) As shown in FIG. 16A, first semiconductor structure **102** can further include a bonding layer **1008** at bonding interface **103** and above and in contact with interconnect layer **1628**. Bonding layer **1008** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, the bonding contacts of bonding layer **1008** include Cu. The remaining area of bonding layer **1008** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. The bonding contacts and surrounding dielectrics in bonding layer **1008** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO₂-to-SiO₂) bonding simultaneously.

(165) As shown in FIG. 16A, second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a back-to-face manner at bonding interface **103**. Second semiconductor structure **104** can also include a bonding layer **1010** at bonding interface **103**, e.g., on the opposite side of bonding interface **103** with respect to bonding layer **1008** in first semiconductor structure **102**. Bonding layer **1010** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials, such as Cu. The remaining area of bonding layer **1010** can be formed with dielectric

materials, such as silicon oxide. The bonding contacts and surrounding dielectrics in bonding layer **1010** can be used for hybrid bonding. In some implementations, bonding interface **103** is the place at which bonding layers **1008** and **1010** are met and bonded. In practice, bonding interface **103** can be a layer with a certain thickness that includes the top surface of bonding layer **1008** of first semiconductor structure **102** and the bottom surface of bonding layer **1010** of second semiconductor structure **104**.

(166) As shown in FIG. **16A**, second semiconductor structure **104** can further include semiconductor layer **1004** having semiconductor materials. Bonding layer **1010** can be formed on the backside of semiconductor layer **1004** and disposed vertically between semiconductor layer **1004** and bonding interface **103**. In some implementations, semiconductor layer **1004** is a thinned silicon substrate having single crystalline silicon. Second semiconductor structure **104** can also include a device layer **1614** above and in contact with semiconductor layer **1004**. In some implementations, device layer **1614** includes a third peripheral circuit **1616** and a fourth peripheral circuit **1618** above and in contact with semiconductor layer **1004**. In some implementations, semiconductor layer **1004** is disposed vertically between bonding interface **103** and device layer **1614** having peripheral circuits **1616** and **1618**. Third peripheral circuit **1616** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **1618** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **1616** includes a plurality of transistors **1620**, and fourth peripheral circuit **1618** includes a plurality of transistors **1622** as well. Transistors **1620** and **1622** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **1620** or **1622** includes a gate dielectric, and the thickness of the gate dielectric of transistor **1620** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **1622** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **1620** than transistor **1622**.

(167) Moreover, the different voltages applied to different transistors **1620**, **1622**, **1608**, and **1610** in second and first semiconductor structures **104** and **102** can lead to differences of device dimensions between second and first semiconductor structures **104** and **102**. In some implementations, the thickness of the gate dielectric of transistor **1608** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **1620** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **1608** than transistor **1620**. In some implementations, the thickness of the gate dielectric of transistor **1622** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **1610** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **1622** and transistor **1610**. In some implementations, the thickness of semiconductor layer **1002** in which transistor **1608** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **1620** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **1608** than transistor **1620**.

(168) As shown in FIG. **16A**, second semiconductor structure **104** can further include an interconnect layer **1626** above and in contact with device layer **1614** to transfer electrical signals to and from transistors **1620** and **1622** of peripheral circuits **1616** and **1618**. Interconnect layer **1626** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. Interconnect layer **1626** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **1626** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **1626** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **1626** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget

(incompatible with high-temperature processes), since the fabrication of interconnect layer **1626** can occur after the high-temperature processes in forming device layer **1614** in second semiconductor structure **104**, as well as being separated from the high-temperature processes in forming first semiconductor structure **102**, the interconnects of interconnect layer **1626** having Cu can become feasible.

(169) As shown in FIG. **16A**, second semiconductor structure **104** can also include one or more contacts **1630** extending vertically through semiconductor layer **1004**. In some implementations, contact **1630**, in conjunction with bonding contacts across bonding interface **103**, couples the interconnects in interconnect layer **1626** to the interconnects in interconnect layer **1628** to make an electrical connection through semiconductor layer **1004** between NAND memory strings **208** and transistors **1620** and **1622**. Contact **1630** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1630** includes Cu. In some implementations, contact **1630** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **1630** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(170) As shown in FIG. **16A**, second semiconductor structure **104** can further include a pad-out interconnect layer **902** above interconnect layer **1626** and transistors **1620** and **1622**. In some implementations, transistors **1620** and **1622** are disposed vertically between pad-out interconnect layer **902** and semiconductor layer **1004**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **1632**, in one or more ILD layers. Pad-out interconnect layer **902** and interconnect layer **1626** can be formed on the same side of semiconductor layer **1004**. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **1600** and external devices, e.g., for pad-out purposes.

(171) As a result, peripheral circuits **1604**, **1606**, **1616**, and **1618** in first and second semiconductor structures **102** and **104** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **1612**, **1626**, and **1628**, bonding layers **1008** and **1010**, as well as contacts **1624** and **1630**. Moreover, peripheral circuits **1604**, **1606**, **1616**, and **1618** and NAND memory strings **208** in 3D memory device **1600** can be further coupled to external devices through contacts **1630** and pad-out interconnect layer **902**.

(172) It is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having peripheral circuit **1616** as shown in FIG. **16A** (corresponding to FIG. **9A**) and may be from first semiconductor structure **102** having peripheral circuit **1604** (corresponding to FIG. **9B**). For example, as shown in FIG. **16B**, a 3D memory device **1601** may include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1002** of first semiconductor structure **102** on which transistors **1608** of peripheral circuit **1604** are formed. In some implementations, first semiconductor structure **102** further includes one or more contacts **1634** extending vertically through semiconductor layer **1002**. In some implementations, contact **1634** couples the interconnects in interconnect layer **1612** in first semiconductor structure **102** to contact pads **1632** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1002**. Contact **1634** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **1634** includes W. In some implementations, contact **1634** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **1634** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(173) As shown in FIG. 16B, second semiconductor structure **104** can further include a passivation layer **1650**, replacing pad-out interconnect layer **902** in FIG. 16A, to protect and encapsulate 3D memory device **1601** from the side of second semiconductor structure **104** without pad-out interconnect layer **902**. Passivation layer **1650** can include dielectric materials, such as silicon nitride and/or silicon oxide. In some implementations, second semiconductor structure **104** in 3D memory device **1601** further includes a handle/carrier substrate **1651** in contact with passivation layer **1650** as the base substrate of 3D memory device **1601** to provide support. It is understood that in some examples, passivation layer **1650** may be omitted or combined with handle substrate **1651** as a single layer for support and protection.

(174) It is also understood that in some examples, first and second semiconductor structures **102** and **104** of 3D memory device **1601** may not include bonding layers **1008** and **1010**, respectively, as shown in FIG. 16B. Bonding interface **103** can result from a transfer bonding process, and semiconductor layer **1004** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of first semiconductor structure **102** by transfer bonding, as described below in detail with respect to the fabrication process. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **103** and semiconductor layer **1004** and/or between bonding interface **105** and interconnect layer **1628** to facilitate the transfer bonding of semiconductor layer **1004** onto interconnect layer **1628**. Thus, it is understood that bonding interface **103** may include the surfaces of the dielectric layer(s) in some examples. It is further understood that the details of the same components (e.g., materials, fabrication process, functions, etc.) in both 3D memory devices **1600** and **1601** are not repeated for ease of description.

(175) FIGS. 17A-17H illustrate a fabrication process for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure. FIG. 19 illustrates a flowchart of a method **1900** for forming the 3D memory devices in FIGS. 15A and 15B, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. 17A-17H and 19 include 3D memory device **1600** depicted in FIG. 16A. FIGS. 17A-17H and 19 will be described together. It is understood that the operations shown in method **1900** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. 19.

(176) Referring to FIG. 19, method **1900** starts at operation **1902**, in which a first transistor is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 17A, a plurality of transistors **1704** and **1706** are formed on a silicon substrate **1702**. Transistors **1704** and **1706** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1702** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1704** and **1706**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1702** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1704** is different from the thickness of gate dielectric of transistor **1706**, for example, by depositing a thicker silicon oxide film in the region of transistor **1704** than the region of transistor **1706**, or by etching back part of the silicon oxide film deposited in the region of transistor **1706**. It is understood that the details of fabricating transistors **1704** and **1706** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(177) In some implementations, an interconnect layer **1708** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 17A, an interconnect layer **1708** can be formed above transistors **1704**

and **1706**. Interconnect layer **1708** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1704** and **1706**. In some implementations, interconnect layer **1708** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1708** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **17A** can be collectively referred to as interconnect layer **1708**. In some implementations, the interconnects in interconnect layer **1708** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(178) Method **1900** proceeds to operation **1904**, as illustrated in FIG. **19**, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. **17B**, a polysilicon layer **1711** is formed above interconnect layer **1708** and transistors **1704** and **1706** on first silicon substrate **1702**. Polysilicon layer **1711** can be formed by depositing polysilicon on interconnect layer **1708** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **1711** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(179) Method **1900** proceeds to operation **1906**, as illustrated in FIG. **19**, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. **17C**, a stack structure, such as a memory stack **1726** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **1711**. To form memory stack **1726**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **1711**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1726** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1726** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1726** and polysilicon layer **1711**.

(180) As illustrated in FIG. **17C**, NAND memory strings **1728** are formed above polysilicon layer **1711**, each of which extends vertically through memory stack **1726** to be in contact with polysilicon layer **1711**. In some implementations, fabrication processes to form NAND memory string **1728** include forming a channel hole through memory stack **1726** (or the dielectric stack) and into polysilicon layer **1711** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1728** may vary depending on the types of channel structures of NAND memory strings **1728** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(181) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. 17C, an interconnect layer **1730** is formed above memory stack **1726** and NAND memory strings **1728**. Interconnect layer **1730** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1728**. In some implementations, interconnect layer **1730** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1730** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 17C can be collectively referred to as interconnect layer **1730**.

(182) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. 17C, one or more contacts **1724** each extending vertically through polysilicon layer **1711** is formed. Contacts **1724** can couple the interconnects in interconnect layers **1730** and **1708**. Contacts **1724** can be formed by first patterning contact holes through polysilicon layer **1711** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(183) In some implementations, a first bonding layer is formed above interconnect layer. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. 12C, a bonding layer **1732** is formed above interconnect layer **1730**. Bonding layer **1732** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **1730** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **1730** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(184) Method **1900** proceeds to operation **1908**, as illustrated in FIG. 19, in which a second transistor is formed on a front side of a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 17D, a plurality of transistors **1714** and **1716** are formed on the front side of a silicon substrate **1710** having single crystalline silicon. Transistors **1714** and **1716** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1710** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1714** and **1716**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1710** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1714** is different from the thickness of gate dielectric of transistor **1716**, for example, by depositing a thicker silicon oxide film in the region of transistor **1714** than the region of transistor **1716**, or by etching back part of the silicon oxide film deposited in the region of transistor **1716**. It is understood that the details of fabricating transistors **1714** and **1716** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for

ease of description.

(185) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 17D, an interconnect layer **1720** can be formed above transistors **1714** and **1716**. Interconnect layer **1720** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1714** and **1716**. In some implementations, interconnect layer **1720** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1720** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 17D can be collectively referred to as interconnect layer **1720**. Different from interconnect layer **1708**, in some implementations, the interconnects in interconnect layer **1720** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **1720** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **1720**.

(186) In some implementations, a contact through the thinned second substrate is formed. As shown in FIG. 17D, a contact **1718** extending vertically into silicon substrate **1710** from the front side of silicon substrate **1710** is formed. Contacts **1718** can be coupled to the interconnects in interconnect layer **1720**. Contacts **1718** can be formed by first patterning contact holes into silicon substrate **1710** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(187) In some implementations, the second substrate is thinned. As illustrated in FIG. 17E, silicon substrate **1710** (shown in FIG. 17D) is thinned to become a semiconductor layer **1709** having single crystalline silicon. Silicon substrate **1710** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. The thickness of semiconductor layer **1709** can be controlled to expose contact **1718** from the backside of thinned silicon substrate **1710**, for example, by controlling the duration of the CMP process. It is understood that in some examples, contacts **1718** may be formed through semiconductor layer **1709** from the backside thereof after the thinning, as opposed to in silicon substrate **1710** before the thinning. In some implementations, before the thinning, a passivation layer **1721** is formed on interconnect layer **1720** by depositing a dielectric material, such as silicon nitride, on interconnect layer **1720** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A handle substrate **1701** can then be attached to passivation layer **1721**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **1710**, such as thinning, contact formation, and bonding.

(188) In some implementations, a second bonding layer is formed on the backside of the thinned second substrate. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. 17F, a bonding layer **1722** is formed on the backside of semiconductor layer **1709** (i.e., thinned silicon substrate **1710**). Bonding layer **1722** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the backside of semiconductor layer **1709** (opposite to the front side on which transistors **1714** and **1716** are formed) by one or more thin film deposition processes including, but not limited to, CVD,

PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **1718** on the backside of thinned silicon substrate **1710** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(189) Method **1900** proceeds to operation **1910**, as illustrated in FIG. **19**, in which the first substrate and the second substrate are bonded in a face-to-back manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(190) As illustrated in FIG. **17G**, silicon substrate **1702** and components formed thereon (e.g., transistors **1704** and **1706** and NAND memory strings **1728**) and thinned silicon substrate **1710** (i.e., semiconductor layer **1709**) and components formed thereon (e.g., transistors **1714** and **1716**) are bonded in a manner that bonding layer **1732** facing up on the front side of silicon substrate **1702** is bonded with bonding layer **1722** facing down on the backside of thinned silicon substrate **1710** (i.e., a face-to-face manner), thereby forming a bonding interface **1712**. That is, silicon substrate **1702** and components formed thereon can be bonded with thinned silicon substrate **1710** and components formed thereon in a face-to-back manner, such that the bonding contacts in bonding layer **1732** are in contact with the bonding contacts in bonding layer **1722** at bonding interface **1712**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **1712** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **1732** and the bonding contacts in bonding layer **1722** are aligned and in contact with one another, such that memory stack **1726** and NAND memory strings **1728** formed therethrough as well as transistors **1704** and **1706** can be coupled to transistors **1714** and **1716** through the bonded bonding contacts across bonding interface **1712**, according to some implementations.

(191) Method **1900** proceeds to optional operation **1912**, as illustrated in FIG. **19**, in which the first substrate is thinned. As illustrated in FIG. **17H**, silicon substrate **1702** (shown in FIG. **17G**) is thinned to become a semiconductor layer **1703** having single crystalline silicon. Silicon substrate **1702** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(192) Method **1900** proceeds to operation **1914**, as illustrated in FIG. **19**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. As illustrated in FIG. **17H**, a pad-out interconnect layer **1736** is formed on semiconductor layer **1703** (the thinned silicon substrate **1702**). Pad-out interconnect layer **1736** can include interconnects, such as contact pads **1938**, formed in one or more ILD layers. Contact pads **1738** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **1734** are formed, extending vertically through semiconductor layer **1703**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **1734** can couple contact pads **1738** in pad-out interconnect layer **1736** to the interconnects in interconnect layer **1708**. It is understood that in some examples, contacts **1734** may be formed in silicon substrate **1702** before thinning (the formation of semiconductor layer **1703**, e.g., shown in FIG. **17G**) and be exposed from the backside of silicon substrate **1702** (where the thinning occurs) after the thinning.

(193) In some implementations, after operation **1910**, optional operation **1912** is skipped, and method **1900** proceeds to operation **1914**, as illustrated in FIG. **19**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. Although not shown in FIG. **17G**, it is understood that in some examples, a pad-out interconnect layer having contact pads may be formed above interconnect layer **1720** and transistors **1716** and **1714** after removing handle substrate **1701** and passivation layer **1721**.

(194) FIGS. **18A-18F** illustrate another fabrication process for forming the 3D memory devices in FIGS. **15A** and **15B**, according to some aspects of the present disclosure. FIG. **20** illustrates a flowchart of another method **2000** for forming the 3D memory devices in FIGS. **15A** and **15B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **18A-18F** and **20** include 3D memory device **1601** depicted in FIG. **16B**. FIGS. **18A-18F** and **20** will be described together. It is understood that the operations shown in method **2000** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **20**.

(195) Referring to FIG. **20**, method **2000** starts at operation **2002**, in which a first transistor is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **18A**, a plurality of transistors **1804** and **1806** are formed on a silicon substrate **1802**. Transistors **1804** and **1806** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **1802** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1804** and **1806**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **1802** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1804** is different from the thickness of gate dielectric of transistor **1806**, for example, by depositing a thicker silicon oxide film in the region of transistor **1804** than the region of transistor **1806**, or by etching back part of the silicon oxide film deposited in the region of transistor **1806**. It is understood that the details of fabricating transistors **1804** and **1806** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(196) In some implementations, an interconnect layer **1808** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18A**, an interconnect layer **1808** can be formed above transistors **1804** and **1806**. Interconnect layer **1808** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1804** and **1806**. In some implementations, interconnect layer **1808** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1808** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **18A** can be collectively referred to as interconnect layer **1808**. In some implementations, the interconnects in interconnect layer **1808** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(197) Method **2000** proceeds to operation **2004**, as illustrated in FIG. **20**, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. **18B**, a polysilicon layer **1811** is formed above interconnect layer **1808** and transistors **1804** and **1806** on first silicon substrate **1802**.

Polysilicon layer **1811** can be formed by depositing polysilicon on interconnect layer **1808** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **1811** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(198) Method **2000** proceeds to operation **2006**, as illustrated in FIG. **20**, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. **18C**, a stack structure, such as a memory stack **1826** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **1811**. To form memory stack **1826**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **1811**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **1826** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **1826** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **1826** and polysilicon layer **1811**.

(199) As illustrated in FIG. **18C**, NAND memory strings **1828** are formed above polysilicon layer **1811**, each of which extends vertically through memory stack **1826** to be in contact with polysilicon layer **1811**. In some implementations, fabrication processes to form NAND memory string **1828** include forming a channel hole through memory stack **1826** (or the dielectric stack) and into polysilicon layer **1811** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **1828** may vary depending on the types of channel structures of NAND memory strings **1828** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(200) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18C**, an interconnect layer **1830** is formed above memory stack **1826** and NAND memory strings **1828**. Interconnect layer **1830** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **1828**. In some implementations, interconnect layer **1830** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1830** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **18C** can be collectively referred to as interconnect layer **1830**.

(201) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. **18C**, one or more contacts **1817** each extending vertically through polysilicon layer **1811** is

formed. Contacts **1817** can couple the interconnects in interconnect layers **1830** and **1808**. Contacts **1817** can be formed by first patterning contact holes through polysilicon layer **1811** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(202) Method **2000** proceeds to operation **2008**, as illustrated in FIG. **20**, in which a semiconductor layer is formed above the array of NAND memory strings. The semiconductor layer can include single crystalline silicon. In some implementations, to form the semiconductor layer, another substrate and the first substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(203) As illustrated in FIG. **18D**, a semiconductor layer **1810**, such as a single crystalline silicon layer, is formed above interconnect layer **1830** and NAND memory strings **1828**. Semiconductor layer **1810** can be attached above interconnect layer **1830** to form a bonding interface **1812** vertically between semiconductor layer **1810** and interconnect layer **1830**. In some implementations, to form semiconductor layer **1810**, another silicon substrate (not shown in FIG. **18D**) and silicon substrate **1802** are bonded in a face-to-face manner (having the components formed on silicon substrate **1802**, such as NAND memory strings **1828** and transistors **1804** and **1806**, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface **1812**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **1810** attached above interconnect layer **1830**.

(204) FIGS. **34A-34D** illustrate a fabrication process of transfer bonding, according to some aspects of the present disclosure. As illustrated in FIG. **34A**, a function layer **3404** can be formed on a base substrate **3402**. Function layer **3404** can include device layers, interconnect layers, and/or any suitable layers disclosed herein, such as transistors **1804** and **1806**, polysilicon layer **106**, memory stack **1826**, NAND memory strings **1828**, and interconnect layers **1808** and **1830** in FIG. **18C**. A transfer substrate **3406**, such as a silicon substrate having single crystalline silicon, is provided. In some implementations, transfer substrate **3406** is a single crystalline silicon substrate. As illustrated in FIG. **34B**, transfer substrate **3406** and base substrate **3402** (and function layer **3404** formed thereon) can be bonded in a face-to-face manner using any suitable substrate/wafer bonding processes including, for example, anodic bonding and fusion (direct) bonding, thereby forming a bonding interface **3410** between transfer substrate **3406** and base substrate **3402**. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **3410**. For example, silicon oxide layers may be formed on the top surfaces of both transfer substrate **3406** and function layer **3404** to allow SiO₂—SiO₂ bonding using fusion bonding; or, silicon oxide layer may be formed only on function layer **3404** to allow SiO₂—Si bonding using anodic bonding or fusion bonding. In some implementations in which a silicon oxide layer is formed on transfer substrate **3406** (e.g., shown in FIG. **34B**), transfer substrate **3406** can be flipped upside, such that the silicon oxide layer on transfer substrate **3406** faces down toward base substrate **3402** before the bonding.

(205) As illustrated in FIG. **34C**, a cut layer **3412** can be formed in transfer substrate **3406**, for example, using ion implantation. In some implementations, light elements, such as hydrogen ions, are implanted into transfer substrate **3406** to a desired depth, for example, by controlling the energy of the ion impanation process, to form cut layer **3412**. As illustrated in FIG. **34D**, transfer substrate **3406** can be thinned to leave only a semiconductor layer **3414** vertically between cut layer **3412**

and bonding interface **3410**. In some implementations, transfer substrate **3406** is split at cut layer **3412** by applying a mechanical force to transfer substrate **3406**, i.e., peeling off the remainder of transfer substrate **3406** from semiconductor layer **3414**. It is understood that transfer substrate **3406** may be split at cut layer **3412** by any suitable means, not limited to mechanical force alone, such as thermal means, acoustic means, etc., or any combination thereof. As a result, semiconductor layer **3414** can be transferred from transfer substrate **3406** and bonded onto base substrate **3402** (and function layer **3404**) using a transfer bonding process. In some implementations, a planarization process, such as chemical mechanical polishing (CMP), is performed on semiconductor layer **3412** to polish and smooth the top surface of semiconductor layer **3412** and adjust the thickness of semiconductor layer **3412**. Semiconductor layer **3414** thus can have the same material as transfer substrate **3406**, such as single crystalline silicon. The thickness of semiconductor layer **3414** can be determined by the depth of cut layer **3412**, for example, by adjusting the implantation energy, and/or by the planarization process. Moreover, the remainder of transfer substrate **3406** can be re-used in the same manner to form semiconductor layers bonded onto other base substrates, thereby reducing the material cost of the transfer bonding process.

(206) FIGS. **35A-35D** illustrate another fabrication process of transfer bonding, according to some aspects of the present disclosure. As illustrated in FIG. **35A**, function layer **3404** can be formed on base substrate **3402**. Function layer **3404** can include device layers, interconnect layers, and/or any suitable layers disclosed herein, such as transistors **1804** and **1806**, polysilicon layer **106**, memory stack **1826**, NAND memory strings **1828**, and interconnect layers **1808** and **1830** in FIG. **18C**. An SOI substrate **3502**, including a base/handle layer **3504**, a buried oxide layer (BOx) **3506**, and a device layer **3508**, can be flipped upside down facing toward base substrate **3402**. As illustrated in FIG. **35B**, SOI substrate **3502** and base substrate **3402** (and function layer **3404** formed thereon) can be bonded in a face-to-face manner using any suitable substrate/wafer bonding processes including, for example, anodic bonding and fusion (direct) bonding, thereby forming a bonding interface **3512** between SOI substrate **3502** and base substrate **3402**. In one example, fusion bonding may be performed between layers of silicon and silicon, silicon and silicon oxide, or silicon oxide and silicon oxide with pressure and heat. In another example, anodic bonding may be performed between layers of silicon oxide (in an ionic glass) and silicon with voltage, pressure, and heat. It is understood that depending on the bonding process, dielectric layers (e.g., silicon oxide layers) may be formed on one or both sides of bonding interface **3512**. For example, silicon oxide layers may be formed on the top surfaces of both SOI substrate **3502** and function layer **3404** to allow SiO.sub.2—SiO.sub.2 bonding using fusion bonding. Or silicon oxide layer may be formed only on function layer **3404** to allow SiO.sub.2—Si bonding using anodic bonding or fusion bonding.

(207) As illustrated in FIGS. **35C** and **35D**, SOI substrate **3502** (shown in FIG. **35B**) can be thinned by sequentially removing base/handle layer **3504** and buried oxide layer **3506**, for example, using wet/dry etching and/or CMP processes, to leave only device layer **3508** (as a semiconductor layer) at bonding interface **3512**. As a result, device layer **3508** can be transferred from SOI substrate **3502** and bonded onto base substrate **3402** (and function layer **3404**) as a semiconductor layer using another transfer bonding process. The transferred semiconductor layer thus can have the same material as device layer **3508**, such as single crystalline silicon. The thickness of the semiconductor layer can be the same as the thickness of device layer **3508**. It is understood that in some examples, device layer **3508** may be further thinned using wet/dry etching and/or CMP processes, such that the transferred semiconductor layer may be thinner than device layer **3508**.

(208) Referring back to FIG. **20**, method **2000** proceeds to operation **2008**, in which a second transistor is formed on the semiconductor layer. As illustrated in FIG. **18E**, a plurality of transistors **1814** and **1816** are formed on semiconductor layer **1810** having single crystalline silicon.

Transistors **1814** and **1816** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any

other suitable processes. In some implementations, doped regions are formed in semiconductor layer **1810** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **1814** and **1816**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **1810** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **1814** is different from the thickness of gate dielectric of transistor **1816**, for example, by depositing a thicker silicon oxide film in the region of transistor **1814** than the region of transistor **1816**, or by etching back part of the silicon oxide film deposited in the region of transistor **1816**. It is understood that the details of fabricating transistors **1814** and **1816** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(209) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **18E**, an interconnect layer **1820** can be formed above transistors **1814** and **1816**. Interconnect layer **1820** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **1814** and **1816**. In some implementations, interconnect layer **1820** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **1820** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **17D** can be collectively referred to as interconnect layer **1820**. Different from interconnect layer **1808**, in some implementations, the interconnects in interconnect layer **1820** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **1820** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **1820**.

(210) In some implementations, a contact vertically through the semiconductor layer is formed. Contact **1818** can extend vertically through semiconductor layer **1810** from the front side thereof. Contacts **1818** can be coupled to the interconnects in interconnect layer **1820**. Contact **1818** can extend further through a dielectric layer (if any) on the backside of semiconductor layer **1810** to be aligned and in contact with the interconnects in interconnect layer **1830** at bonding interface **1812**. Thus, contact **1818** couples the interconnects in interconnect layer **1820** to the interconnects in interconnect layer **1830** through semiconductor layer **1810** and across bonding interface **1812**. Contacts **1818** can be formed by first patterning contact holes into semiconductor layer **1810** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(211) Method **2000** skips optional operation **2012** and proceeds to operation **2014**, as illustrated in FIG. **20**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor. As illustrated in FIG. **18F**, a pad-out interconnect layer **1836** is formed above interconnect layer **1820** and transistors **1814** and **1816** on semiconductor layer **1810**. Pad-out interconnect layer **1836** can include interconnects, such as contact pads **1838**, formed in one or more ILD layers. Contact pads **1838** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon

oxynitride, low-k dielectrics, or any combination thereof.

(212) In some implementations, to form a pad-out interconnect layer on the first substrate, after operation **2010**, method **2000** proceeds to optional operation **2012**, as illustrated in FIG. **20**, in which the first substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **1802** (shown in FIG. **18E**) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **1802**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **1802** before thinning and be exposed from the backside of silicon substrate **1802** (where the thinning occurs) after the thinning.

(213) Method **2000** proceeds to operation **2014**, as illustrated in FIG. **20**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **1802**.

(214) FIGS. **21A** and **21B** illustrate schematic views of cross-sections of 3D memory devices **2100** and **2101** having two stacked semiconductor structures, according to various aspects of the present disclosure. 3D memory devices **2100** and **2101** may be examples of 3D memory device **101** in FIG. **1B** in which the peripheral circuits of first semiconductor structure **102** are disposed vertically between the memory cell array of first semiconductor structure **102** and the peripheral circuits of second semiconductor structure **104**. That is, the two separate portions of the peripheral circuits can be disposed adjacently in the vertical direction. As shown in FIGS. **21A** and **21B**, second semiconductor structure **104** including some of the peripheral circuits is bonded to first semiconductor structure **102** on the side having the peripheral circuits to form bonding interface **105** between second semiconductor structure **104** and the peripheral circuits of first semiconductor structure **102**, according to some implementations.

(215) Moreover, as shown in FIGS. **21A** and **21B**, 3D memory device **2100** or **2101** can further include a pad-out interconnect layer **902** for pad-out purposes, i.e., interconnecting with external devices using contact pads on which bonding wires can be soldered. In one example shown in FIG. **21B**, second semiconductor structure **104** including some of the peripheral circuits on one side of 3D memory device **2101** may include pad-out interconnect layer **902**, such that 3D memory device **2101** may be pad-out from the peripheral circuit side to reduce the interconnect distance between contact pads and the peripheral circuits, thereby decreasing the parasitic capacitance from the interconnects and improving the electrical performance of 3D memory device **2101**. In another example shown in FIG. **21A**, first semiconductor structure **102** including the memory cell array and some of the peripheral circuits on another side of 3D memory device **2100** may include pad-out interconnect layer **902**, such that 3D memory device **901** may be pad-out from the memory cell array side.

(216) FIGS. **22A** and **22B** illustrate schematic views of cross-sections of the 3D memory devices in FIGS. **21A** and **21B**, according to various aspects of the present disclosure. 3D memory devices **2200** and **2201** may be examples of 3D memory devices **2100** and **2101** in FIGS. **21A** and **21B**. As shown in FIG. **22A**, 3D memory device **2200** can include stacked first and second semiconductor structures **102** and **104**. In some implementations, first semiconductor structure **102** includes semiconductor layer **1002**, a bonding layer **1014**, a memory cell array, some of the peripheral circuits vertically between semiconductor layer **1002** and polysilicon layer **106**, and polysilicon layer **106** vertically between the memory cell array and the peripheral circuits.

(217) The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with polysilicon layer **106** (e.g., as shown in FIG. **8**). Polysilicon layer **106** can be a

deposited polysilicon layer (e.g., either N-type doped, P-type doped, or undoped), which is suitable for “floating gate” type of NAND memory strings or certain designs of channel structures (e.g., channel structure **812** in FIG. **8**) in “charge trap” type of NAND memory strings, for example, that are suitable for GIDL erase operations. Through contacts (e.g., ILVs/TSVs) through polysilicon layer **106** can make direct, short-distance (e.g., submicron- or micron-level) electrical connections between the memory cell array and the peripheral circuits in first semiconductor structure **102**. Bonding layer **1014** can include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(218) In some implementations, the peripheral circuits in first semiconductor structure **102** are in contact with semiconductor layer **1002**, but not polysilicon layer **106**. That is, the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1002** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. The peripheral circuits and bonding layer **1014** can be formed on opposite sides of semiconductor layer **1002**, such that semiconductor layer **1002** is disposed vertically between the peripheral circuits and bonding layer **1014**. In some implementations, the transistors of the peripheral circuits are formed on the front side of semiconductor layer **1002**, and the bonding contacts of bonding layer **1014** are formed on the backside of semiconductor layer **1002**.

(219) In some implementations, second semiconductor structure **104** includes a semiconductor layer **1004**, a bonding layer **1012**, and some of the peripheral circuits of the memory cell array vertically between semiconductor layer **1004** and bonding layer **1012**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1004**. Similar to semiconductor layer **1002**, semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance.

(220) Similar to bonding layer **1014** in first semiconductor structure **102**, bonding layer **1012** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **105** is vertically between and in contact with bonding layers **1012** and **1014**, respectively, according to some implementations. That is, bonding layers **1012** and **1014** can be disposed on opposite sides of bonding interface **105**, and the bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **105**, in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layer **1002**, can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(221) It is understood that in some examples, first and second semiconductor structures **102** and **104** may not include bonding layers **1014** and **1012**, respectively, disposed on opposite sides of bonding interface **105** as shown in FIG. **22A**. In FIG. **22B**, semiconductor layer **1002** in first semiconductor structure **102** of 3D memory device **2201** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of second semiconductor structure **104** by transfer bonding, and bonding interface **105** between first and second semiconductor structures **102** and **104** can result from transfer bonding, as opposed to

hybrid bonding. Through contacts (e.g., ILVs/TSVs) through semiconductor layer **1002** vertically between first and second semiconductor structures **102** and **104** can make direct, short-distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(222) As shown in FIGS. **22A** and **22B**, since first and second semiconductor structures **102** and **104** are bonded in a back-to-face manner (e.g., semiconductor layers **1002** and **1004** being disposed on the bottom sides of first and second semiconductor structures **102**, respectively, in FIGS. **22A** and **22B**), the transistors of peripheral circuits in first and second semiconductor structures **102** and **104** are disposed facing toward the same direction (e.g., the positive y-direction in FIGS. **12A** and **12B**), according to some implementations. Moreover, within first semiconductor structure **102**, since polysilicon layer **106** is vertically between the memory cell array and the peripheral circuits, and the memory cell array and the peripheral circuits formed on polysilicon layer **106** and semiconductor layer **1002**, respectively, the memory cell array and the peripheral circuits face toward the same direction (e.g., in the positive y-direction in FIGS. **22A** and **22B**). It is understood that pad-out interconnect layer **902** in FIGS. **21A** and **21B** is omitted from 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B** for ease of illustration and may be included in 3D memory devices **2200** and **2201** as described above with respect to FIGS. **21A** and **21B**.

(223) As described above, first and second semiconductor structures **102** and **104** can have peripheral circuits having transistors with different applied voltages. For example, first semiconductor structure **102** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and second semiconductor structure **104** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1002** and **1004** in first and second semiconductor structures **102** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and first semiconductor structure **102** may include LLV circuits **402**, and the thickness of semiconductor layer **1002** in first semiconductor structure **102** may be smaller than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in first and second semiconductor structures **102** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, second semiconductor structure **104** may include HV circuits **406** and first semiconductor structure **102** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in second semiconductor structure **104** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in first semiconductor structure **102**.

(224) FIGS. **23A** and **23B** illustrate side views of various examples of 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B**, according to various aspects of the present disclosure. As shown in FIG. **23A**, as one example of 3D memory devices **2200** and **2201** in FIGS. **22A** and **22B**, 3D memory device **2300** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104**, which are stacked over one another in different planes in the vertical direction (e.g., they-direction in FIG. **23A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **105** therebetween, according to some implementations.

(225) As shown in FIG. **23A**, second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a silicon substrate having single crystalline silicon. Second semiconductor structure **104** can also include a device layer **2302** above and in contact with semiconductor layer **1004**. In some implementations, device layer **2302** includes a first peripheral circuit **2304** and a second peripheral circuit **2306**. First peripheral circuit **2304** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line

driver **306**), and second peripheral circuit **2306** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **2304** includes a plurality of transistors **2308** in contact with semiconductor layer **1004**, and second peripheral circuit **2306** includes a plurality of transistors **2310** in contact with semiconductor layer **1004**. Transistors **2308** and **2310** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2308** or **2310** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2308** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2310** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **2308** than transistor **2310**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **2308** and **2310**) can be formed on or in semiconductor layer **1004** as well.

(226) In some implementations, second semiconductor structure **104** further includes an interconnect layer **2312** above device layer **2302** to transfer electrical signals to and from peripheral circuits **2306** and **2304**. As shown in FIG. **23A**, interconnect layer **2312** can be vertically between bonding interface **105** and device layer **2302** (including transistors **2308** and **2310** of peripheral circuits **2304** and **2306**). Interconnect layer **2312** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnects in interconnect layer **2312** can be coupled to transistors **2308** and **2310** of peripheral circuits **2304** and **2306** in device layer **2302**. Interconnect layer **2312** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **2312** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **2302** are coupled to one another through the interconnects in interconnect layer **2312**. For example, peripheral circuit **2304** may be coupled to peripheral circuit **2306** through interconnect layer **2312**. The interconnects in interconnect layer **2312** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2312** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **2312** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(227) As shown in FIG. **23A**, first semiconductor structure **102** can be bonded on top of second semiconductor structure **104** in a back-to-face manner at bonding interface **105**. First semiconductor structure **102** can further include semiconductor layer **1002** having semiconductor materials. In some implementations, bonding interface **105** is the place at which semiconductor layer **1002** of first semiconductor structure **102** and interconnect layer **2312** of second semiconductor structure **104** are met and bonded. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of interconnect layer **2312** of second semiconductor structure **104** and the bottom surface of semiconductor layer **1002** of first semiconductor structure **102**. Bonding interface **105** can result from a transfer bonding process, and semiconductor layer **1002** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the top surface of first semiconductor structure **102** by transfer bonding, as described below in detail with respect to the fabrication process. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1002** and/or between bonding interface **105** and interconnect layer **2312** to facilitate the transfer bonding of semiconductor layer **1002** onto interconnect layer **2312**. Thus, it is understood that bonding interface **105** may include the surfaces of the dielectric layer(s) in some examples.

(228) As shown in FIG. **23A**, first semiconductor structure **102** can also include a device layer **2314** above and in contact with semiconductor layer **1002**. In some implementations, device layer

2314 includes a third peripheral circuit **2316** and a fourth peripheral circuit **2318** above and in contact with semiconductor layer **1002**. In some implementations, semiconductor layer **1002** is disposed vertically between bonding interface **105** and device layer **2314** having peripheral circuits **2316** and **2318**. Third peripheral circuit **2316** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and fourth peripheral circuit **2318** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **2316** includes a plurality of transistors **2320**, and fourth peripheral circuit **2318** includes a plurality of transistors **2322** as well. Transistors **2320** and **2322** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2320** or **2322** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2320** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **2322** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **2320** than transistor **2322**.

(229) Moreover, the different voltages applied to different transistors **2320**, **2322**, **2308**, and **2310** in first and second semiconductor structures **102** and **104** can lead to differences of device dimensions between first and second semiconductor structures **102** and **104**. In some implementations, the thickness of the gate dielectric of transistor **2308** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2320** (e.g., in LLV circuit **402**) due to the higher voltage applied to transistor **2308** than transistor **2320**. In some implementations, the thickness of the gate dielectric of transistor **2322** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **2310** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **2322** and transistor **2310**. In some implementations, the thickness of semiconductor layer **1004** in which transistor **2308** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1002** in which transistor **2320** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **2308** than transistor **2320**.

(230) As shown in FIG. **23A**, first semiconductor structure **102** can further include an interconnect layer **2326** above and in contact with device layer **2314** to transfer electrical signals to and from transistors **2320** and **2322** of peripheral circuits **2316** and **2318**. Interconnect layer **2326** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. Interconnect layer **2326** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2326** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2326** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **2326** include W, which has a relatively high thermal budget (compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(231) As shown in FIG. **23A**, first semiconductor structure **102** can also include one or more contacts **2330** extending vertically through semiconductor layer **1002**. In some implementations, contact **2330** extends further through the dielectric layer (if any) on the backside of semiconductor layer **1002** to be in contact with the interconnects of interconnect layer **2312** at bonding interface **105**. Contact **2330** can thus couple the interconnects in interconnect layer **2326** to the interconnects in interconnect layer **2312** to make an electrical connection through semiconductor layer **1004** and across bonding interface **105** between first and second semiconductor structures **102** and **104**. Contact **2330** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2330** includes W. In some implementations, contact **2330** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **2330** can be an ILV having a depth in the submicron-level

(e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(232) As shown in FIG. 23A, first semiconductor structure can further include polysilicon layer **106** above and in contact with interconnect layer **2326**. Polysilicon layer **106** is a doped polysilicon layer on interconnect layer **2326**, as described below in detail with respect to the fabrication process, according to some implementations. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in polysilicon layer **106** as well.

(233) As shown in FIG. 23A, first semiconductor structure **102** can also include a memory cell array, such as an array of NAND memory strings **208** above and in contact with polysilicon layer **106**. The sources of the NAND memory strings **208** can be in contact with polysilicon layer **106**. In some implementations, polysilicon layer **106** is vertically between NAND memory strings **208** and device layer **2314** including transistors **2320** and **2322**. In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as channel structure **812**, described above in detail with respect to FIG. 8. In some implementations, NAND memory string **208** is a “floating gate” type of NAND memory string, and polysilicon layer **106** is the source plate of the floating gate type of NAND memory strings.

(234) Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **2327**. Memory stack **2327** may be an example of memory stack **804** in FIG. 8, and the conductive layer and dielectric layer in memory stack **2327** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **2327** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **2327**.

(235) As shown in FIG. 23A, first semiconductor structure **102** can further include an interconnect layer **2328** above and in contact with NAND memory strings **208** to transfer electrical signals to and from NAND memory strings **208**. In some implementations, memory stack **2327** and NAND memory strings **208** are vertically between interconnect layer **2328** and polysilicon layer **106**. Interconnect layer **2328** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **2328** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer **2328** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2328** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2328** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(236) As shown in FIG. 23A, first semiconductor structure **102** can further include one or more contacts **2324** extending vertically through polysilicon layer **106**. In some implementations, contact **2324** couples the interconnects in interconnect layer **2328** to the interconnects in interconnect layer **2326** to make an electrical connection through polysilicon layer **106** between NAND memory strings **208** and transistors **2320** and **2322**. Contact **2324** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2324** includes W or Cu. In some implementations, contact **2324** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from polysilicon layer **106**. Depending on the thickness of polysilicon layer **106**, contact **2324** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having

a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(237) As shown in FIG. 23A, first semiconductor structure **102** can further include a pad-out interconnect layer **902** above interconnect layer **2328** and NAND memory strings **208**. In some implementations, NAND memory strings **208** are disposed vertically between pad-out interconnect layer **902** and polysilicon layer **106**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **2332**, in one or more ILD layers. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **2300** and external devices, e.g., for pad-out purposes.

(238) As a result, peripheral circuits **2304**, **2306**, **2316**, and **2318** in second and first semiconductor structures **104** and **102** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **2312**, **2326**, and **2328**, as well as contacts **2324** and **2330**. Moreover, peripheral circuits **2304**, **2306**, **2316**, and **2318** and NAND memory strings **208** in 3D memory device **2300** can be further coupled to external devices through contacts **2330** and pad-out interconnect layer **902**.

(239) It is understood that the pad-out of 3D memory devices is not limited to from first semiconductor structure **102** having NAND memory strings **208** and peripheral circuit **2316** as shown in FIG. 23A (corresponding to FIG. 21A) and may be from second semiconductor structure **104** having peripheral circuit **2304** (corresponding to FIG. 21B). For example, as shown in FIG. 23B, a 3D memory device **2301** may include pad-out interconnect layer **902** in second semiconductor structure **104**. Pad-out interconnect layer **902** can be in contact with semiconductor layer **1004** of second semiconductor structure **104** on which transistors **2308** of peripheral circuit **2304** are formed. In some implementations, second semiconductor structure **104** further includes one or more contacts **2334** extending vertically through semiconductor layer **1004**. In some implementations, contact **2334** couples the interconnects in interconnect layer **2312** in second semiconductor structure **104** to contact pads **2332** in pad-out interconnect layer **902** to make an electrical connection through semiconductor layer **1004**. Contact **2334** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2334** includes W. In some implementations, contact **2334** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **2334** can be an ILV having a thickness in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(240) As shown in FIG. 23B, first semiconductor structure **102** can further include a passivation layer **2350**, replacing pad-out interconnect layer **902** in FIG. 23A, to protect and encapsulate 3D memory device **2301** from the side of first semiconductor structure **102** without pad-out interconnect layer **902**. Passivation layer **2350** can include dielectric materials, such as silicon nitride and/or silicon oxide. In some implementations, first semiconductor structure **102** in 3D memory device **2301** further includes a handle/carrier substrate **2351** in contact with passivation layer **2350** as the base substrate of 3D memory device **2301** to provide support. It is understood that in some examples, passivation layer **2350** may be omitted or combined with handle substrate **2351** as a single layer for support and protection.

(241) It is also understood that in some examples, first and second semiconductor structures **102** and **104** of 3D memory device **2301** may further include bonding layers **1012** and **1014**, respectively, at bonding interface **105** (on opposite sides of bonding interface **105**), as shown in FIG. 23B. That is, bonding interface **105** can result from hybrid bonding, as opposed to transfer bonding. Bonding layer **1012** can be disposed between bonding interface **105** and interconnect layer **2312**, and bonding layer **1014** can be disposed between bonding interface **105** and semiconductor layer **1002**. In some implementations, bonding layer **1014** is formed on the backside of semiconductor layer **1002** (e.g., a thinned silicon substrate) opposite to the front side on which device layer **2314** is formed. In practice, bonding interface **105** can be a layer with a certain

thickness that includes the top surface of bonding layer **1012** of second semiconductor structure **104** and the bottom surface of bonding layer **1014** of first semiconductor structure **102**.

(242) Bonding layers **1012** and **1014** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, the bonding contacts of bonding layers **1012** and **1014** include Cu. The remaining area of bonding layers **1012** and **1014** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. The bonding contacts and surrounding dielectrics in bonding layers **1012** and **1014** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO₂-to-SiO₂) bonding simultaneously.

(243) FIGS. **24A-24F** illustrate a fabrication process for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. FIG. **26** illustrates a flowchart of a method **2600** for forming the 3D memory devices in FIGS. **22A** and **22B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **24A-24F** and **26** include 3D memory device **2300** depicted in FIG. **23A**. FIGS. **24A-24F** and **26** will be described together. It is understood that the operations shown in method **2600** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **26**.

(244) Referring to FIG. **26**, method **2600** starts at operation **2602**, in which a first transistor is formed on a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **24A**, a plurality of transistors **2404** and **2406** are formed on a silicon substrate **2402**. Transistors **2404** and **2406** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2402** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2404** and **2406**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2402** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2404** is different from the thickness of gate dielectric of transistor **2406**, for example, by depositing a thicker silicon oxide film in the region of transistor **2404** than the region of transistor **2406**, or by etching back part of the silicon oxide film deposited in the region of transistor **2406**. It is understood that the details of fabricating transistors **2404** and **2406** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(245) In some implementations, an interconnect layer **2408** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **24A**, an interconnect layer **2408** can be formed above transistors **2404** and **2406**. Interconnect layer **2408** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2404** and **2406**. In some implementations, interconnect layer **2408** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2408** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any

combination thereof. The ILD layers and interconnects illustrated in FIG. 24A can be collectively referred to as interconnect layer **2408**. In some implementations, the interconnects in interconnect layer **2408** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(246) Method **2600** proceeds to operation **2604**, as illustrated in FIG. 26, in which a semiconductor layer is formed above the first transistor. The semiconductor layer can include single crystalline silicon. In some implementations, to form the semiconductor layer, another substrate and the first substrate are bonded in a face-to-face manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(247) As illustrated in FIG. 24B, a semiconductor layer **2410**, such as a single crystalline silicon layer, is formed above interconnect layer **2408** and transistors **2404** and **2406**. Semiconductor layer **2410** can be attached above interconnect layer **2408** to form a bonding interface **2412** vertically between semiconductor layer **2410** and interconnect layer **2408**. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed between bonding interface **2412** and semiconductor layer **2410** and/or between bonding interface **2412** and interconnect layer **2408** to facilitate the transfer bonding of semiconductor layer **2410**. In some implementations, to form semiconductor layer **2410**, another silicon substrate (not shown in FIG. 24B) and silicon substrate **2402** are bonded in a face-to-face manner (having the components formed on silicon substrate **2402**, such as transistors **2404** and **2406**, facing toward the other silicon substrate) using transfer bonding, thereby forming bonding interface **2412**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **2410** attached above interconnect layer **2408**. The details of various transfer bonding processes are described above with respect to FIGS. 34A-34D and FIGS. 35A-35D and thus, are not repeated for ease of description.

(248) Method **2600** proceeds to operation **2606**, in which a second transistor is formed on the semiconductor layer. As illustrated in FIG. 24C, a plurality of transistors **2414** and **2416** are formed on semiconductor layer **2410** having single crystalline silicon. Transistors **2414** and **2416** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **2410** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2414** and **2416**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **2410** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2414** is different from the thickness of gate dielectric of transistor **2416**, for example, by depositing a thicker silicon oxide film in the region of transistor **2414** than the region of transistor **2416**, or by etching back part of the silicon oxide film deposited in the region of transistor **2416**. It is understood that the details of fabricating transistors **2414** and **2416** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(249) In some implementations, an interconnect layer is formed above the transistor. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 24C, an interconnect layer **2420** can be formed above transistors **2414** and **2416**.

Interconnect layer **2420** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2414** and **2416**. In some implementations, interconnect layer **2420** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2420** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin

film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 24C can be collectively referred to as interconnect layer **2420**. In some implementations, the interconnects in interconnect layer **2420** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(250) In some implementations, a contact through the semiconductor layer is formed. Contact **2418** can extend vertically through semiconductor layer **2410** from the front side thereof. Contacts **2418** can be coupled to the interconnects in interconnect layer **2420**. Contact **2418** can extend further through a dielectric layer (if any) on the backside of semiconductor layer **2410** to be aligned and in contact with the interconnects in interconnect layer **2408** at bonding interface **2412**. Thus, contact **2418** couples the interconnects in interconnect layer **2408** to the interconnects in interconnect layer **2420** through semiconductor layer **2410** and across bonding interface **2412**. Contacts **2418** can be formed by first patterning contact holes into semiconductor layer **2410** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(251) Method **2600** proceeds to operation **2608**, as illustrated in FIG. 26, in which a polysilicon layer is formed above the second transistor. As illustrated in FIG. 24D, a polysilicon layer **2411** is formed above interconnect layer **2420** and transistors **2414** and **2416** on semiconductor layer **2410**. Polysilicon layer **2411** can be formed by depositing polysilicon on interconnect layer **2420** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **2411** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(252) Method **2600** proceeds to operation **2010**, as illustrated in FIG. 26, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. 24E, a stack structure, such as a memory stack **2426** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **2411**. To form memory stack **2426**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **2411**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **2426** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **2426** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **2426** and polysilicon layer **2411**.

(253) As illustrated in FIG. 24E, NAND memory strings **2428** are formed above polysilicon layer **2411**, each of which extends vertically through memory stack **2426** to be in contact with polysilicon layer **2411**. In some implementations, fabrication processes to form NAND memory string **2428** include forming a channel hole through memory stack **2426** (or the dielectric stack) and into polysilicon layer **2411** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that

the details of fabricating NAND memory strings **2428** may vary depending on the types of channel structures of NAND memory strings **2428** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(254) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **24E**, an interconnect layer **2430** is formed above memory stack **2426** and NAND memory strings **2428**. Interconnect layer **2430** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **2428**. In some implementations, interconnect layer **2430** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2430** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **24E** can be collectively referred to as interconnect layer **2430**.

(255) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. **24E**, one or more contacts **2417** each extending vertically through polysilicon layer **2411** is formed. Contacts **2417** can couple the interconnects in interconnect layers **2430** and **2420**. Contacts **2417** can be formed by first patterning contact holes through polysilicon layer **2411** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(256) Method **2600** skips optional operation **2612** and proceeds to operation **2614**, as illustrated in FIG. **26**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the array of NAND memory strings. As illustrated in FIG. **24F**, a pad-out interconnect layer **2436** is formed above interconnect layer **2430** and NAND memory strings **2428** on polysilicon layer **2411**. Pad-out interconnect layer **2436** can include interconnects, such as contact pads **2438**, formed in one or more ILD layers. Contact pads **2438** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(257) In some implementations, to form a pad-out interconnect layer on the first substrate, after operation **2610**, method **2600** proceeds to optional operation **2612**, as illustrated in FIG. **26**, in which the first substrate is thinned. It is understood that although not shown, in some examples, silicon substrate **2402** (shown in FIG. **24E**) may be thinned to become a semiconductor layer having single crystalline silicon using processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. After the thinning, contacts may be formed extending vertically through the thinned silicon substrate **2402**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. It is understood that in some examples, the contacts may be formed in silicon substrate **2402** before thinning and be exposed from the backside of silicon substrate **2402** (where the thinning occurs) after the thinning.

(258) Method **2600** proceeds to operation **2614**, as illustrated in FIG. **26**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned first substrate. It is understood that although not shown, in some examples, a pad-out interconnect layer having contact pads may be formed on the thinned silicon substrate **2402**.

(259) FIGS. **25A-25G** illustrate another fabrication process for forming the 3D memory devices in

FIGS. 22A and 22B, according to some aspects of the present disclosure. FIG. 27 illustrates a flowchart of another method **2700** for forming the 3D memory devices in FIGS. 22A and 22B, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. 25A-25G and 27 include 3D memory devices **2301** depicted in FIG. 23B. FIGS. 25A-25G and 27 will be described together. It is understood that the operations shown in method **2700** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. 27.

(260) Referring to FIG. 27, method **2700** starts at operation **2702**, in which a first transistor is formed on a front side of a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 25A, a plurality of transistors **2514** and **2516** are formed on the front side of a silicon substrate **2510**. Transistors **2514** and **2516** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2510** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2514** and **2516**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2510** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2514** is different from the thickness of gate dielectric of transistor **2516**, for example, by depositing a thicker silicon oxide film in the region of transistor **2514** than the region of transistor **2516**, or by etching back part of the silicon oxide film deposited in the region of transistor **2516**. It is understood that the details of fabricating transistors **2514** and **2516** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(261) In some implementations, an interconnect layer **2520** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 25A, an interconnect layer **2520** can be formed above transistors **2514** and **2516**. Interconnect layer **2520** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2514** and **2516**. In some implementations, interconnect layer **2520** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2520** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 25A can be collectively referred to as interconnect layer **2520**. In some implementations, the interconnects in interconnect layer **2520** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(262) In some implementations, a contact through the thinned first substrate is formed. As shown in FIG. 25A, a contact **2518** extending vertically into silicon substrate **2510** from the front side of silicon substrate **2510** is formed. Contacts **2518** can be coupled to the interconnects in interconnect layer **2520**. Contacts **2518** can be formed by first patterning contact holes into silicon substrate **2510** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(263) Method **2700** proceeds to operation **2704**, as illustrated in FIG. 27, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. 25B, a polysilicon layer **2511** is

formed above interconnect layer **2520** and transistors **2514** and **2516** on first silicon substrate **2510**. Polysilicon layer **2511** can be formed by depositing polysilicon on interconnect layer **2520** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **2511** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(264) Method **2700** proceeds to operation **2706**, as illustrated in FIG. **27**, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. **25C**, a stack structure, such as a memory stack **2526** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **2511**. To form memory stack **2526**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **2511**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **2526** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **2526** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **2526** and polysilicon layer **2511**.

(265) As illustrated in FIG. **25C**, NAND memory strings **2528** are formed above polysilicon layer **2511**, each of which extends vertically through memory stack **2526** to be in contact with polysilicon layer **2511**. In some implementations, fabrication processes to form NAND memory string **2528** include forming a channel hole through memory stack **2526** (or the dielectric stack) and into polysilicon layer **2511** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **2528** may vary depending on the types of channel structures of NAND memory strings **2528** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(266) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **25C**, an interconnect layer **2530** is formed above memory stack **2526** and NAND memory strings **2528**. Interconnect layer **2530** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **2528**. In some implementations, interconnect layer **2530** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2530** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **25C** can be collectively referred to as interconnect layer **2530**.

(267) In some implementations, a contact through the polysilicon layer is formed. As illustrated in

FIG. 25C, one or more contacts **2517** each extending vertically through polysilicon layer **2511** is formed. Contacts **2517** can couple the interconnects in interconnect layers **2530** and **2520**. Contacts **2517** can be formed by first patterning contact holes through polysilicon layer **2511** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(268) In some implementations, the first substrate is thinned. As illustrated in FIG. 25E, silicon substrate **2510** (shown in FIG. 25C) is thinned to become a semiconductor layer **2509** having single crystalline silicon. Silicon substrate **2510** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. The thickness of semiconductor layer **2509** can be controlled to expose contact **2518** from the backside of thinned silicon substrate **2510**, for example, by controlling the duration of the CMP process. It is understood that in some examples, contacts **2518** may be formed through semiconductor layer **2509** from the backside thereof after the thinning, as opposed to in silicon substrate **2510** before the thinning. In some implementations, before the thinning, a passivation layer **2523** is formed on interconnect layer **2530** by depositing a dielectric material, such as silicon nitride, on interconnect layer **2530** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A handle substrate **2501** can then be attached to passivation layer **2523**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **2510**, such as thinning, contact formation, and bonding.

(269) In some implementations, a first bonding layer is formed on the backside of the thinned first substrate. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. 25E, a bonding layer **2522** is formed on the backside of semiconductor layer **2509** (i.e., thinned silicon substrate **2510**). Bonding layer **2522** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the backside of semiconductor layer **2509** (opposite to the front side on which transistors **2514** and **2516** are formed) by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **2518** on the backside of thinned silicon substrate **2510** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(270) Method **2700** proceeds to operation **2708**, as illustrated in FIG. 27, in which a second transistor is formed on a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. 25D, a plurality of transistors **2504** and **2506** are formed on a silicon substrate **2502** having single crystalline silicon. Transistors **2504** and **2506** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **2502** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **2504** and **2506**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **2502** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **2504** is different from the thickness of gate dielectric of transistor **2506**, for example, by depositing a thicker silicon oxide film in the region of transistor **2504** than the region of transistor **2506**, or by etching back part of the silicon oxide film deposited in the region of transistor **2506**. It is understood that the details of fabricating transistors **2504** and **2506** may vary

depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(271) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **25D**, an interconnect layer **2508** can be formed above transistors **2504** and **2506**. Interconnect layer **2508** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **2504** and **2506**. In some implementations, interconnect layer **2508** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **2508** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **25D** can be collectively referred to as interconnect layer **2520**. Different from interconnect layer **2520**, in some implementations, the interconnects in interconnect layer **2508** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **2508** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **2508**.

(272) In some implementations, a second bonding layer is formed above interconnect layer. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **25D**, a bonding layer **2521** is formed above interconnect layer **2508**. Bonding layer **2521** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the top surface of interconnect layer **2508** by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with the interconnects in interconnect layer **2508** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(273) Method **2700** proceeds to operation **2710**, as illustrated in FIG. **27**, in which the first substrate and the second substrate are bonded in a back-to-face manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(274) As illustrated in FIG. **25F**, thinned silicon substrate **2510** (i.e., semiconductor layer **2509**) and components formed thereon (e.g., transistors **2514** and **2516** and NAND memory strings **2528**) and silicon substrate **2502** and components formed thereon (e.g., transistors **2504** and **2506**) are bonded in a manner that bonding layer **2521** facing up on the front side of silicon substrate **2502** is bonded with bonding layer **2522** facing down on the backside of thinned silicon substrate **2510** (i.e., a face-to-face manner), thereby forming a bonding interface **2524**. That is, thinned silicon substrate **2510** and components formed thereon can be bonded with silicon substrate **2502** and components formed thereon in a back-to-face manner, such that the bonding contacts in bonding layer **2521** are in contact with the bonding contacts in bonding layer **2522** at bonding interface **2524**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding surfaces prior to bonding. As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **2524** can be inter-mixed.

After the bonding, the bonding contacts in bonding layer **2521** and the bonding contacts in bonding layer **2522** are aligned and in contact with one another, such that memory stack **2526** and NAND memory strings **2528** formed therethrough as well as transistors **2514** and **2516** can be coupled to transistors **2504** and **2506** through the bonded bonding contacts across bonding interface **2524**, according to some implementations.

(275) Method **2700** proceeds to optional operation **2712**, as illustrated in FIG. 27, in which the second substrate is thinned. As illustrated in FIG. 25F, silicon substrate **2502** (shown in FIG. 25E) is thinned to become a semiconductor layer **2503** having single crystalline silicon. Silicon substrate **2502** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof.

(276) Method **2700** proceeds to operation **2714**, as illustrated in FIG. 27, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed on the thinned second substrate. As illustrated in FIG. 25F, a pad-out interconnect layer **2536** is formed on semiconductor layer **2503** (the thinned silicon substrate **2502**). Pad-out interconnect layer **2536** can include interconnects, such as contact pads **2538**, formed in one or more ILD layers. Contact pads **2538** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, after the bonding and thinning, contacts **2534** are formed, extending vertically through semiconductor layer **2503**, for example, by wet/dry etching followed by depositing dielectric materials as spacers and conductive materials as conductors. Contacts **2534** can couple contact pads **2538** in pad-out interconnect layer **2536** to the interconnects in interconnect layer **2508**. It is understood that in some examples, contacts **2534** may be formed in silicon substrate **2502** before thinning (the formation of semiconductor layer **2503**, e.g., shown in FIG. 25F) and be exposed from the backside of silicon substrate **2502** (where the thinning occurs) after the thinning.

(277) In some implementations, after operation **2710**, optional operation **2712** is skipped, and method **2700** proceeds to operation **2714**, as illustrated in FIG. 27, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the array of NAND memory strings. Although not shown in FIG. 25F, it is understood that in some examples, a pad-out interconnect layer having contact pads may be formed above interconnect layer **2530** and NAND memory strings **2528** after removing handle substrate **2501** and passivation layer **2523**.

(278) FIGS. 28A and 28B illustrate schematic views of cross-sections of the 3D memory devices in FIGS. 21A and 21B, according to various aspects of the present disclosure. 3D memory devices **2800** and **2801** may be examples of 3D memory devices **2100** and **2101** in FIGS. 21A and 21B. As shown in FIG. 28A, 3D memory device **2800** can include stacked first and second semiconductor structures **102** and **104**. In some implementations, first semiconductor structure **102** includes semiconductor layer **1002**, a bonding layer **1014**, a memory cell array, some of the peripheral circuits vertically between semiconductor layer **1002** and polysilicon layer **106**, and polysilicon layer **106** vertically between the memory cell array and the peripheral circuits.

(279) The memory cell array can include an array of NAND memory strings (e.g., NAND memory strings **208** disclosed herein), and the sources of the array of NAND memory strings can be in contact with polysilicon layer **106** (e.g., as shown in FIG. 8). Polysilicon layer **106** can be a deposited polysilicon layer (e.g., either N-type doped, P-type doped, or undoped), which is suitable for “floating gate” type of NAND memory strings or certain designs of channel structures (e.g., channel structure **812** in FIG. 8) in “charge trap” type of NAND memory strings, for example, that are suitable for GIDL erase operations. Through contacts (e.g., ILVs/TSVs) through polysilicon layer **106** can make direct, short-distance (e.g., submicron- or micron-level) electrical connections between the memory cell array and the peripheral circuits in first semiconductor structure **102**. Bonding layer **1014** can include conductive bonding contacts (not shown) and dielectrics

electrically isolating the bonding contacts, which can be used, for example, for hybrid bonding as described below in detail.

(280) In some implementations, the peripheral circuits in first semiconductor structure **102** are in contact with semiconductor layer **1002**, but not polysilicon layer **106**. That is, the transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with semiconductor layer **1002**. Semiconductor layer **1002** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1002** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. The peripheral circuits and bonding layer **1014** can be formed on opposite sides of semiconductor layer **1002**, such that semiconductor layer **1002** is disposed vertically between the peripheral circuits and bonding layer **1014**. In some implementations, the transistors of the peripheral circuits are formed on the front side of semiconductor layer **1002**, and the bonding contacts of bonding layer **1014** are formed on the backside of semiconductor layer **1002**.

(281) In some implementations, second semiconductor structure **104** includes bonding layer **1012**, some of the peripheral circuits of the memory cell array, and semiconductor layer **1004** vertically between the peripheral circuits and bonding layer **1012**. The transistors (e.g., planar transistors **500** and 3D transistors **600**) of the peripheral circuits can be in contact with the front side of semiconductor layer **1004**. Similar to semiconductor layer **1002**, semiconductor layer **1004** can include semiconductor materials, such as single crystalline silicon (e.g., a silicon substrate or a thinned silicon substrate). It is understood that in some examples, different from polysilicon layer **106** in first semiconductor structure **102**, semiconductor layer **1004** on which the transistors are formed may include single crystalline silicon, but not polysilicon, due to the superior carrier mobility of single crystalline silicon that is desirable for transistors' performance. The peripheral circuits and bonding layer **1012** can be formed on opposite sides of semiconductor layer **1004**, such that semiconductor layer **1004** is disposed vertically between the peripheral circuits and bonding layer **1012**. In some implementations, the transistors of the peripheral circuits are formed on the front side of semiconductor layer **1004**, and the bonding contacts of bonding layer **1012** are formed on the backside of semiconductor layer **1004**.

(282) Similar to bonding layer **1014** in first semiconductor structure **102**, bonding layer **1012** can also include conductive bonding contacts (not shown) and dielectrics electrically isolating the bonding contacts. Bonding interface **105** is vertically between and in contact with bonding layers **1012** and **1014**, respectively, according to some implementations. That is, bonding layers **1012** and **1014** can be disposed on opposite sides of bonding interface **105**, and the bonding contacts of bonding layer **1012** can be in contact with the bonding contacts of bonding layer **1014** at bonding interface **105**. As a result, a large number (e.g., millions) of bonding contacts across bonding interface **105**, in conjunction with through contacts (e.g., ILVs/TSVs) through semiconductor layers **1002** and **1004**, can make direct, short-distance (e.g., micron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(283) It is understood that in some examples, first and second semiconductor structures **102** and **104** may not include bonding layers **1014** and **1012**, respectively, disposed on opposite sides of bonding interface **105** as shown in FIG. 28A. In FIG. 28B, semiconductor layer **1004** in second semiconductor structure **104** of 3D memory device **2801** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the bottom surface of first semiconductor structure **102** by transfer bonding, and bonding interface **105** between first and second semiconductor structures **102** and **104** can result from transfer bonding, as opposed to hybrid bonding. Through contacts (e.g., ILVs/TSVs) through semiconductor layers **1002** and **1004** vertically between first and second semiconductor structures **102** and **104** can make direct, short-

distance (e.g., submicron-level) electrical connections between adjacent semiconductor structures **102** and **104**.

(284) As shown in FIGS. **28A** and **28B**, since first and second semiconductor structures **102** and **104** are bonded in a back-to-back manner (e.g., semiconductor layers **1002** being disposed on the bottom side of first semiconductor structure **102**, while semiconductor layers **1004** being disposed on the top side of second semiconductor structure **104** in FIGS. **28A** and **28B**), the transistors of peripheral circuits in first and second semiconductor structures **102** and **104** are disposed facing away from each other, according to some implementations. Moreover, within first semiconductor structure **102**, since polysilicon layer **106** is vertically between the memory cell array and the peripheral circuits, and the memory cell array and the peripheral circuits formed on polysilicon layer **106** and semiconductor layer **1002**, respectively, the memory cell array and the peripheral circuits face toward the same direction (e.g., in the positive y-direction in FIGS. **28A** and **28B**). It is understood that pad-out interconnect layer **902** in FIGS. **21A** and **21B** is omitted from 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B** for ease of illustration and may be included in 3D memory devices **2800** and **2801** as described above with respect to FIGS. **21A** and **21B**.

(285) As described above, first and second semiconductor structures **102** and **104** can have peripheral circuits having transistors with different applied voltages. For example, second semiconductor structure **104** may be one example of semiconductor structure **408** including LLV circuits **402** (and LV circuits **404** in some examples) in FIG. **4B**, and first semiconductor structure **102** may be one example of semiconductor structure **410** including HV circuits **406** (and LV circuits **404** in some examples) in FIG. **4B**, or vice versa. Thus, in some implementations, semiconductor layers **1002** and **1004** in first and second semiconductor structures **102** and **104** have different thicknesses to accommodate the transistors with different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of semiconductor layer **1002** in first semiconductor structure **102** may be larger than the thickness of semiconductor layer **1004** in second semiconductor structure **104**. Moreover, in some implementations, the gate dielectrics of the transistors in first and second semiconductor structures **102** and **104** have different thicknesses as well to accommodate the different applied voltages. In one example, first semiconductor structure **102** may include HV circuits **406** and second semiconductor structure **104** may include LLV circuits **402**, and the thickness of the gate dielectrics of the transistors in first semiconductor structure **102** may be larger (e.g., at least 5-fold) than the thickness of the gate dielectrics of the transistors in second semiconductor structure **104**.

(286) FIGS. **29A** and **29B** illustrate side views of various examples of 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B**, according to various aspects of the present disclosure. As shown in FIG. **29A**, as one example of 3D memory devices **2800** and **2801** in FIGS. **28A** and **28B**, 3D memory device **2900** is a bonded chip including first semiconductor structure **102** and second semiconductor structure **104**, which are stacked over one another in different planes in the vertical direction (e.g., the y-direction in FIG. **29A**), according to some implementations. First and second semiconductor structures **102** and **104** are bonded at bonding interface **105** therebetween, according to some implementations.

(287) As shown in FIG. **29A**, second semiconductor structure **104** can include semiconductor layer **1004** having semiconductor materials. In some implementations, semiconductor layer **1004** is a silicon substrate having single crystalline silicon. Second semiconductor structure **104** can also include a device layer **2902** above and in contact with semiconductor layer **1004**. In some implementations, device layer **2902** includes a first peripheral circuit **2904** and a second peripheral circuit **2906**. First peripheral circuit **2904** can include LLV circuits **402**, such as I/O circuits (e.g., in interface **316** and data bus **318**), and second peripheral circuit **2906** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, first peripheral circuit **2904** includes a

plurality of transistors **2908** in contact with semiconductor layer **1004**, and second peripheral circuit **2906** includes a plurality of transistors **2910** in contact with semiconductor layer **1004**. Transistors **2908** and **2910** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2908** or **2910** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2908** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **2910** (e.g., in LV circuit **404**) due to the lower voltage applied to transistor **2908** than transistor **2910**. Trench isolations (e.g., STIs) and doped regions (e.g., wells, sources, and drains of transistors **2908** and **2910**) can be formed on or in semiconductor layer **1004** as well.

(288) In some implementations, second semiconductor structure **104** further includes an interconnect layer **2912** above device layer **2902** to transfer electrical signals to and from peripheral circuits **2906** and **2904**. As shown in FIG. **29A**, device layer **2902** (including transistors **2908** and **2910** of peripheral circuits **2904** and **2906**) can be vertically between bonding interface **105** and interconnect layer **2912**. Interconnect layer **2912** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnects in interconnect layer **2912** can be coupled to transistors **2908** and **2910** of peripheral circuits **2904** and **2906** in device layer **2902**. Interconnect layer **2912** can further include one or more ILD layers in which the lateral lines and vias can form. That is, interconnect layer **2912** can include lateral lines and vias in multiple ILD layers. In some implementations, the devices in device layer **2902** are coupled to one another through the interconnects in interconnect layer **2912**. For example, peripheral circuit **2904** may be coupled to peripheral circuit **2906** through interconnect layer **2912**. The interconnects in interconnect layer **2912** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2912** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(289) In some implementations, the interconnects in interconnect layer **2912** include Cu, which has a relatively low resistivity (better electrical performance) among conductive metal materials. As described below with respect to the fabrication process, although Cu has a relatively low thermal budget (incompatible with high-temperature processes), since the fabrication of interconnect layer **2912** can occur after the high-temperature processes in forming device layer **2902** and NAND memory strings **208** in first semiconductor structure **102**, the interconnects of interconnect layer **1126** having Cu can become feasible.

(290) As shown in FIG. **29A**, second semiconductor structure **104** can further include one or more contacts **2931** extending vertically through semiconductor layer **1004**. In some implementations, contact **2931** extends further through the dielectric layer (if any) on the backside of semiconductor layer **1004** to bonding interface **105**. Contact **2931** can be coupled to the interconnects in interconnect layer **2912**. Contact **2931** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2931** includes W or Cu. In some implementations, contact **2931** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1004**. Depending on the thickness of semiconductor layer **1004**, contact **2931** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μ m and 100 μ m).

(291) As shown in FIG. **29A**, first semiconductor structure **102** can be bonded on top of second semiconductor structure **104** in a back-to-back manner at bonding interface **105**. First semiconductor structure **102** can further include semiconductor layer **1002** having semiconductor materials. In some implementations, bonding interface **105** is the place at which semiconductor layer **1002** of first semiconductor structure **102** and semiconductor layer **1004** of second semiconductor structure **104** are met and bonded. In practice, bonding interface **105** can be a layer

with a certain thickness that includes the top surface of semiconductor layer **1004** of second semiconductor structure **104** and the bottom surface of semiconductor layer **1002** of first semiconductor structure **102**. Bonding interface **105** can result from a transfer bonding process, and semiconductor layer **1004** can be a layer of single crystalline silicon transferred from a silicon substrate or an SOI substrate and attached to the bottom surface of first semiconductor structure **102** by transfer bonding, as described below in detail with respect to the fabrication process. In some implementations, dielectric layer(s) (e.g., silicon oxide layer) are formed vertically between bonding interface **105** and semiconductor layer **1002** and/or between bonding interface **105** and semiconductor layer **1004** to facilitate the transfer bonding of semiconductor layer **1004** onto semiconductor layer **1002**. Thus, it is understood that bonding interface **105** may include the surfaces of the dielectric layer(s) in some examples.

(292) As shown in FIG. **29A**, first semiconductor structure **102** can also include a device layer **2914** below and in contact with semiconductor layer **1002**. In some implementations, device layer **2914** includes a third peripheral circuit **2916** and a fourth peripheral circuit **2918** below and in contact with semiconductor layer **1002**. In some implementations, semiconductor layer **1002** is disposed vertically between bonding interface **105** and device layer **2914** having peripheral circuits **2916** and **2918**. Third peripheral circuit **2916** can include HV circuits **406**, such as driving circuits (e.g., string drivers **704** in row decoder/word line driver **308** and drivers in column decoder/bit line driver **306**), and fourth peripheral circuit **2918** can include LV circuits **404**, such as page buffer circuits (e.g., page buffer circuits **702** in page buffer **304**) and logic circuits (e.g., in control logic **312**). In some implementations, third peripheral circuit **2916** includes a plurality of transistors **2920**, and fourth peripheral circuit **2918** includes a plurality of transistors **2922** as well. Transistors **2920** and **2922** can include any transistors disclosed herein, such as planar transistors **500** and 3D transistors **600**. As described above in detail with respect to transistors **500** and **600**, in some implementations, each transistor **2920** or **2922** includes a gate dielectric, and the thickness of the gate dielectric of transistor **2920** (e.g., in HV circuit **406**) is larger than the thickness of the gate dielectric of transistor **2922** (e.g., in LV circuit **404**) due to the higher voltage applied to transistor **2920** than transistor **2922**.

(293) Moreover, the different voltages applied to different transistors **2920**, **2922**, **2908**, and **2910** in first and second semiconductor structures **102** and **104** can lead to differences of device dimensions between first and second semiconductor structures **102** and **104**. In some implementations, the thickness of the gate dielectric of transistor **2908** (e.g., in LLV circuit **402**) is smaller than the thickness of the gate dielectric of transistor **2920** (e.g., in HV circuit **406**) due to the lower voltage applied to transistor **2908** than transistor **2920**. In some implementations, the thickness of the gate dielectric of transistor **2922** (e.g., in LV circuit **404**) is the same as the thickness of the gate dielectric of transistor **2910** (e.g., in LV circuit **404**) due to the same voltage applied to transistor **2922** and transistor **2910**. In some implementations, the thickness of semiconductor layer **1002** in which transistor **2920** (e.g., in HV circuit **406**) is formed is larger than the thickness of semiconductor layer **1004** in which transistor **2908** (e.g., in LLV circuit **402**) is formed due to the higher voltage applied to transistor **2920** than transistor **2908**.

(294) As shown in FIG. **29A**, first semiconductor structure **102** can further include an interconnect layer **2926** below and in contact with device layer **2914** to transfer electrical signals to and from transistors **2920** and **2922** of peripheral circuits **2916** and **2918**. Interconnect layer **2926** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. Interconnect layer **2926** can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2926** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2926** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. In some implementations, the interconnects in interconnect layer **2926** include W, which has a relatively high thermal budget

(compatible with high-temperature processes) and good quality (fewer defects, e.g., voids) among conductive metal materials.

(295) As shown in FIG. 29A, first semiconductor structure **102** can also include one or more contacts **2930** extending vertically through semiconductor layer **1002**. In some implementations, contact **2930** extends further through the dielectric layer (if any) on the backside of semiconductor layer **1002** to be in contact with contact **2931** at bonding interface **105**. Contact **2930** can be coupled to the interconnects in interconnect layer **2926**. Contacts **2930** and **2931** can thus couple the interconnects in interconnect layer **2926** to the interconnects in interconnect layer **2912** to make an electrical connection through semiconductor layers **1002** and **1004** and across bonding interface **105** between first and second semiconductor structures **102** and **104**. Contact **2930** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2930** includes W. In some implementations, contact **2930** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from semiconductor layer **1002**. Depending on the thickness of semiconductor layer **1002**, contact **2930** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μ m), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μ m and 100 μ m).

(296) As shown in FIG. 23A, first semiconductor structure can further include polysilicon layer **106** below and in contact with interconnect layer **2926**. Polysilicon layer **106** is a doped polysilicon layer on interconnect layer **2926**, as described below in detail with respect to the fabrication process, according to some implementations. It is understood that in some examples, trench isolations and doped regions (not shown) may be formed in polysilicon layer **106** as well.

(297) As shown in FIG. 29A, first semiconductor structure **102** can also include a memory cell array, such as an array of NAND memory strings **208** below and in contact with polysilicon layer **106**. The sources of the NAND memory strings **208** can be in contact with polysilicon layer **106**. In some implementations, polysilicon layer **106** is vertically between NAND memory strings **208** and device layer **2914** including transistors **2920** and **2922**. In some implementations, each NAND memory string **208** is a “charge trap” type of NAND memory string including any suitable channel structures disclosed herein, such as channel structure **812**, described above in detail with respect to FIG. 8. In some implementations, NAND memory string **208** is a “floating gate” type of NAND memory string, and polysilicon layer **106** is the source plate of the floating gate type of NAND memory strings.

(298) Each NAND memory string **208** extends vertically through a plurality of pairs each including a conductive layer and a dielectric layer, according to some implementations. The stacked and interleaved conductive layers and dielectric layers are also referred to herein as a stack structure, e.g., a memory stack **2927**. Memory stack **2927** be an example of memory stack **804** in FIG. 8, and the conductive layer and dielectric layer in memory stack **2927** may be examples of gate conductive layers **806** and dielectric layer **808**, respectively, in memory stack **804**. The interleaved conductive layers and dielectric layers in memory stack **2927** alternate in the vertical direction, according to some implementations. Each conductive layer can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of the conductive layer can extend laterally as a word line, ending at one or more staircase structures of memory stack **2927**.

(299) As shown in FIG. 29A, first semiconductor structure **102** can further include an interconnect layer **2928** below and in contact with NAND memory strings **208** to transfer electrical signals to and from NAND memory strings **208**. In some implementations, memory stack **2927** and NAND memory strings **208** are vertically between interconnect layer **2928** and polysilicon layer **106**. Interconnect layer **2928** can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. In some implementations, the interconnects in interconnect layer **2928** also include local interconnects, such as bit line contacts and word line contacts. Interconnect layer

2928 can further include one or more ILD layers in which the lateral lines and vias can form. The interconnects in interconnect layer **2928** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in interconnect layer **2928** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(300) As shown in FIG. **29A**, first semiconductor structure **102** can further include one or more contacts **2924** extending vertically through polysilicon layer **106**. In some implementations, contact **2924** couples the interconnects in interconnect layer **2928** to the interconnects in interconnect layer **2926** to make an electrical connection through polysilicon layer **106** between NAND memory strings **208** and transistors **2920** and **2922**. Contact **2924** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, contact **2924** includes W or Cu. In some implementations, contact **2924** includes a via surrounded by a dielectric spacer (e.g., having silicon oxide) to electrically separate the via from polysilicon layer **106**. Depending on the thickness of polysilicon layer **106**, contact **2924** can be an ILV having a depth in the submicron-level (e.g., between 10 nm and 1 μm), or a TSV having a depth in the micron- or tens micron-level (e.g., between 1 μm and 100 μm).

(301) As shown in FIG. **29A**, second semiconductor structure **104** can further include a pad-out interconnect layer **902** above interconnect layer **2912** and transistors **2908** and **2910**. In some implementations, transistors **2908** and **2910** are disposed vertically between pad-out interconnect layer **902** and semiconductor layer **1004**. Pad-out interconnect layer **902** can include interconnects, e.g., contact pads **2932**, in one or more ILD layers. In some implementations, the interconnects in pad-out interconnect layer **902** can transfer electrical signals between 3D memory device **2900** and external devices, e.g., for pad-out purposes.

(302) As a result, peripheral circuits **2904**, **2906**, **2916**, and **2918** in second and first semiconductor structures **104** and **102** can be coupled to NAND memory strings **208** in first semiconductor structure **102** through various interconnection structures, including interconnect layers **2912**, **2926**, and **2928**, as well as contacts **2924**, **2930**, and **2931**. Moreover, peripheral circuits **2904**, **2906**, **2916**, and **2918** and NAND memory strings **208** in 3D memory device **2900** can be further coupled to external devices through contacts **2930** and pad-out interconnect layer **902**.

(303) As shown in FIG. **29A**, first semiconductor structure **102** can further include a passivation layer **2950** to protect and encapsulate 3D memory device **2900** from the side of first semiconductor structure **102** without pad-out interconnect layer **902**. Passivation layer **2950** can include dielectric materials, such as silicon nitride and/or silicon oxide. In some implementations, first semiconductor structure **102** in 3D memory device **2900** further includes a handle/carrier substrate **2951** in contact with passivation layer **2950** as the base substrate of 3D memory device **2900** to provide support. It is understood that in some examples, passivation layer **2950** may be omitted or combined with handle substrate **2951** as a single layer for support and protection.

(304) It is understood that the pad-out of 3D memory devices is not limited to from second semiconductor structure **104** having peripheral circuit **2904** as shown in FIG. **29A** (corresponding to FIG. **21B**) and may be from first semiconductor structure **102** having NAND memory strings **208** and peripheral circuit **2916** (corresponding to FIG. **21A**). For example, as shown in FIG. **29B**, a 3D memory device **2901** may include pad-out interconnect layer **902** in first semiconductor structure **102**. Pad-out interconnect layer **902** can be above interconnect layer **2928** and NAND memory strings **208**. As shown in FIG. **29B**, second semiconductor structure **104** can further include passivation layer **2950**, replacing pad-out interconnect layer **902**, to protect and encapsulate 3D memory device **2901** from the side of second semiconductor structure **104** without pad-out interconnect layer **902**. In some implementations, second semiconductor structure **104** in 3D memory device **2901** further includes handle/carrier substrate **2951** in contact with passivation layer **2950** as the base substrate of 3D memory device **2901** to provide support. It is understood that in some examples, passivation layer **2950** may be omitted or combined with handle substrate

2951 as a single layer for support and protection.

(305) It is also understood that in some examples, first and second semiconductor structures **102** and **104** of 3D memory device **2901** may further include bonding layers **1012** and **1014**, respectively, at bonding interface **105** (on opposite sides of bonding interface **105**), as shown in FIG. **29B**. That is, bonding interface **105** can result from hybrid bonding, as opposed to transfer bonding. Bonding layer **1012** can be disposed between bonding interface **105** and semiconductor layer **1004**, and bonding layer **1014** can be disposed between bonding interface **105** and semiconductor layer **1002**. In some implementations, bonding layer **1014** is formed on the backside of semiconductor layer **1002** (e.g., a thinned silicon substrate) opposite to the front side on which device layer **2914** is formed. Similarly, bonding layer **1012** is formed on the backside of semiconductor layer **1004** (e.g., a thinned silicon substrate) opposite to the front side on which device layer **2902** is formed, according to some implementations. In practice, bonding interface **105** can be a layer with a certain thickness that includes the top surface of bonding layer **1012** of second semiconductor structure **104** and the bottom surface of bonding layer **1014** of first semiconductor structure **102**.

(306) Bonding layers **1012** and **1014** can include a plurality of bonding contacts and dielectrics electrically isolating the bonding contacts. The bonding contacts can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. In some implementations, the bonding contacts of bonding layers **1012** and **1014** include Cu. The remaining area of bonding layers **1012** and **1014** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. The bonding contacts and surrounding dielectrics in bonding layers **1012** and **1014** can be used for hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal (e.g., Cu-to-Cu) bonding and dielectric-dielectric (e.g., SiO₂-to-SiO₂) bonding simultaneously.

(307) FIGS. **30A-30G** illustrate a fabrication process for forming the 3D memory devices in FIGS. **29A** and **29B**, according to some aspects of the present disclosure. FIG. **32** illustrates a flowchart of a method **3200** for forming the 3D memory devices in FIGS. **29A** and **29B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **30A-30G** and **32** include 3D memory device **2900** depicted in FIG. **29A**. FIGS. **30A-30G** and **32** will be described together. It is understood that the operations shown in method **3200** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **32**.

(308) Referring to FIG. **32**, method **3200** starts at operation **3202**, in which a first transistor is formed on a front side of a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **30A**, a plurality of transistors **3014** and **3016** are formed on the front side of a silicon substrate **3010**. Transistors **3014** and **3016** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3010** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3014** and **3016**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3010** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3014** is different from the thickness of gate dielectric of transistor **3016**, for example, by depositing a thicker silicon oxide film in the region of transistor **3014** than the region of transistor **3016**, or by etching back part of the silicon oxide film deposited in the region of transistor **3016**. It is understood that the details of fabricating transistors **3014** and **3016** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and

6B) and thus, are not elaborated for ease of description.

(309) In some implementations, an interconnect layer **3020** is formed above the transistor on the first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **30A**, an interconnect layer **3020** can be formed above transistors **3014** and **3016**. Interconnect layer **3020** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3014** and **3016**. In some implementations, interconnect layer **3020** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3020** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **30A** can be collectively referred to as interconnect layer **3020**. In some implementations, the interconnects in interconnect layer **3020** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(310) In some implementations, a contact through the thinned first substrate is formed. As shown in FIG. **30A**, a contact **3018** extending vertically into silicon substrate **3010** from the front side of silicon substrate **3010** is formed. Contacts **3018** can be coupled to the interconnects in interconnect layer **3020**. Contacts **3018** can be formed by first patterning contact holes into silicon substrate **3010** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(311) Method **3200** proceeds to operation **3204**, as illustrated in FIG. **32**, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. **30B**, a polysilicon layer **3011** is formed above interconnect layer **3020** and transistors **3014** and **3016** on first silicon substrate **3010**. Polysilicon layer **3011** can be formed by depositing polysilicon on interconnect layer **3020** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **3011** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(312) Method **3200** proceeds to operation **3206**, as illustrated in FIG. **32**, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. **30C**, a stack structure, such as a memory stack **3026** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **3011**. To form memory stack **3026**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **3011**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **3026** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **3026** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including

silicon oxide is formed between memory stack **3026** and polysilicon layer **3011**.

(313) As illustrated in FIG. **30C**, NAND memory strings **3028** are formed above polysilicon layer **3011**, each of which extends vertically through memory stack **3026** to be in contact with polysilicon layer **3011**. In some implementations, fabrication processes to form NAND memory string **3028** include forming a channel hole through memory stack **3026** (or the dielectric stack) and into polysilicon layer **3011** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **3028** may vary depending on the types of channel structures of NAND memory strings **3028** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(314) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **30C**, an interconnect layer **3030** is formed above memory stack **3026** and NAND memory strings **3028**. Interconnect layer **3030** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **3028**. In some implementations, interconnect layer **3030** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3030** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **30C** can be collectively referred to as interconnect layer **3030**.

(315) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. **30C**, one or more contacts **3017** each extending vertically through polysilicon layer **3011** is formed. Contacts **3017** can couple the interconnects in interconnect layers **3030** and **3020**. Contacts **3017** can be formed by first patterning contact holes through polysilicon layer **3011** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(316) In some implementations, the first substrate is thinned. As illustrated in FIG. **30D**, silicon substrate **3010** (shown in FIG. **30C**) is thinned to become a semiconductor layer **3009** having single crystalline silicon. Silicon substrate **3010** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. The thickness of semiconductor layer **3009** can be controlled to expose contact **3018** from the backside of thinned silicon substrate **3010**, for example, by controlling the duration of the CMP process. It is understood that in some examples, contacts **3018** may be formed through semiconductor layer **3009** from the backside thereof after the thinning, as opposed to in silicon substrate **3010** before the thinning. In some implementations, before the thinning, a passivation layer **3021** is formed on interconnect layer **3030** by depositing a dielectric material, such as silicon nitride, on interconnect layer **3030** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A handle substrate **3001** can then be attached to passivation layer **3021**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **3010**, such as thinning, contact formation, and bonding.

(317) Method **3200** proceeds to operation **3208**, as illustrated in FIG. **32**, in which a semiconductor

layer is formed on a backside of the first substrate. The semiconductor layer can include single crystalline silicon. In some implementations, to form the semiconductor layer, another substrate and the first substrate are bonded in a face-to-back manner, and the other substrate is thinned to leave the semiconductor layer. The bonding can include transfer bonding. The other substrate can be a silicon substrate having single crystalline silicon.

(318) As illustrated in FIG. 30E, a semiconductor layer **3002**, such as a single crystalline silicon layer, is formed on the backside of semiconductor layer **3009** (i.e., thinned silicon substrate **3010**). Semiconductor layer **3002** can be attached to the backside of semiconductor layer **3009** to form a bonding interface **3012** vertically between semiconductor layer **3009** and semiconductor layer **3002**. In some implementations, to form semiconductor layer **3002**, another silicon substrate (not shown in FIG. 30E) and thinned silicon substrate **3010** are bonded in a face-to-back manner (having the components formed on thinned silicon substrate **3010**, such as transistors **3014** and **3016** and NAND memory strings **3028**, facing away from the other silicon substrate) using transfer bonding, thereby forming bonding interface **3012**. The other silicon substrate can then be thinned using any suitable processes to leave semiconductor layer **3002** attached to the backside of thinned silicon substrate **3010**. The details of various transfer bonding processes are described above with respect to FIGS. 34A-34D and FIGS. 35A-35D and thus, are not repeated for ease of description.

(319) Method **3200** proceeds to operation **3210**, in which a second transistor is formed on the semiconductor layer. As illustrated in FIG. 30F, a plurality of transistors **3004** and **3006** are formed on semiconductor layer **3002** having single crystalline silicon. Transistors **3004** and **3006** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in semiconductor layer **3002** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3004** and **3006**. In some implementations, isolation regions (e.g., STIs) are also formed in semiconductor layer **3002** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3004** is different from the thickness of gate dielectric of transistor **3006**, for example, by depositing a thicker silicon oxide film in the region of transistor **3004** than the region of transistor **3006**, or by etching back part of the silicon oxide film deposited in the region of transistor **3006**. It is understood that the details of fabricating transistors **3004** and **3006** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. 5A, 5B, 6A, and 6B) and thus, are not elaborated for ease of description.

(320) In some implementations, an interconnect layer is formed above the transistor. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 30F, an interconnect layer **3008** can be formed above transistors **3004** and **3006**.

Interconnect layer **3008** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3004** and **3006**. In some implementations, interconnect layer **3008** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3008** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 24C can be collectively referred to as interconnect layer **3008**. Different from interconnect layer **3020**, in some implementations, the interconnects in interconnect layer **3008** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **3008** may become feasible since there are no more high-

temperature processes after the fabrication of interconnect layer **3008**.

(321) In some implementations, a contact through the semiconductor layer is formed. Contact **3019** can extend vertically through semiconductor layer **3002** from the front side thereof. Contacts **3019** can be coupled to the interconnects in interconnect layer **3008**. Contact **3019** can extend further through a dielectric layer (if any) on the backside of semiconductor layer **3002** to be aligned and in contact with the contact **3018** at bonding interface **3012**. Thus, contacts **3018** and **3018** can couple the interconnects in interconnect layer **3020** to the interconnects in interconnect layer **3008** through semiconductor layers **3009** and **3002** and across bonding interface **3012**. Contacts **3019** can be formed by first patterning contact holes into semiconductor layer **3002** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(322) Method **3200** proceeds to operation **3212**, as illustrated in FIG. **32**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor or the array of NAND memory strings. As illustrated in FIG. **30G**, a pad-out interconnect layer **3036** is formed above interconnect layer **3008** and transistor **3004** on semiconductor layer **3002**. Pad-out interconnect layer **3036** can include interconnects, such as contact pads **3038**, formed in one or more ILD layers. Contact pads **3038** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. It is understood that although not shown, in some examples, passivation layer **3021** and handle substrate **3001** may be removed to expose interconnect layer **3030**, and pad-out interconnect layer **3036** may be formed above interconnect layer **3030** and NAND memory strings **3028** on polysilicon layer **3011**.

(323) FIGS. **31A-31H** illustrate another fabrication process for forming the 3D memory devices in FIGS. **29A** and **29B**, according to some aspects of the present disclosure. FIG. **33** illustrates a flowchart of another method **3300** for forming the 3D memory devices in FIGS. **29A** and **29B**, according to some aspects of the present disclosure. Examples of the 3D memory devices depicted in FIGS. **31A-31H** and **33** include 3D memory devices **2901** depicted in FIG. **29B**. FIGS. **31A-31H** and **33** will be described together. It is understood that the operations shown in method **3300** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **33**.

(324) Referring to FIG. **33**, method **3300** starts at operation **3302**, in which a first transistor is formed on a front side of a first substrate. The first substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **31A**, a plurality of transistors **3114** and **3116** are formed on the front side of a silicon substrate **3110**. Transistors **3114** and **3116** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3110** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3114** and **3116**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3110** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3114** is different from the thickness of gate dielectric of transistor **3116**, for example, by depositing a thicker silicon oxide film in the region of transistor **3114** than the region of transistor **3116**, or by etching back part of the silicon oxide film deposited in the region of transistor **3116**. It is understood that the details of fabricating transistors **3114** and **3116** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(325) In some implementations, an interconnect layer **3120** is formed above the transistor on the

first substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. 31A, an interconnect layer **3120** can be formed above transistors **3114** and **3116**. Interconnect layer **3120** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3114** and **3116**. In some implementations, interconnect layer **3120** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3120** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. 31A can be collectively referred to as interconnect layer **3120**. In some implementations, the interconnects in interconnect layer **3120** include W, which has a relatively high thermal budget among conductive metal materials to sustain later high-temperature processes.

(326) In some implementations, a contact through the thinned first substrate is formed. As shown in FIG. 31A, a contact **3118** extending vertically into silicon substrate **3110** from the front side of silicon substrate **3110** is formed. Contacts **3118** can be coupled to the interconnects in interconnect layer **3120**. Contacts **3118** can be formed by first patterning contact holes into silicon substrate **3110** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(327) Method **3300** proceeds to operation **3304**, as illustrated in FIG. 33, in which a polysilicon layer is formed above the first transistor. As illustrated in FIG. 31B, a polysilicon layer **3111** is formed above interconnect layer **3120** and transistors **3114** and **3116** on first silicon substrate **3110**. Polysilicon layer **3111** can be formed by depositing polysilicon on interconnect layer **3120** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some implementations, polysilicon layer **3111** is doped with P-type or N-type dopant using an in-situ doping process during the deposition process or ion implantation/diffusion process after the deposition process.

(328) Method **3300** proceeds to operation **3306**, as illustrated in FIG. 33, in which an array of NAND memory strings is formed on the polysilicon layer. In some implementations, to form the array of NAND memory strings, a memory stack is formed on the polysilicon layer. As illustrated in FIG. 31C, a stack structure, such as a memory stack **3126** including interleaved conductive layers and dielectric layers, is formed on polysilicon layer **3111**. To form memory stack **3126**, in some implementations, a dielectric stack (not shown) including interleaved sacrificial layers (not shown) and the dielectric layers is formed on polysilicon layer **3111**. In some implementations, each sacrificial layer includes a layer of silicon nitride, and each dielectric layer includes a layer of silicon oxide. The interleaved sacrificial layers and dielectric layers can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. Memory stack **3126** can then be formed by a gate replacement process, e.g., replacing the sacrificial layers with the conductive layers using wet/dry etch of the sacrificial layers selective to the dielectric layers and filling the resulting recesses with the conductive layers. In some implementations, each conductive layer includes a metal layer, such as a layer of W. It is understood that memory stack **3126** may be formed by alternately depositing conductive layers (e.g., doped polysilicon layers) and dielectric layers (e.g., silicon oxide layers) without the gate replacement process in some examples. In some implementations, a pad oxide layer including silicon oxide is formed between memory stack **3126** and polysilicon layer **3111**.

(329) As illustrated in FIG. 31C, NAND memory strings **3128** are formed above polysilicon layer

3111, each of which extends vertically through memory stack **3126** to be in contact with polysilicon layer **3111**. In some implementations, fabrication processes to form NAND memory string **3128** include forming a channel hole through memory stack **3126** (or the dielectric stack) and into polysilicon layer **3111** using dry etching/and or wet etching, such as DRIE, followed by subsequently filling the channel hole with a plurality of layers, such as a memory film (e.g., a tunneling layer, a storage layer, and a blocking layer) and a semiconductor layer, using thin film deposition processes such as ALD, CVD, PVD, or any combination thereof. It is understood that the details of fabricating NAND memory strings **3128** may vary depending on the types of channel structures of NAND memory strings **3128** (e.g., channel structure **812** in FIG. **8**) and thus, are not elaborated for ease of description.

(330) In some implementations, an interconnect layer is formed above the array of NAND memory strings. The interconnect layer can include a first plurality of interconnects in one or more ILD layers. As illustrated in FIG. **31C**, an interconnect layer **3130** is formed above memory stack **3126** and NAND memory strings **3128**. Interconnect layer **3130** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with NAND memory strings **3128**. In some implementations, interconnect layer **3130** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3130** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **31C** can be collectively referred to as interconnect layer **3130**.

(331) In some implementations, a contact through the polysilicon layer is formed. As illustrated in FIG. **31C**, one or more contacts **3117** each extending vertically through polysilicon layer **3111** is formed. Contacts **3117** can couple the interconnects in interconnect layers **3130** and **3120**. Contacts **3117** can be formed by first patterning contact holes through polysilicon layer **3111** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., W or Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(332) In some implementations, the first substrate is thinned. As illustrated in FIG. **31D**, silicon substrate **3110** (shown in FIG. **31C**) is thinned to become a semiconductor layer **3109** having single crystalline silicon. Silicon substrate **3110** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. The thickness of semiconductor layer **3109** can be controlled to expose contact **3118** from the backside of thinned silicon substrate **3110**, for example, by controlling the duration of the CMP process. It is understood that in some examples, contacts **3118** may be formed through semiconductor layer **3109** from the backside thereof after the thinning, as opposed to in silicon substrate **3110** before the thinning. In some implementations, before the thinning, a passivation layer **3123** is formed on interconnect layer **3130** by depositing a dielectric material, such as silicon nitride, on interconnect layer **3130** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A handle substrate **3101** can then be attached to passivation layer **3123**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **3110**, such as thinning, contact formation, and bonding.

(333) In some implementations, a first bonding layer is formed on the backside of the thinned first substrate. The first bonding layer can include a plurality of first bonding contacts. As illustrated in FIG. **31D**, a bonding layer **3122** is formed on the backside of semiconductor layer **3109** (i.e.,

thinned silicon substrate **3110**). Bonding layer **3122** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the backside of semiconductor layer **3109** (opposite to the front side on which transistors **3114** and **3116** are formed) by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **3118** on the backside of thinned silicon substrate **3110** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(334) Method **3300** proceeds to operation **3308**, as illustrated in FIG. **33**, in which a second transistor is formed on a front side of a second substrate. The second substrate can be a silicon substrate having single crystalline silicon. As illustrated in FIG. **31E**, a plurality of transistors **3104** and **3106** are formed on a silicon substrate **3102** having single crystalline silicon. Transistors **3104** and **3106** can be formed by a plurality of processes including, but not limited to, photolithography, dry/wet etch, thin film deposition, thermal growth, implantation, CMP, and any other suitable processes. In some implementations, doped regions are formed in silicon substrate **3102** by ion implantation and/or thermal diffusion, which function, for example, as wells and source/drain regions of transistors **3104** and **3106**. In some implementations, isolation regions (e.g., STIs) are also formed in silicon substrate **3102** by wet/dry etch and thin film deposition. In some implementations, the thickness of gate dielectric of transistor **3104** is different from the thickness of gate dielectric of transistor **3106**, for example, by depositing a thicker silicon oxide film in the region of transistor **3104** than the region of transistor **3106**, or by etching back part of the silicon oxide film deposited in the region of transistor **3106**. It is understood that the details of fabricating transistors **3104** and **3106** may vary depending on the types of the transistors (e.g., planar transistors **500** or 3D transistors **600** in FIGS. **5A**, **5B**, **6A**, and **6B**) and thus, are not elaborated for ease of description.

(335) In some implementations, an interconnect layer is formed above the transistor on the second substrate. The interconnect layer can include a plurality of interconnects in one or more ILD layers. As illustrated in FIG. **31F**, an interconnect layer **3108** can be formed above transistors **3104** and **3106**. Interconnect layer **3108** can include interconnects of MEOL and/or BEOL in a plurality of ILD layers to make electrical connections with transistors **3104** and **3106**. In some implementations, interconnect layer **3108** includes multiple ILD layers and interconnects therein formed in multiple processes. For example, the interconnects in interconnect layer **3108** can include conductive materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, electroplating, electroless plating, or any combination thereof. Fabrication processes to form interconnects can also include photolithography, CMP, wet/dry etch, or any other suitable processes. The ILD layers can include dielectric materials deposited by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The ILD layers and interconnects illustrated in FIG. **31F** can be collectively referred to as interconnect layer **3108**. Different from interconnect layer **3120**, in some implementations, the interconnects in interconnect layer **3108** include Cu, which has a relatively low resistivity among conductive metal materials. It is understood that although Cu has a relatively low thermal budget (incompatible with high-temperature processes), using Cu as the conductive materials of the interconnects in interconnect layer **3108** may become feasible since there are no more high-temperature processes after the fabrication of interconnect layer **3108**.

(336) In some implementations, a contact through the thinned second substrate is formed. As shown in FIG. **31E**, a contact **3119** extending vertically into silicon substrate **3102** from the front side of silicon substrate **3102** is formed. Contacts **3119** can be coupled to the interconnects in

interconnect layer **3108**. Contacts **3119** can be formed by first patterning contact holes into silicon substrate **3102** using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing a spacer (e.g., a silicon oxide layer) before depositing the conductor.

(337) In some implementations, the second substrate is thinned. As illustrated in FIG. **31F**, silicon substrate **3102** (shown in FIG. **31E**) is thinned to become a semiconductor layer **3103** having single crystalline silicon. Silicon substrate **3102** can be thinned by processes including, but not limited to, wafer grinding, dry etch, wet etch, CMP, any other suitable processes, or any combination thereof. The thickness of semiconductor layer **3103** can be controlled to expose contact **3119** from the backside of thinned silicon substrate **3102**, for example, by controlling the duration of the CMP process. It is understood that in some examples, contacts **3119** may be formed through semiconductor layer **3103** from the backside thereof after the thinning, as opposed to in silicon substrate **3102** before the thinning. In some implementations, before the thinning, a passivation layer **3140** is formed on interconnect layer **3108** by depositing a dielectric material, such as silicon nitride, on interconnect layer **3108** using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. A handle substrate **3141** can then be attached to passivation layer **3140**, for example, using adhesive bonding, prior to the thinning to allow the subsequent backside processes on silicon substrate **3102**, such as thinning, contact formation, and bonding.

(338) In some implementations, a second bonding layer is formed on the backside of the thinned second substrate. The second bonding layer can include a plurality of second bonding contacts. As illustrated in FIG. **31F**, a bonding layer **3121** is formed on the backside of semiconductor layer **3103** (i.e., thinned silicon substrate **3102**). Bonding layer **3121** can include a plurality of bonding contacts surrounded by dielectrics. In some implementations, a dielectric layer is deposited on the backside of semiconductor layer **3103** (opposite to the front side on which transistors **3104** and **3106** are formed) by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. The bonding contacts can then be formed through the dielectric layer and in contact with contacts **3119** on the backside of thinned silicon substrate **3102** by first patterning contact holes through the dielectric layer using patterning process (e.g., photolithography and dry/wet etch of dielectric materials in the dielectric layer). The contact holes can be filled with a conductor (e.g., Cu). In some implementations, filling the contact holes includes depositing an adhesion (glue) layer, a barrier layer, and/or a seed layer before depositing the conductor.

(339) Method **3300** proceeds to operation **3310**, as illustrated in FIG. **33**, in which the first substrate and the second substrate are bonded in a back-to-back manner. The first bonding contact in the first bonding layer can be in contact with the second bonding contact in the second bonding layer at a bonding interface after bonding the first and second substrates. The bonding can include hybrid bonding.

(340) As illustrated in FIG. **31G**, thinned silicon substrate **3110** (i.e., semiconductor layer **3109**) and components formed thereon (e.g., transistors **3114** and **3116** and NAND memory strings **3128**) and thinned silicon substrate **3102** (i.e., semiconductor layer **3103**) and components formed thereon (e.g., transistors **3104** and **3106**) are bonded in a manner that bonding layer **3121** facing up on the backside of thinned silicon substrate **3102** is bonded with bonding layer **3122** facing down on the backside of thinned silicon substrate **3110** (i.e., a back-to-back manner), thereby forming a bonding interface **3112**. That is, thinned silicon substrate **3110** and components formed thereon can be bonded with thinned silicon substrate **3102** and components formed thereon in a back-to-back manner, such that the bonding contacts in bonding layer **3121** are in contact with the bonding contacts in bonding layer **3122** at bonding interface **3112**. In some implementations, a treatment process, e.g., plasma treatment, wet treatment and/or thermal treatment, is applied to bonding

surfaces prior to bonding. As a result of the bonding, e.g., hybrid bonding, the bonding contacts on opposite sides of bonding interface **3112** can be inter-mixed. After the bonding, the bonding contacts in bonding layer **3121** and the bonding contacts in bonding layer **3122** are aligned and in contact with one another, such that memory stack **3126** and NAND memory strings **3128** formed therethrough as well as transistors **3114** and **3116** can be coupled to transistors **3104** and **3106** through the bonded bonding contacts across bonding interface **3112**, according to some implementations.

(341) Method **3300** proceeds to operation **3312**, as illustrated in FIG. **33**, in which a pad-out interconnect layer is formed. The pad-out interconnect layer can be formed above the second transistor or the array of NAND memory strings. As illustrated in FIG. **31H**, passivation layer **3123** and handle substrate **3101** (shown in FIG. **13G**) are removed to expose interconnect layer **3130**, and a pad-out interconnect layer **3136** is formed above interconnect layer **3130** and NAND memory strings **3128** on polysilicon layer **3111**. Pad-out interconnect layer **3136** can include interconnects, such as contact pads **3138**, formed in one or more ILD layers. Contact pads **3138** can include conductive materials including, but not limited to, W, Co, Cu, Al, doped silicon, silicides, or any combination thereof. The ILD layers can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. It is understood that although not shown, in some examples, passivation layer **3140** and handle substrate **3141** may be removed to expose interconnect layer **3108**, and pad-out interconnect layer **3136** may be formed above interconnect layer **3108** and transistors **3104** and **3106** on semiconductor layer **3103**.

(342) FIG. **36** illustrates a block diagram of a system **3600** having a memory device, according to some aspects of the present disclosure. System **3600** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **36**, system **3600** can include a host **3608** and a memory system **3602** having one or more memory devices **3604** and a memory controller **3606**. Host **3608** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **3608** can be configured to send or receive the data to or from memory devices **3604**.

(343) Memory device **3604** can be any memory devices disclosed herein, such as 3D memory devices **100** and **101**. In some implementations, each memory device **3604** includes an array of memory cells, a first peripheral circuit of the array of memory cells, and a second peripheral circuit of the array of memory cells, which are stacked over one another in different planes, as described above in detail.

(344) Memory controller **3606** is coupled to memory device **3604** and host **3608** and is configured to control memory device **3604**, according to some implementations. Memory controller **3606** can manage the data stored in memory device **3604** and communicate with host **3608**. In some implementations, memory controller **3606** is designed for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, memory controller **3606** is designed for operating in a high duty-cycle environment SSDs or embedded multi-media-cards (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. Memory controller **3606** can be configured to control operations of memory device **3604**, such as read, erase, and program operations. In some implementations, memory controller **3606** is configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit. Memory controller **3606** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **3604**.

including, but not limited to bad-block management, garbage collection, logical-to-physical address conversion, wear leveling, etc. In some implementations, memory controller **3606** is further configured to process error correction codes (ECCs) with respect to the data read from or written to memory device **3604**. Any other suitable functions may be performed by memory controller **3606** as well, for example, formatting memory device **3604**. Memory controller **3606** can communicate with an external device (e.g., host **3608**) according to a particular communication protocol. For example, memory controller **3606** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

(345) Memory controller **3606** and one or more memory devices **3604** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, memory system **3602** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. 37A, memory controller **3606** and a single memory device **3604** may be integrated into a memory card **3702**. Memory card **3702** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc. Memory card **3702** can further include a memory card connector **3704** coupling memory card **3702** with a host (e.g., host **3608** in FIG. 36). In another example as shown in FIG. 37B, memory controller **3606** and multiple memory devices **3604** may be integrated into an SSD **3706**. SSD **3706** can further include an SSD connector **3708** coupling SSD **3706** with a host (e.g., host **3608** in FIG. 36). In some implementations, the storage capacity and/or the operation speed of SSD **3706** is greater than those of memory card **3702**.

(346) According to one aspect of the present disclosure, a 3D memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes an array of NAND memory strings, a first peripheral circuit of the array of NAND memory strings including a first transistor, a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, and a first semiconductor layer in contact with the first transistor. The polysilicon layer is in contact with sources of the array of NAND memory strings. The second semiconductor structure includes a second peripheral circuit of the array of NAND memory strings including a second transistor, and a second semiconductor layer in contact with the second transistor. The second semiconductor layer is between the bonding interface and the second peripheral circuit. The polysilicon layer is between the first semiconductor layer and the second semiconductor layer.

(347) In some implementations, the first peripheral circuit is between the first semiconductor layer and the polysilicon layer.

(348) In some implementations, each of the first and second semiconductor layers includes single crystalline silicon.

(349) In some implementations, a thickness of the first semiconductor layer is greater than a thickness of the second semiconductor layer.

(350) In some implementations, the first transistor includes a first gate dielectric, the second transistor includes a second gate dielectric, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(351) In some implementations, a difference between the thicknesses of the first and second gate dielectrics is at least 5-fold.

(352) In some implementations, the first semiconductor structure further includes a third peripheral

circuit of the array of NAND memory strings, and the third peripheral circuit includes a third transistor including a third gate dielectric. In some implementations, the second semiconductor structure further includes a fourth peripheral circuit of the array of NAND memory strings, and the fourth peripheral circuit including a fourth transistor including a fourth gate dielectric.

(353) In some implementations, the third and fourth gate dielectrics have a same thickness.

(354) In some implementations, the thickness of the third and fourth gate dielectrics is between the thicknesses of the first and second gate dielectrics.

(355) In some implementations, the third and fourth peripheral circuits include at least one of a page buffer circuit or a logic circuit.

(356) In some implementations, the first semiconductor structure further includes a first interconnect layer between the polysilicon layer and the first peripheral circuit, and the first interconnect layer includes a first interconnect coupled to the first transistor. In some implementations, the second semiconductor structure further includes a second interconnect layer such that the second peripheral circuit is between the bonding interface and the second interconnect layer, and the second interconnect layer includes a second interconnect coupled to the second transistor.

(357) In some implementations, the first interconnect includes tungsten, and the second interconnect includes copper.

(358) In some implementations, the first semiconductor structure further includes a third interconnect layer between the bonding interface and the array of NAND memory strings, and a first contact through the polysilicon layer and coupling the third interconnect to the first interconnect. In some implementations, the second semiconductor structure further includes a second contact through the second semiconductor layer and coupling the third interconnect to the second interconnect.

(359) In some implementations, the first semiconductor structure further includes a first pad-out interconnect layer in contact with the first semiconductor layer, or the second semiconductor structure further includes a second pad-out interconnect layer above the second transistor.

(360) In some implementations, the first peripheral circuit includes a driving circuit, and the second peripheral circuit includes an I/O circuit.

(361) In some implementations, the 3D memory device further includes a first voltage source coupled to the first peripheral circuit and configured to provide a first voltage to the first peripheral circuit, and a second voltage source coupled to the second peripheral circuit and configured to provide a second voltage to the second peripheral circuit. In some implementations, the first voltage is greater than the second voltage.

(362) In some implementations, the first semiconductor structure further includes a first bonding layer including a first bonding contact such that the array of NAND memory strings is between the first bonding layer and the polysilicon layer. In some implementations, the second semiconductor structure further includes a second bonding layer including a second bonding contact such that the second semiconductor layer is between the second bonding layer and the second peripheral circuit. In some implementations, the first bonding contact is in contact with the second bonding contact at the bonding interface.

(363) In some implementations, the first transistor is formed on the first semiconductor layer, and the second transistor is formed on the second semiconductor layer.

(364) According to another aspect of the present disclosure, a system includes a memory device configured to store data. The memory device includes a first semiconductor structure, a second semiconductor structure, and a bonding interface between the first semiconductor structure and the second semiconductor structure. The first semiconductor structure includes an array of NAND memory strings, a first peripheral circuit of the array of NAND memory strings including a first transistor, a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, and a first semiconductor layer in contact with the first transistor. The polysilicon layer is in

contact with sources of the array of NAND memory strings. The second semiconductor structure includes a second peripheral circuit of the array of NAND memory strings including a second transistor, and a second semiconductor layer in contact with the second transistor. The second semiconductor layer is between the bonding interface and the second peripheral circuit. The polysilicon layer is between the first semiconductor layer and the second semiconductor layer. The system also includes a memory controller coupled to the memory device and configured to control the array of memory cells through the first peripheral circuit and the second peripheral circuit.

(365) According to still another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. A first transistor is formed on a first substrate. A polysilicon layer is formed above the first transistor on the first substrate. An array of NAND memory strings is formed on the polysilicon layer. A second transistor is formed on a front side a second substrate. The first substrate and the second substrate are bonded in a face-to-back manner.

(366) In some implementations, a pad-out interconnect layer is formed the second transistor after bonding the first and second substrates.

(367) In some implementations, the first substrate is thinned after bonding the first and second substrates, and a pad-out interconnect layer is formed on the thinned first substrate.

(368) In some implementations, bonding the first and second substrates includes hybrid bonding.

(369) In some implementations, a first bonding layer including a first bonding contact is formed above the array of NAND memory strings, a second bonding layer including a second bonding contact is formed on a backside of the second substrate, and the first bonding contact is in contact with the second bonding contact at a bonding interface after bonding the first and second substrates.

(370) In some implementations, a first contact is formed through the polysilicon layer.

(371) In some implementations, the second substrate is thinned before bonding the first and second substrates, and a second contact is formed through the thinned second substrate.

(372) In some implementations, to form the first transistor, a first gate dielectric is formed, to form the second transistor, a second gate dielectric is formed, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(373) According to yet another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. A first transistor is formed on a first substrate. A polysilicon layer is formed above the first transistor on the first substrate. An array of NAND memory strings is formed on the polysilicon layer. A semiconductor layer is formed on above the array of NAND memory strings. The semiconductor layer includes single crystalline silicon. A second transistor is formed on the semiconductor layer.

(374) In some implementations, a pad-out interconnect layer is formed above the second transistor.

(375) In some implementations, the first substrate is thinned after forming the second transistor, and a pad-out interconnect layer is formed on the thinned first substrate.

(376) In some implementations, a first contact is formed through the polysilicon layer before forming the semiconductor layer.

(377) In some implementations, a second contact is formed through the semiconductor layer.

(378) In some implementations, to form the semiconductor layer, a second substrate and the first substrate are bonded, and the second substrate is thinned to leave the semiconductor layer.

(379) In some implementations, bonding the second and first substrates includes transfers bonding.

(380) In some implementations, to form the first transistor, a first gate dielectric is formed, to form the second transistor, a second gate dielectric is formed, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

(381) The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.

(382) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary implementations, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A three-dimensional (3D) memory device, comprising: a first semiconductor structure, comprising: an array of NAND memory strings; a first peripheral circuit of the array of NAND memory strings, the first peripheral circuit comprising a first transistor; a polysilicon layer between the array of NAND memory strings and the first peripheral circuit, the polysilicon layer being in contact with sources of the array of NAND memory strings; and a first semiconductor layer in contact with the first transistor; a second semiconductor structure, comprising: a second peripheral circuit of the array of NAND memory strings, the second peripheral circuit comprising a second transistor; and a second semiconductor layer in contact with the second transistor; and a bonding interface between the first semiconductor structure and the second semiconductor structure, wherein the second semiconductor layer is between the second peripheral circuit and the bonding interface; the second semiconductor layer and the array of NAND memory strings are between the second peripheral circuit and the polysilicon layer; the polysilicon layer is between the first semiconductor layer and the second semiconductor layer; the first transistor comprises a first gate dielectric; the second transistor comprises a second gate dielectric; and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.
2. The 3D memory device of claim 1, wherein the first peripheral circuit is between the first semiconductor layer and the polysilicon layer.
3. The 3D memory device of claim 1, wherein each of the first and second semiconductor layers comprises single crystalline silicon.
4. The 3D memory device of claim 1, wherein a thickness of the first semiconductor layer is greater than a thickness of the second semiconductor layer.
5. The 3D memory device of claim 1, wherein a difference between the thicknesses of the first and second gate dielectrics is between 5-fold and 50-fold.
6. The 3D memory device of claim 1, wherein the first semiconductor structure further comprises a third peripheral circuit of the array of NAND memory strings, the third peripheral circuit comprising a third transistor comprising a third gate dielectric; and the second semiconductor structure further comprises a fourth peripheral circuit of the array of NAND memory strings, the fourth peripheral circuit comprising a fourth transistor comprising a fourth gate dielectric.
7. The 3D memory device of claim 6, wherein the third and fourth gate dielectrics have a same thickness.
8. The 3D memory device of claim 7, wherein the thickness of the third and fourth gate dielectrics is between the thicknesses of the first and second gate dielectrics.
9. The 3D memory device of claim 6, wherein the third and fourth peripheral circuits comprise at least one of a page buffer circuit or a logic circuit.
10. The 3D memory device of claim 1, wherein the first semiconductor structure further comprises a first interconnect layer between the polysilicon layer and the first peripheral circuit, the first interconnect layer comprising a first interconnect coupled to the first transistor; and the second semiconductor structure further comprises a second interconnect layer such that the second peripheral circuit is between the bonding interface and the second interconnect layer, the second interconnect layer comprising a second interconnect coupled to the second transistor.
11. The 3D memory device of claim 10, wherein the first interconnect comprises tungsten, and the second interconnect comprises copper.
12. The 3D memory device of claim 10, wherein the first semiconductor structure further comprises: a third interconnect layer between the bonding interface and the array of NAND

memory strings, the third interconnect layer comprising a third interconnect coupled to the array of NAND memory strings; and a first contact through the polysilicon layer and coupling the third interconnect to the first interconnect; and the second semiconductor structure further comprises a second contact through the second semiconductor layer and coupling the third interconnect to the second interconnect.

13. The 3D memory device of claim 1, wherein the first semiconductor structure further comprises a first pad-out interconnect layer in contact with the first semiconductor layer; or the second semiconductor structure further comprises a second pad-out interconnect layer above the second transistor.

14. The 3D memory device of claim 1, wherein the first peripheral circuit comprises a driving circuit, and the second peripheral circuit comprises an input/output (I/O) circuit.

15. The 3D memory device of claim 1, further comprising: a first voltage source coupled to the first peripheral circuit and configured to provide a first voltage to the first peripheral circuit; and a second voltage source coupled to the second peripheral circuit and configured to provide a second voltage to the second peripheral circuit, wherein the first voltage is greater than the second voltage.

16. The 3D memory device of claim 1, wherein the first semiconductor structure further comprises a first bonding layer such that the array of NAND memory strings is between the first bonding layer and the polysilicon layer, the first bonding layer comprising a first bonding contact; the second semiconductor structure further comprises a second bonding layer such that the second semiconductor layer is between the second bonding layer and the second peripheral circuit, the second bonding layer comprising a second bonding contact; and the first bonding contact is in contact with the second bonding contact at the bonding interface.

17. The 3D memory device of claim 1, wherein the first transistor is formed on the first semiconductor layer, and the second transistor is formed on the second semiconductor layer.

18. A three-dimensional (3D) memory device, comprising: a first semiconductor structure, comprising: NAND memory strings, first transistors on a first semiconductor layer, and a polysilicon layer between the NAND memory strings and the first transistors, the polysilicon layer functions as a common source plate of the NAND memory strings; and a second semiconductor structure, comprising second transistors on a second semiconductor layer, wherein the first semiconductor structure is bonded with the second semiconductor structure, the second semiconductor layer is between the second transistors and the NAND memory strings, the second semiconductor layer and the NAND memory strings are between the second transistors and the polysilicon layer, the polysilicon layer is between the first semiconductor layer and the second semiconductor layer, each first transistor comprises a first gate dielectric, each second transistor comprises a second gate dielectric, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.

19. A three-dimensional (3D) memory device, comprising: NAND memory strings on a polysilicon layer, the polysilicon layer functioning as a common source plate of the NAND memory strings; a first peripheral circuit of the NAND memory strings, comprising a first transistor on a first semiconductor layer; a second peripheral circuit of the NAND memory strings, comprising a second transistor on a second semiconductor layer; and a contact structure penetrating the polysilicon layer and the second semiconductor layer and coupled between the first transistor and the second transistor, wherein the NAND memory strings are between the first peripheral circuit and the second peripheral circuit, the second semiconductor layer is between the second peripheral circuit and the NAND memory strings, the second semiconductor layer and the NAND memory strings are between the second peripheral circuit and the polysilicon layer, the first transistor comprises a first gate dielectric, the second transistor comprises a second gate dielectric, and a thickness of the first gate dielectric is greater than a thickness of the second gate dielectric.
